Working, but in Slums: Climate Migration and Integration in Urban Africa

Mélanie Gittard*and Eleanor Wiseman†

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Abstract

Does climate-induced migration contribute to the growth of slums in African cities? We build bilateral yearly migration flows between districts of eight African countries from harmonized census microdata covering 1981–2018, and instrument migration with drought intensity at origin, selected by post-IV LASSO among candidate climate variables. Drought-induced migration raises the inflow into low-quality housing: of every hundred migrants that drought pushes into a city, about twenty-eight settle in low-quality housing, which over the period amounts to roughly 110,000 additional people. Four fifths of the effect operates through the number of arrivals rather than through a change in who they are. These migrants are not economically excluded but absorbed into employment: 85% of those who end up in low-quality housing are employed, against 61% of those who do not. The cost of climate migration is thus residential rather than occupational, and it is a matter of destination, as the working poor that climate migration adds land on small and secondary cities. Drought does not displace the least-educated adults, who remain trapped at origin, and amongst those who migrate the least educated end up more often in slums. These results shift the policy focus toward residential absorption in the fast-growing secondary cities, a place-based lever that reaches climate migrants without having to identify them.

Keywords: Climate Change, Migration, Low-Quality Housing, Urbanization, Labor Markets

JEL codes: Q54, R23, O15, O18, J61

*Stanford University, Center on Food Security and the Environment (FSE). mgittard@stanford.edu

†University of Wisconsin-Madison. wiseman3@wisc.edu.

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1 Introduction

African cities stand at the intersection of two major transitions: rapid urbanization and intensifying climate change and variability. By 2050, an estimated 68% of the continent's population will reside in urban areas, up from 41% in 2020 (Un-Habitat, 2020). This urban transition comes with major deficits in both housing and employment: 62.7% of Sub-Saharan Africa's urban population currently lives in slums (Un-Habitat, 2022), and a substantial share of urban residents remain employed in agriculture rather than in manufacturing or services (Henderson and Kriticos, 2018). The growth of such settlements is among the central challenges of African urban policy. Slums are hard to service with water, sanitation and electricity, difficult to tax and govern, and leave residents without secure property rights and exposed to crime, fire, and climate hazards such as flooding and extreme heat (Marx et al., 2019). At the same time, climate variability increasingly affects rural livelihoods: droughts deteriorate agricultural productivity, deplete household incomes, and push rural populations to migrate as a coping strategy (Henderson et al., 2017; Mueller et al., 2020). Yet where these climate-driven migrants settle, under what conditions, and whether they find work at destination remain poorly understood. Whether climate migration contributes to the growth of slums in African cities, where the pressure lands, and through which mechanism, is therefore a question of first-order policy relevance.

This paper asks whether climate migration fuels the growth of African slums, and whether it does so because of who the migrants are or because of where they land. We study two margins of integration, housing and employment. We identify the causal effect of migration by instrumenting bilateral migration flows between an origin and a destination — a corridor — with drought intensity at origin, in a yearly panel built from census microdata across eight African countries. The first challenge is measurement: studying migrants' outcomes at destination requires knowing how many migrants move along each corridor in each year, bilateral flows at annual frequency that existing data for Africa seldom provide. A first contribution is to build such a panel from IPUMS census microdata: retrospective questions recording each migrant's administrative unit of origin and year of arrival let us reconstruct yearly cohorts of migrants for every origin–destination pair at the admin-2 level. This cohort-bilateral panel, built from censuses over 1990–2019 with migration windows covering 1981–2018, is matched at the household level to a multi-criteria measure of housing quality based on the UN-Habitat slum definition and to each migrant's employment status. The second main contribution of this paper is to investiage what happens to climate migrants at destination for the first time in the African context, which has been abled thanks to the construction of this data. The second challenge is endogeneity: corridors with more migration differ systematically in ways correlated with destination housing. We instrument bilateral migration with drought intensity at origin, selected by a post-LASSO procedure among candidate climate variables to remove the discretionary choice of instrument. Drought strongly predicts migration, and the identifying assumption is that it affects destination housing only through migration along the corridor, conditional on corridor and arrival-year fixed effects.

The first result of the paper is the first stage, showing that drought moves people to cities. Comparing the same corridor across years, we find that when an origin's growing season is unusually dry, more people migrate to urban areas than in normal years. A one-

standard-deviation increase in drought at origin raises the corridor's urban migration rate by about 7% of its mean. The response is specific to the dry tail of the rainfall distribution; symmetric wet shocks do not move people, consistent with an agricultural push rather than generic weather variation. Heterogeneity and mechanism analyses confirm this reading: the response is borne mainly by the most rural and most agricultural origins — those with the highest shares of harvested land — and drought is shown to lower crop yields at origin. This first stage is the variation on which all subsequent results rest.

The main finding of the paper concerns where these additional migrants end up: the inflow of climate migrants contributes to the growth of low-quality housing in the cities that receive them. A one percentage-point (pp) increase in the drought-induced migration rate raises the inflow into low-quality housing at destination by 0.28 pp: of every hundred migrants whom drought pushes into a city — people who, absent the dry shock, would have stayed at origin — 28 settle in low-quality housing. In the counterfactual without drought, neither these migrants nor their housing demand would be in the city at all. Drought also shifts the composition of the cohort that migrates: within a corridor, a one-standard-deviation dry shock raises the share of arrivals living in low-quality housing BY 0.34 p.p, from 40.6 to 41.0 percent. Decomposing the total effect into these two margins, the scale channel — more migrants arriving, sorting into housing roughly as those already flowing along the corridor do — accounts for about 79% of the additional inflow into low-quality housing, and the composition channel for the remaining 21%. Climate shocks therefore fuel urban slums mainly through volume. A back-of-the-envelope aggregation puts magnitudes on these estimates: over 1981–2018, drought generated roughly 390,000 of the 11.6 million urban moves recorded in our corridors (3.3%), and thereby placed about 110,000 additional people in low-quality housing.

The second main finding is that climate migration strains cities residentially rather than economically: drought-induced migrants are absorbed into work, but into low-quality housing. Applying the same design to labor-force status, of every hundred adult migrants whom drought adds to a city, 68 arrive into employment, 13 into unemployment and 20 into inactivity. As at the housing margin, this composition is not detectably different from that of the migrants who would have moved anyway: climate migrants resemble the cohort they join on both dimensions taken separately. What distinguishes them is where the two margins meet. Among the adults whom drought places in low-quality housing, 85% are employed, against 61% of those it places in adequate housing. The people the shock pushes into the worst housing are the ones who are working, and the margin of exclusion is residential rather than occupational. Aggregating over the eight countries and the 1981–2018 window gives the order of magnitude. Of 11.6 million observed urban in-migrants, some 387,700 are attributable to drought at origin, and 110,300 of these enter low-quality housing. Of the latter, 58,400 arrived as adults; among the adults, 49,800 were employed. Drought thus added on the order of fifty thousand working poor – employed city residents housed badly – and a comparable number of children growing up in low-quality housing.

Climate migrants hold the same kinds of jobs as other arrivals, but in different sectors. Of those who work, 60% are in wage or salaried employment — the same share as among the migrants they join — and they are if anything less likely to be in unpaid work, so the jobs themselves are not of a lesser kind. What differs is the sector: services still take two thirds of them, as of other arrivals, but a quarter work in urban agriculture against 8% of the

migrants they join, and half as many are in industry. The reallocation is toward the activity closest to the one they left, though our instrument identifies this margin only weakly.

The burden of these working poor falls on small and secondary cities, and the conversion of climate-induced migrants into slums is more a question of places than of who migrates. That conversion into slums varies with the size of the receiving city, while absorption into employment remains stable across the distribution. The sorting into low-quality housing concentrates in towns below 100,000 and in secondary cities of 100,000 to 1 million, which convert about 20 more of every 100 marginal migrants into slums than metropolises do: 30 per 100 against 10. These destinations are also the worse housed to begin with — 59% of their residents in low-quality housing against 46% in metropolises. The burden concentrates there not because drought sends most migrants to the smaller cities but because these cities convert arrivals into low-quality housing three times as often. Drought redirects migration toward these smaller destinations by opening new corridors that lead to them. Corridors that carry no flow in the pre-period and open only under drought direct 55% of their migrants to towns below 100,000, against 11% on established corridors. Climate change thus fuels urban slums less by changing who migrates than by redirecting migration toward the cities least equipped to house the arrivals.

The remaining heterogeneity is about how many migrants a destination receives rather than what becomes of them once there: drought-induced migration is drawn to cities that are close by, fast-expanding, already connected to the origin by an established corridor and not yet saturated with low-quality housing - defined by baseline urban population slum shares - which receive between 1.7 and 6 times more of it than their counterparts. Not one of these characteristics shifts the conversion rate, as the number of arrivals entering low-quality housing is indistinguishable across every split, so an established migrant network raises the number of arrivals without buffering any of them into better housing.

Climate migration is driven mainly by an agricultural-push mechanism, and origin agricultural characteristics predict it at both ends of the causal chain. Climate migrants come disproportionately from the most cultivated districts: ranking origins by the share of their area under harvested cropland, the first stage rises from 0.99 to 1.56 to 7.42 per thousand across terciles, some seven times stronger out of the most cultivated third. These same origins see their harvests hit by drought intensity : a one-standard-deviation increase in drought lowers maize yield per hectare by 3.1%, while leaving harvested area unchanged, so the loss falls on the harvest itself rather than on land taken out of production. What predicts climate migration is cropland, not rurality: within the rural world the response is flat, because the census urban share captures where people live while the cropland share captures their exposure to rainfall.

Who migrates under drought also matters, but not in a way that sets climate migrants apart from other arrivals. Drought does not displace the least-educated adults at all: their migration response is indistinguishable from zero and significantly below that of the secondary-educated, a trapped-population pattern, and those who do migrate are if anything more educated than the average arrival on the same corridor. Among the adults drought displaces, the less educated fare worse on both margins at once: 39 of every 100 enter low-quality housing against 22, but only 79 find work against 91. This is a uniform shortfall rather than the decoupling of work from housing that defines the working poor. This educational gradient is identical in size on the two margins and is the one education

already commands among all urban in-migrants. Climate migration adds arrivals to this landscape rather than altering it.

We subject these results to an extensive robustness analysis. The main threat to identification is that drought at origin could affect destination housing through channels other than migration along the corridor; the exclusion restriction is dsicussed showing no results of changes on food prices at destination and conflict-induced displacement, as well a weather shocks at destination. We also investigate crowding out effects on past migrants. The estimates are stable across fixed-effects structures and across the alternative deprivation margins of the UN-Habitat definition, including a continuous count of deprived dimensions. Section 7 further reports checks on the instrument itself - alternative drought measures, out-of-sample validation of the LASSO selection - as well as alternative population denominators, an extension to fifteen countries using five-year migration brackets, and a leave-one-country-out exercise.

This paper gives the first causal estimate of how climate shocks translate into urban low-quality housing. Using drought as an instrument, we identify that drought shocks raise the inflow into low-quality housing in African cities, but not by displacing a different kind of person.

This paper makes four contributions. First, it provides the first causal evidence linking climate-induced migration to the growth of low-quality housing in Sub-Saharan Africa. A large literature establishes that climate shocks drive migration, from rural-rural moves (Gittard, 2024; McGuirk and Nunn, 2021; Gray and Mueller, 2012) to rural-urban (Henderson et al., 2017; Cattaneo and Peri, 2016; Mueller et al., 2020) and cross-border migration (Beine and Parsons, 2016; Missirian and Schlenker, 2017), while a separate literature debates whether informal settlements are transitional or persistent poverty traps and documents the vulnerability of their residents (Marx et al., 2013, 2019). This paper connects the two: it shifts the focus from out-migration at origin to the housing absorption of climate migrants at destination, and decomposes the effect into a scale and a propensity channel. Second, it characterizes the labor-market integration of these migrants. Whereas existing work emphasizes barriers that keep rural-urban migrants in low-productivity activities (Gollin et al., 2014; Bryan and Morten, 2019; Alvarez, 2020; Hamory et al., 2021) and negative labor-market spillovers of climate migration (Kleemans and Magruder, 2018; Gittard, 2024), we find that drought-induced migrants are absorbed into employment: the margin of exclusion i mainly residential, not occupational, a distinction that matters for targeting policy. Measured on the same individuals, 85% of those whom drought places in low-quality housing are employed, against 61% of those it places in adequate housing — the working poor the shock adds to the city. Destination heterogeneity sharpens the distinction: the decoupling of work from housing is geographic, arising in the secondary tier and not in the metropolises, while education degrades both margins together rather than separating them, and the least-educated adults are not displaced at all — the trapped-population margin documented by Cattaneo and Peri (2016). Third, it contributes to the literature on African urbanization and structural transformation (Henderson and Kriticos, 2018; McMillan et al., 2014; Jedwab et al., 2017; Gollin et al., 2016) by showing that the housing pressure of climate migration concentrates in small and secondary cities, shifting attention from primate cities to the tier where urban growth is fastest. Fourth, methodologically, it illustrates a disciplined treatment of instrument choice in climate econometrics: a post-IV LASSO selection among 25

candidate climate variables (Belloni et al., 2014), validated by a wet-side placebo, replaces the discretionary choice of climate instrument.

The remainder of the paper is organized as follows. Section 2 describes the data and the construction of the migration, weather, and housing-quality variables. Section 3 presents the identification strategy, including the post-IV LASSO selection of the instrument. Section 4 reports the main results on housing and employment, the scale-propensity decomposition, and the aggregate magnitude. Section 6 examines heterogeneity by destination city size and by migrant profile. Section 7 presents robustness checks, Section 8 develops a two-margin model of post-migration integration that organizes the findings, and Section 9 concludes.

2 Data

This section describes the construction of the weather, migration, urbanization, housing-quality and employment status variables. Appendix Table 15 reports descriptive statistics for all variables used in the analysis, and Appendix Figures 7–9 map the climate exposure of origins and the migration, urbanization, and migrant outcomes of destinations. Appendix Figure 10 maps the observed urban in-migration corridors across the eight countries, illustrating the bilateral structure of the data: each origin–destination pair contributes a yearly flow, and identification rests on within-corridor variation in the timing of droughts at origin.

2.1 Weather Variables

Overview - The empirical model selects among a large set of candidate weather variables constructed from raw climate data, using a post-IV LASSO procedure (Belloni et al., 2014). The choice of candidate variables is grounded in recent climate-econ damage function literature (Hultgren et al., 2025; Auffhammer, 2018), which identifies the most influential weather terms for agricultural production. All weather variables are computed on a common historical reference period of 1983–2023. We distinguish temperature and precipitation terms. Weather variables are defined over the growing season, denoted by a superscript g (e.g. Precip^g), with yearly versions denoted Precip^y used in robustness checks. Growing seasons are defined at the national level using the main rainy season identified from historical monthly precipitation distributions for each country. Precipitation variables are derived from CHIRPS daily data (Funk et al., 2015). Temperature variables and evapotranspiration are derived from ERA5-Land reanalysis data (Muñoz-Sabater et al., 2021).

Precipitation variables - We construct several candidate precipitation variables, all derived from growing-season daily precipitation aggregated to the admin unit.[1] Cumulative precipitation Precip^g_{cum} is the growing-season mean; its square $(\text{Precip}^g_{cum})^2$ captures nonlinear responses. The drought dummies Drought^g_{p10} and Drought^g_{p15} equal 1 if Precip^g_{cum} falls below the 10th and 15th percentile of its historical distribution. The number of dry days N^g_{dry}

[1]Following Hsiang (2016), strongly nonlinear transformations (squared precipitation, squared temperature, GDD) are computed at the pixel level before spatial aggregation to preserve local nonlinear responses. Threshold-based drought variables (Drought^g_{pJ} and drought_int_z) use admin-unit-level historical precipitation distributions, consistent with the spatial scale of our migration outcomes.

counts days with daily precipitation below 1 mm. Finally, the standardized drought intensity $\text{drought_int_z}_{o,t} = \max(0, -\tilde{z}_{o,t})$, with $\tilde{z}_{o,t} = (\text{Precip}^g_{o,t} - \overline{\text{Precip}}^g_o)/\sigma^g_{\text{Precip},o}$ the seasonal precipitation anomaly, captures drought asymmetrically: it activates only for below-average rainfall and measures the number of standard deviations below the historical mean. For an extreme-precipitation analysis conducted separately, we construct two symmetric extreme-wet variables: ExtremeWet^g_{pJ} is a dummy equal to 1 if Precip^g_{cum} exceeds the 85th or 90th percentile of the historical distribution, and $N^g_{extreme}$ counts days with daily precipitation above 20 mm at the pixel level, spatially averaged at the admin-unit level.

Temperature variables - We construct three temperature candidate variables, computed at the pixel level before spatial aggregation to the admin unit.[2] The seasonal averages of daily minimum and mean 2m air temperatures, T^g_{min} and T^g_{mean}, enter along with their squares $(T^g_{min})^2$ and $(T^g_{mean})^2$. Growing Degree Days GDD^g are computed daily as $\max\left(0, \frac{1}{2}(\tilde{T}_{min,d} + \tilde{T}_{max,d}) - 8\right)$, where $\tilde{T}_{min,d}$ and $\tilde{T}_{max,d}$ are daily minimum and maximum 2m temperatures capped at 30°C, then summed over the growing season.

Moisture variables - Moisture-related variables include potential evapotranspiration PET^g and actual evapotranspiration ET^g, both extracted from ERA5-Land at the daily level and aggregated over the growing season. PET captures atmospheric moisture demand (irrigation requirements), while ET captures realized evaporation.

LASSO selection - The candidate set used in the post-IV LASSO selection comprises thirteen contemporaneous weather variables, all measured at the admin-unit level over the growing season - the drought dummies Drought^g_{p10} and Drought^g_{p15}, the standardized drought intensity drought_int_z, N^g_{dry}, the precipitation and temperature levels and polynomials, and the degree-day and evapotranspiration measures described above - augmented with the lags and three-season means of the drought-family variables, for 25 candidates in total (Section 3.3). The procedure selects drought_int_z as the unique relevant instrument for bilateral migration. We use this single-variable selection for the main analysis to maintain clarity in interpretation. The selection is probed in several directions (Section 3.3 and Appendix Figure 13). It is stable across LASSO penalty values, from the theoretical benchmark of Belloni et al. (2014) to more permissive thresholds; among lagged candidates, the three-season mean of drought intensity dominates, consistent with the cumulative instrument used in robustness checks; and a wet-side placebo - re-running the exercise on symmetric excess-rainfall candidates - finds essentially no predictive power, confirming that the instrument's strength is specific to the dry tail of the rainfall distribution.

2.2 Migration data

Sources - Both migration and housing quality data come from IPUMS census microdata (Minnesota Population Center, 2019) for fifteen African countries over the period 1987–2019. The sample includes exhaustive population censuses from Benin (2002, 2013), Burkina Faso

[2]Squared terms are computed after conversion to Celsius, before spatial averaging, to capture nonlinear thermal responses.

(1996, 2006), Cameroon (1987, 2005), Egypt (1986, 1996, 2006), Ghana (2000, 2010), Kenya (1989, 1999, 2009, 2019), Malawi (1998, 2008), Mali (1998, 2009), Mauritius (1990, 2000, 2011), Mozambique (1997, 2007, 2017), Senegal (1988, 2002, 2013), Sierra Leone (2015), South Africa (1996, 2001, 2011), Togo (2010), Uganda (2002, 2014), and Zambia (1990, 2000, 2010). These censuses provide harmonized individual-level information on demographic characteristics, migration history (place of birth, year of arrival, previous residence), and housing conditions (dwelling type, access to basic services, urban/rural classification at the enumeration area level). One contribution of this paper is the construction of *bilateral yearly migration flows* at the IPUMS administrative level for 8 countries with available year-of-arrival information, and *five-year migration brackets* for the remaining 7 countries. Table 17 gives the coverage of the origin-destination panels by country and census year for the 8 countries for which we have build the yearly bilateral migration.

Migrant identification - We define an individual i observed at destination d in census year t_c as a migrant from origin o in year t if (i) the IPUMS variable `migyrs1` reports a years-since-arrival value τ such that $t = t_c - \tau$, and (ii) the IPUMS variable for previous administrative unit of residence (or, when unavailable, place of birth [3]) identifies origin o. The destination d is the administrative unit where the individual resides at the census date. For all censuses, we trace migration histories back twelve years prior to each census for the yearly migration analysis, capping the window in multi-census countries so that it does not overlap the preceding census. The individual migration indicator is:

$$\text{InMigrants}_{o,d,t} = \sum_i \mathbf{1}\{i \text{ at } d \text{ in } t_c,\ \text{arrived from } o \text{ in } t\} \cdot \text{perwt}_i,$$

where perwt_i is the IPUMS person-weight and the sum runs over all individuals observed at destination d in census t_c.

Urban Migrant - We use the IPUMS variable `urban` to identify whether an individual resides in an urban or rural enumeration area within their destination administrative unit. This classification is finer than the admin unit level: a single administrative unit may contain both urban and rural enumeration areas. Urban migrants are individuals classified as residing in urban enumeration areas at destination (`urban` = 2 in IPUMS).

Population denominators - We construct two destination population measures. The *census-year population* $\text{Pop}^{\text{census}}_{d,T_c}$ is the sum of person-weights observed at destination d in census year T_c. The *urban census-year population* $\text{Pop}^{\text{census,urban}}_{d,T_c}$ restricts to individuals classified as urban residents (`urban` = 2). For robustness, we also reconstruct annual destination populations $\text{Pop}_{d,t}$ backward from the census year via the accounting identity:

$$\text{Pop}_{d,t} = \text{Pop}_{d,T_c} - \text{InMig}_{d,[t+1,T_c]} - \text{Births}_{d,[t+1,T_c]} + \text{OutMig}_{d,[t+1,T_c]} + \text{Deaths}_{d,[t+1,T_c]},$$

[3] Countries for which previous administrative unit of residnce is proxied by the place of birth are Kenya and Zambia

where births are estimated from age or birth-year information in the census, and deaths are derived from country-specific mortality rates in the corresponding period [4]. Our main results use the urban census-year population, which is more stable; the annual reconstruction is used in robustness checks.

Migration rates. - The bilateral migration rate from origin o to destination d in year t is:

$$\text{MigRate}_{o,d,t} = \frac{\text{InMigrants}_{o,d,t}}{\text{Pop}^{\text{census}}_{d,T_c}} \times 100,$$

expressed in percentage points. The urban migration rate restricts the numerator to urban-resident migrants and uses the urban census-year population as denominator:

$$\text{MigRate}^{\text{urban}}_{o,d,t} = \frac{\text{InMigrants}^{\text{urban}}_{o,d,t}}{\text{Pop}^{\text{census,urban}}_{d,T_c}} \times 100.$$

For countries with only five-year migration brackets, we construct the equivalent five-year migration rate as the sum of migrants arriving over $[t-4, t]$, normalized by the same census-year denominator. In robustness checks, we re-estimate the main results using the annual reconstructed destination population $\text{Pop}_{d,t}$ as denominator in place of the census-year population, to verify that our results are not driven by the choice of a fixed denominator.

2.3 Urbanization data

Admin-unit urbanization. - For our heterogeneity and robustness analyses, we classify IPUMS administrative units by their urban population share at the most recent census:

$$\text{UrbanShare}_a = \frac{\sum_{i \in a, T_c} \text{perwt}_i \cdot \mathbf{1}\{\text{urban}_i = 2\}}{\sum_{i \in a, T_c} \text{perwt}_i}.$$

This admin-unit classification is distinct from the individual-level urban variable: a rural admin unit may contain some urban residents, and a city admin unit some rural enumeration areas. For the cities and rural sample restrictions, we label an admin unit as a *city* if $\text{UrbanShare}_a \geq 0.55$ and as *rural* otherwise.[5] For the urbanization heterogeneity analysis, we use three groups: rural-like ($\text{UrbanShare}_a < 0.15$), urbanizing cities ($[0.15, 0.55)$), and main cities ($\geq 0.55$).

We validate this classification against the Africapolis database (OECD/SWAC, 2024), which delineates all urban agglomerations above 10,000 inhabitants in Africa. Matching each destination to its largest agglomeration confirms that the classification orders city sizes correctly: the median largest agglomeration rises from about 47,000 inhabitants in rural-like destinations to 86,000 in urbanizing cities and 880,000 in main cities, and the share of

[4] Death rates are derived fro, country-specific mortality rates in the correspondiong period, from the annual national crude death rates from the World Bank Indicator.

[5] The 55% threshold is calibrated on the median urban share across our sample of African administrative units; the rolling-window estimates in Figure S4 trace the estimates continuously along the urban-share distribution, showing that no single threshold drives the results.

destinations hosting a city of at least 100,000 inhabitants rises from 28% to 43% to 70%. The main-cities group contains the principal cities of each country (Nairobi, Kampala, Yaoundé, Bamako), the urbanizing-cities group classic secondary cities (Nakuru, Garoua, Fort Portal), and the rural-like group mostly small towns. Appendix 10.1.3 reports the full correspondence (Figure 12 and Table 19).

2.4 Housing Quality

Individual definition of housing quality - The household-level slum classification follows the UN-Habitat multi-criteria slum definition (UN-Habitat, 2003).[6] The classification relies on IPUMS variables covering five housing dimensions grouped in two categories. Structural quality of the dwelling captures the materials of the floor, walls, and roof. Access to basic services and living conditions captures sanitation and household crowding (persons per room).[7] In the main definition, metal-sheet roofing is classified as durable, consistent with the UN-Habitat durability criterion that durable materials protect occupants from climatic conditions; the robustness definition instead counts metal-sheet roofing as deficient, reflecting its use as a low-cost material in the Sub-Saharan African urban context.[8]

A household is classified as residing in a slum - or a low-quality housing - if at least one of these dimensions is deficient, following the UN-Habitat operational slum definition, under which a household lacking any one of the deprivation criteria is classified as slum (a logical OR across dimensions) (UN-Habitat, 2003). As a conservative robustness definition, we also use a stricter threshold requiring at least two deficient dimensions. The coding of each dimension - which materials, services, and crowding levels are classified as deficient, in harmonized IPUMS categories identical across countries, with dimension availability varying by census - is reported in Appendix Table 18.

From the household-level classification, two individual-level housing-quality measures are constructed: the binary indicator $\text{deprived}_i \in \{0, 1\}$ equal to 1 if individual i resides in a household classified as low-quality, and the continuous count $\text{deprived_count}_i \in \{0, 1, ..., 5\}$ equal to the number of deficient housing dimensions for individual i's household.

Aggregate housing quality outcomes - The main analysis focuses on origin-destination-year cohorts of migrants. From the individual-level housing-quality classification above, two aggregate outcomes are constructed at the corridor level. The first outcome, used in the main

[6] The census-based classification captures the material dimension of housing precarity - structural quality and access to basic services - but not the legal or tenure dimension that defines informal settlements in a stricter sense (unauthorized land occupation or unregulated construction). Throughout the paper, slum refers to the UN-Habitat statistical slum definition rather than to the legal sense of informal settlement.

[7] Water access is not included. The UN-Habitat criterion counts boreholes and protected wells as improved sources, but the harmonized IPUMS water-supply variable in these censuses distinguishes types of piped supply rather than improved from unimproved sources, so it cannot reliably separate deficient from adequate access in this setting.

[8] The UN-Habitat definition comprises five deprivations (durable housing, sufficient living area, improved water, improved sanitation, and secure tenure); we adapt it to the dimensions the harmonized census microdata can measure reliably, and cannot capture tenure security, water access, or, in some countries, sanitation. Our measure therefore reflects the material and service dimensions of housing precarity rather than the full UN-Habitat set.

results to preserve the same estimation sample between first and second stages, is the *scale* outcome $S_{o,d,t}$, the low-quality housing inflow rate at destination:

$$S_{o,d,t} = \frac{\sum_i \text{perwt}_i \cdot \mathbf{1}\{i \text{ at } d \text{ in } t_c,\ \text{arrived from } o \text{ in } t, \text{deprived}_i = 1\}}{\text{Pop}^{\text{census}}_{d,T_c}} \times 100,$$

the share of urban destination population accounted for by urban migrants in low-quality housing arriving from origin o in year t. The second outcome, designed to isolate the propensity channel, is the *propensity* outcome $P_{o,d,t}$, the share of migrants residing in low-quality housing in each corridor:

$$P_{o,d,t} = \frac{\sum_i \text{perwt}_i \cdot \mathbf{1}\{i \text{ at } d \text{ in } t_c,\ \text{arrived from } o \text{ in } t, \text{deprived}_i = 1\}}{\sum_i \text{perwt}_i \cdot \mathbf{1}\{i \text{ at } d \text{ in } t_c,\ \text{arrived from } o \text{ in } t\}}.$$

This second outcome is defined only on cells with at least one migrant, which yields a smaller estimation sample than $S_{o,d,t}$. The analysis is on urban-restricted versions, $S^{\text{urb}}_{o,d,t}$ and $P^{\text{urb}}_{o,d,t}$, restrict the numerator to migrants residing in urban enumeration areas at destination and use the urban census-year population as denominator.

Because $S^{\text{urb}}_{o,d,t}$ is a rate per urban resident, destinations with a very small urban population produce rates that are arithmetically valid but economically meaningless. We therefore restrict the sample to destinations with at least 2,000 urban residents, the conventional minimum size of an urban locality and the threshold underlying the Africapolis classification used below.[9] The estimation panel contains 241,540 corridor-years.

Decomposing scale and propensity. - The scale outcome embeds two distinct mechanisms by construction. The accounting identity

$$S_{o,d,t} = \text{MigRate}_{o,d,t} \times P_{o,d,t}$$

makes explicit that a causal effect of drought-induced migration on the low-quality housing inflow can operate through two channels, that could be happening at the same time. The scale channel reflects the mechanical contribution of drought-induced migrants arriving at the baseline propensity to reside in low-quality housing. The propensity shift reflects a change in that propensity among the migrants themselves: within a given corridor, drought-induced migrants may be more or less likely to settle in low-quality housing than those who would have migrated regardless. Distinguishing the two matters for policy: whether climate change fuels slum growth through sheer demographic pressure—more arrivals at the baseline rate (the scale channel)—or through a shift in their propensity to settle in low-quality housing (the propensity shift) points to different policy implications.

A first-order decomposition of the 2SLS coefficient on the scale outcome yields:

$$\widehat{\beta}^{2SLS}_S \approx \underbrace{\overline{P}}_{\beta^{\text{scale}}} + \underbrace{\overline{\text{MigRate}} \times \widehat{\beta}^{2SLS}_P}_{\beta^{\text{shift}}},$$

[9]The restriction is stated ex ante and applies uniformly across countries. It binds on eight destinations (3,752 corridor-years, 1.5% of the panel), five of which are census-to-census reclassification artefacts: their recorded urban population collapses by 77-99% between consecutive censuses while total population grows, reflecting a change in what the census codes as urban rather than a change in settlement.

where β^{scale} is the effect that would obtain if the propensity were fixed at its sample mean, and β^{shift} the additional contribution of a shift in that propensity. Both terms are dimensionless: $P_{o,d,t}$ is a share and the migration rate is already expressed per hundred urban residents.

Estimating the propensity shift directly requires a separate 2SLS specification with $P_{o,d,t}$ as the outcome, which raises three difficulties. First, small-sample ratio bias: $P_{o,d,t}$ is a ratio, and on corridor-years with few migrants its expectation departs from the underlying share even when both counts are unbiased. We therefore define it only on corridor-years with a positive urban migrant flow ($N = 72{,}723$).[10] Second, the shared denominator: the migrant count in the denominator of $P_{o,d,t}$ is also the numerator of the endogenous migration rate, so classical measurement error in that count would bias an OLS regression of P on the migration rate downward. Instrumenting with origin drought addresses this, since the instrument moves the true flow rather than its measurement error—an additional rationale for the IV strategy. Third, interpretive ambiguity: a nonzero coefficient establishes that the migrants drought adds are housed differently from those who would have moved regardless, but not why. Origin-destination fixed effects rule out a reweighting of the flow across destinations as the source; what remains is selection on migrant characteristics—which Section 6.2 explores (and does not detect on education, age, or gender)—and differences in the conditions or timing of arrival within a corridor, which we cannot separate from it.

2.5 Other Variables

Aridity index for heterogeneity analyses - For heterogeneity analyses by climate baseline, we use the CGIAR-CSI Global Aridity Index (Trabucco and Zomer, 2019), computed as the long-run mean ratio of precipitation to potential evapotranspiration. This index is extracted from the CGIAR-CSI global 30-arcsec dataset and spatially averaged at the IPUMS administrative level. Lower values indicate more arid conditions; we use within-country medians to classify origins as more arid or less arid, avoiding conflation between aridity and country fixed effects.

Built-up surface for the construction-capacity analysis - To characterise how fast a destination was able to build before our migration flows, we use the Global Human Settlement Layer's built-up surface product, GHS-BUILT-S release R2023A (Pesaresi and Politis, 2023). The product measures, in a 100-metre grid derived from Landsat and Sentinel-2 composites, the surface in square metres covered by buildings in each cell, in twelve epochs at five-year intervals from 1975 to 2030. We aggregate the grid to IPUMS administrative units by summing built-up surface over the cells falling within each `GEOLEVEL2` polygon. For each unit d and epoch t we observe the total built-up surface $B_{d,t}$ in square metres and the polygon area. All 351 destinations of the estimation panel are matched, and the built-up sample is therefore the full canonical panel ($N = 241{,}540$).

Our measure of construction capacity is the log growth of built-up surface over the fifteen years ending in 1990, $\text{BuiltGrowth}_d = \log B_{d,1990} - \log B_{d,1975}$, reported in the figures as the implied proportional increase, $B_{d,1990}/B_{d,1975} - 1$: the median destination expanded its built-up surface by 99%, with a tenth-to-ninetieth percentile range of 46% to 177%. The window

[10] In the robustness section, results are weighted by corridor size, so that thin cells do not enter on the same footing as dense ones.

closes before the migration flows we study and the droughts that instrument them, so the measure is predetermined [11], and it captures a flow of construction rather than a stock, which is what a capacity interpretation requires.[12] As with aridity, destinations are split at the within-country median, since the level of built-up surface mostly reflects national development.

We emphasise that this variable is not a housing-supply elasticity in the sense of Saiz (2010): the data contain no house prices, no measure of undevelopable land and no construction-cost variation. It is a proxy for persistent urban dynamism, which mixes the capacity to build with the demand that was already being accommodated, and Section **??** reports the pattern it produces together with that caveat.

Additional sources for the exclusion restriction and the mechanism tests - Three further datasets enter the robustness and mechanism analysis rather than the main specification. Conflict comes from the ACLED (Armed Conflict Location and Event) Data project (Raleigh et al., 2010), whose geo-located events we aggregate to IPUMS administrative units and years; coverage begins in 1997 and reaches 70% of the estimation panel, which is why the conflict tests are run on that sub-sample against a matched no-control benchmark. Cropping patterns come from CROPGRIDS v1.08 (Tang et al., 2024), a global 0.05° grid of harvested area by crop for 2020, from which we compute, for each administrative unit, the share of its land area under each of six staples – maize, cassava, wheat, sorghum, rice and soybean. World prices come from the World Bank's monthly commodity price series, annualized, for a basket of nine traded crops: the four traded staples above plus cotton, coffee, tea, tobacco and cocoa, with Arabica coffee used for Kenya and Robusta elsewhere. Administrative crop statistics for the yield mechanism come from HarvestStat Africa v1.0 (Lee et al., 2025), matched to our origin districts through its areal crosswalk to IPUMS boundaries; coverage is six of the eight countries and 2,535 district-years, and Appendix Table 24 sets out its limits.

2.6 Descriptive Overview

Migration patterns - Appendix Figure 6 displays the average urban in-migration rate over time in the eight countries of the estimation sample, constructed from the yearly bilateral panel. Rates differ by an order of magnitude across countries, and within each census window they rise toward the census year - the mechanical recall and onward-migration attrition of earlier cohorts that the distance-to-census control absorbs in estimation (Section 3). The corresponding five-year-bracket rates for the full set of fifteen IPUMS countries are shown in Supplementary Appendix Figure S2.[13]

[11] For Zambia's 1990 census and the first five years of Mali's 1998 window - together 15% of the panel - the measurement period and the arrival years overlap, so the measure is predetermined for the bulk of the sample rather than for all of it.

[12] The stock analogue - the share of the unit's area built up in 1990 - has a median of only 0.19% at this spatial scale and carries no signal (Section **??**).

[13] A potential concern with retrospective migration data is recall bias: individuals may report their year of arrival with diminishing accuracy as the gap between migration year t and census year t_c widens. All specifications include a linear control for the survey-to-arrival gap $|t - t_c|$, absorbing systematic differences in measurement quality across migration cohorts.

Spatial variation in climate and migration - Beyond temporal patterns, the data reveal substantial spatial heterogeneity within and across countries. Appendix Figure 7 maps mean growing-season precipitation and mean drought intensity by administrative unit, illustrating the diversity of agricultural climates spanned by the sample - from the arid regions of northern Mali and Kenya (below 500 mm per season) to the humid coastal zones of southern Cameroon (exceeding 2,500 mm). Figure 1 shows both the spatial and temporal variation of the drought intensity instrument.

Appendix Figures 8 and 10 map urban in-migration rates at the administrative unit level, revealing within-country variation in destination patterns, with concentrated flows toward major coastal and capital cities. This spatial granularity allows us to construct origin-destination migration matrices at a fine geographic scale and to match each origin district with its specific climate exposure.

Slums - Low-quality housing is widespread among the urban in-migrants we study: across the estimation panel, 41 percent of urban in-migrants in the average corridor reside in housing classified as low-quality under the main UN-Habitat definition, and 36 percent once corridors are weighted by their migrant flow (Appendix Table 15), a magnitude consistent with the prevalence UN-Habitat reports for Sub-Saharan Africa's urban population (Un-Habitat, 2022). Appendix Figure 9 maps this share by destination, showing variation both across and within countries. Appendix Figure 11 reports the index dimension by dimension, for urban residents and for in-migrants separately, and Supplementary Appendix Figures S17 and S20 show which material and facility categories fall on the deprived side of each rule. Cross-country levels partly reflect census vintage - Malawi and Mali, the highest in the sample, are measured in 2008 and 2009, while Zambia's are measured in its 1990 and 2000 censuses, the earliest we observe - and are absorbed by the corridor fixed effects in estimation.

3 Identification Strategy

3.1 Empirical Model

This paper investigates whether weather-shock-induced migration increases migrant inflows into low-quality housing in African cities, and how those migrants are absorbed. Three questions structure the analysis. First, does drought-induced migration raise the number of migrants arriving into deprived housing at destination? Second, are the same migrants absorbed into urban employment, so that the integration gap is residential rather than occupational? Third, does the housing effect operate through the number of arrivals, each settling at the usual rate - the scale channel – or through a shift in the rate at which arrivals settle in low-quality housing, which would mean the migrants drought adds differ from those who would have moved regardless?

To address these questions, we implement an instrumental-variable strategy using the cohort-bilateral panel of migration flows (o, d, t). The first stage isolates variation in bilateral migration generated by exogenous weather shocks at origin:

$$\text{MigRate}_{o,d,t} = \alpha_1 Z_{o,t} + \gamma_1 |t - t_c| + \delta_{o,d} + \lambda_t + \varepsilon_{o,d,t}, \tag{1}$$

where $Z_{o,t}$ denotes the weather shock at origin o in year t, selected from a set of candidate climatic variables using post-IV LASSO (Section 3.3). The selected instrument is standardized drought intensity,

$$Z_{o,t} = \max\left(0, -\widetilde{z}_{o,t}\right),$$

as defined in Section 2.1. Origin-destination pair fixed effects, $\delta_{o,d}$, absorb all time-invariant corridor characteristics, so identification relies on within-corridor variation in the timing of droughts. Arrival-year fixed effects, λ_t, absorb shocks common to all corridors, while $|t - t_c|$ controls linearly for the distance between migration year t and the reference census year t_c. The distance-to-census control, $|t - t_c|$, addresses three sources of measurement error in retrospective migration data. First, recall error may increase with the time elapsed between migration and the census. Second, recent arrival cohorts may be more fully captured than earlier cohorts. Third, the census records migrants at their place of residence at enumeration, so members of earlier cohorts who subsequently moved to another destination or returned to their origin are no longer observed in the original corridor. Standard errors are clustered at the origin administrative-unit level, where the instrument varies. Robustness checks show that the results are stable across clustering levels (Appendix Table S4).

The second stage estimates the causal effect of bilateral migration on migrant inflows into low-quality housing at destination:

$$S_{o,d,t} = \beta_1 \widehat{\text{MigRate}}_{o,d,t} + \gamma_2 |t - t_c| + \mu_{o,d} + \lambda_t + \nu_{o,d,t}, \tag{2}$$

where $\widehat{\text{MigRate}}_{o,d,t}$ is the fitted migration rate from Equation 1, and the fixed effects and controls are defined as above. The scale outcome $S_{o,d,t}$, the inflow of migrants into low-quality housing per 100 destination residents, is defined as

$$S_{o,d,t} = \frac{\sum_{i \in \text{Migrants}(o,d,t)} \text{perwt}_i \mathbf{1}\{\text{deprived}_i = 1\}}{\text{Pop}^{\text{census}}_{d,t_c}} \times 100.$$

Because both the outcome and the migration rate are expressed per 100 urban residents at destination, the shared denominator cancels and β_1 measures the share of the marginal drought-induced migrant inflow that settles in low-quality housing. This specification identifies the total effect on the low-quality housing inflow, which Section 4.1.3 decomposes into its two channels.

We isolate the propensity shift using an analogous specification in which the outcome is the share $P_{o,d,t}$ of arriving migrants living in deprived housing.

A positive coefficient means that a drought-induced rise in arrivals raises the share of a corridor's migrants living in low-quality housing. Two readings are consistent with it: the migrants drought induces to move may themselves be more housing-vulnerable than those who would have migrated regardless, or the additional arrivals may crowd always-takers of the same cohort out of decent housing. Crowding-out is tested in Section 7.1.

3.2 Discussion

Interpretation of the IV estimate - The 2SLS estimate β_1 in equation (2) identifies a Local Average Treatment Effect: the causal effect of bilateral migration on the low-quality housing

inflow for the migrants whose decision to move from origin o in year t responds to drought at origin – the compliers, as against the always-takers, who would have moved regardless. It does not identify the average treatment effect across all migrants, but the effect on the climate-responsive margin.

Discussion of the compliers - In a standard application, these compliers are a technical by-product of the instrument, and the LATE is often viewed as a limitation relative to an average effect over all migrants. Here the interpretation is more direct: the population the instrument selects - migrants induced to move by drought - is the population of interest, since the paper asks how climate shocks reshape urban housing through migration. The LATE is therefore the estimand of interest rather than a nuisance parameter to be reconciled with a broader average. The alignment also means that the scale coefficient does double duty: it is at once a causal effect and a description of the migrants drought displaces, since it counts how many of every hundred of them settle in low-quality housing.

Validity of the instrument and exclusion restriction - Identification rests on two key assumptions. The relevance condition requires that drought intensity predicts bilateral migration; the strong first-stage relationships reported in Section 3.4 ($F > 20$) confirm this. The exclusion restriction requires that drought at origin o affects the low-quality housing inflow at destination d only through its effect on migration from o to d, conditional on origin-destination pair and arrival-year fixed effects. This assumption is plausible in our setting: drought is a local agricultural shock at origin, and the origin-destination fixed effects absorb any time-invariant correlation between origin climate and destination housing conditions. Potential threats arise if the origin shock reaches destination housing through a channel other than migration: a drought at origin may raise food prices in regionally integrated markets, or feed conflict, at the destination itself. Section 7.1 addresses this by controlling for destination conflict and destination crop-price exposure, and by testing each as a placebo treatment, together with the destination's own rainfall.

Why a within-corridor design rather than a shift-share - A common alternative for destination-side consequences of migration is a shift-share built from predetermined origin shares and origin shocks (Kleemans and Magruder, 2018). Two features of climate-migration data make it infeasible here. Because a destination's origins lie near it, origin drought proxies the destination's own weather (correlation 0.95), which plausibly affects destination outcomes directly and must be controlled for—leaving almost no identifying variation; and genuinely predetermined shares require an earlier census, which five of our eight countries lack. The within-corridor design sidesteps both: identification from the timing of droughts within an origin-destination pair differences out destination weather and every time-invariant destination characteristic, requires no predetermined shares, and extends to all eight countries. The price is that it speaks to corridor-level flows and the migrants travelling them, not to destination-wide aggregates such as built-up expansion or effects on natives.

3.3 Instrument Selection: Post-IV LASSO

The choice of the instrument $Z_{o,t}$ is validated by a data-driven selection procedure based on the post-IV LASSO of Belloni et al. (2014). The procedure runs two LASSO regressions on a candidate set of climate variables at origin - one for the endogenous migration rate and one for the outcome, both on Frisch-Waugh-Lovell residuals that partial out the origin-destination pair and arrival-year fixed effects and the distance-to-census control - and retains the union of the selected sets. This removes the discretionary choice of instrument among ex-ante plausible candidates and provides valid post-selection inference under the conditions of Belloni et al. (2014).

The candidate set comprises thirteen contemporaneous climate variables described in Section 2.1 - drought intensity, drought dummies at two thresholds, dry days, precipitation and temperature polynomials, degree-day and evapotranspiration measures - augmented with the lags and 3-season means of the drought-family variables (25 candidates in total). We implement the selection under three values of the penalty multiplier c, from the theoretical benchmark of Belloni et al. (2014) ($c = 1.1$) to more permissive thresholds ($c = 0.5$, 0.25) that let more instruments enter.

The selection confirms the paper's instrument. Drought intensity is the strongest single candidate on both samples ($F = 23.9$ on the scale sample, $F = 13.0$ on the propensity sample) and is selected at the default penalty in both; among lagged candidates its 3-season mean dominates ($F = 27.4$), consistent with the cumulative instrument used throughout. We use the single strongest candidate, keeping the model just-identified so that Anderson-Rubin inference remains exact. As a placebo, we repeat the exercise on wet-side mirror candidates: excess-rainfall intensity has essentially no predictive power ($F = 4.0$ and $F = 0.03$), and the flood dummies are similarly weak. The instrument's strength is thus specific to the dry tail of the rainfall distribution. Appendix 10.2.1 reports the full selection results (Figure 13).

3.4 First Stage

Table 1 reports the first stage under alternative sets of fixed effects and controls, for the sample of urban migrations to all destinations. The table shows the strength and the stability of the instrument for in-migration towards urban areas. Column (1) is the paper's main specification: origin-destination pair and arrival-year fixed effects with the distance-to-census control. Comparing columns (1) and (2) shows that this control is not driving the first stage: the raw specification in column (2), without any control, already delivers a positive and highly significant coefficient, and adding the distance to the census mainly improves precision. Column (3) confirms that the instrument captures rainfall deficits rather than heat, and column (4) adds temperature at destination to absorb a potential climatic pull factor; column (5) includes all controls jointly. In all three cases the coefficient is essentially unchanged and the first stage remains strong. When accounting for destination-by-year fixed effects (column (6)), the coefficient remains positive and significant at the 5% level, but the first-stage F drops below conventional thresholds, as these fixed effects absorb a large share of the identifying variation, rainfall shocks being spatially correlated across origins and destinations within a country.

Quantitatively, the main specification in column (1) implies that a one standard deviation

Table 1: First Stage: Drought Intensity and Urban Migration

	DV: Urban migration rate					
	(1) Main spec.	(2)	(3)	(4)	(5)	(6)
Drought intensity	0.00362***	0.00244***	0.00396***	0.00389***	0.00400***	0.00159**
	(0.00066)	(0.00068)	(0.00076)	(0.00071)	(0.00077)	(0.00073)
First-stage F	29.65	12.72	26.77	29.67	26.91	4.74
Effect of 1-SD drought (% mean)	+6.8%	+4.6%	+7.4%	+7.3%	+7.5%	+3.0%
Origin FE	No	No	No	No	No	No
Destination FE	No	No	No	No	No	No
Origin-Destination FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Country × Year FE	No	No	No	No	No	No
Destination × Year FE	No	No	No	No	No	Yes
Dist. to census control	Yes	No	Yes	Yes	Yes	No[a]
Destination controls	No	No	No	Yes	Yes	No
Origin controls	No	No	Yes	No	Yes	No
R^2	0.486	0.481	0.486	0.486	0.486	0.512
Within R^2	0.010	0.000	0.010	0.010	0.010	0.000
Mean migration rate	0.026820	0.026820	0.026820	0.026820	0.026820	0.026820
Observations	241,540	241,540	241,540	241,540	241,540	241,540
OD pairs	16,846	16,846	16,846	16,846	16,846	16,846

Notes: Sample: urban migrants, all destinations; 8 countries, 12-year window. Instrument: contemporaneous drought intensity at origin, $\max(0, -\tilde{z})$. Column (1) is the paper's main specification; fixed effects and controls as indicated ([a]`dist_census` mechanically absorbed). First-stage F: cluster-robust Wald statistic on the instrument. Effect of 1-SD drought: coefficient × SD(instrument) ÷ mean migration rate × 100, on each column's own sample. Standard errors, clustered at `admin_origin`, in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

increase in the dryness of the growing season at origin raises in-migration to urban areas by 6.8% of the mean migration rate. Let's consider a destination of 10,000 urban inhabitants. At the mean migration rate, this corridor receives 2.68 migrants in a normal year. A one standard deviation drier growing season raises the migration rate by $0.00362 \times \text{SD}(\text{drought})$, which scaled by the destination population adds about 0.18 migrants — bringing the inflow from 2.68 to 2.86, a 6.8% increase.

This magnitude is in line with existing estimates of drought-induced migration in sub-Saharan Africa (Henderson et al., 2017; Mueller et al., 2020; Gittard, 2024; Hoffmann et al., 2020). Relative to this literature, we focus on in-migration and identify the first stage within origin-destination corridors at annual frequency: a corridor receives more migrants precisely in the years its origin is hit by drought, net of fixed corridor characteristics and common year shocks.

Table 20 in Appendix 10.3 assesses the robustness of the first stage along two dimensions: the sample and the drought exposure window, reporting the paper's main specification throughout.

3.5 Synchronous Shocks

This subsection shows that the migration response depends on the spatio-temporal structure of the shocks and not only on their intensity. It investigates the spatial correlation of the droughts, as a destination draws on several origins. This portfolio of origins, and whether those origins are hit together, changes how many people arrive and the interpretation of the main result.

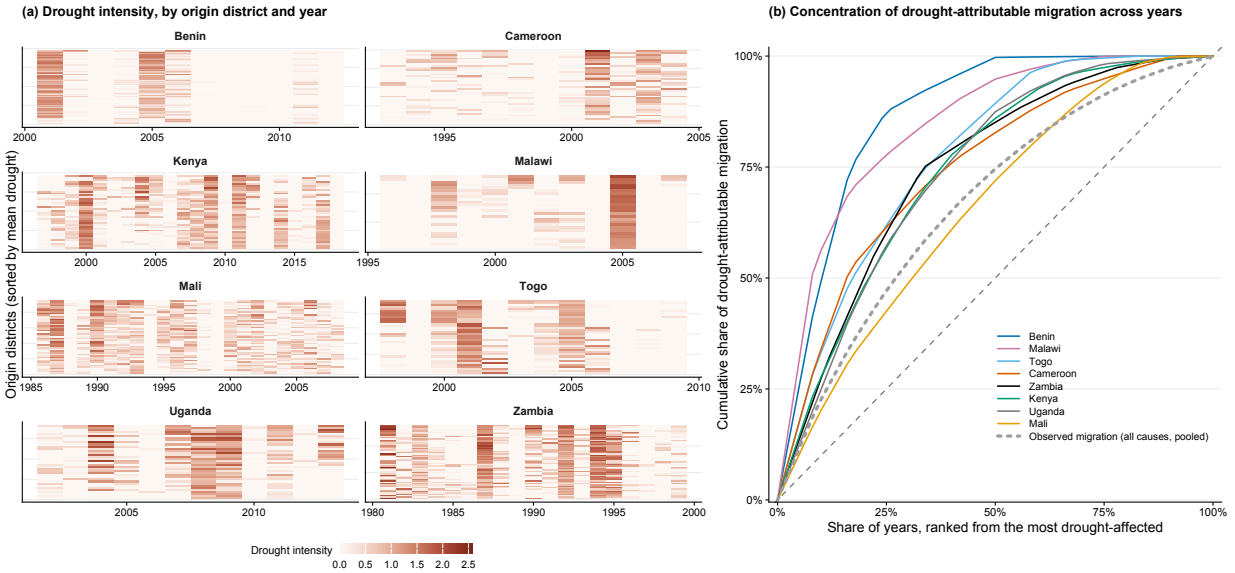
We build three measures of that structure at the destination level, all with predetermined weights: the origin shares w_{od} of each destination's inflow in the three years preceding its recall window, census years -13 to -15, so that no variation from the estimation window enters the classifier.[14] The first is the temporal concentration of the destination's induced inflow, $\log \text{HHI}_d - \log(1/T_d)$, where HHI_d is the Herfindahl–Hirschman index of the year shares, which measures how far the sequence $\sum_o w_{od}\,\text{drought}_{ot}$ departs from being spread evenly across the T_d years of the window.[15] The second is a variance ratio, $\text{Var}_t(\sum_o w_{od}\,\text{drought}_{ot}) \big/ \sum_o w_{od}^2\,\text{Var}_t(\text{drought}_{ot})$, which equals one if the origin shocks are independent and exceeds one when they arrive together. The third is the $w_{od}w_{o'd}$-weighted average pairwise correlation of drought across the destination's origins.

Panel (a) of Figure 1 plots drought intensity for each administrative unit and each year. Some years are national droughts in which nearly every origin is hit at once; others display far more spatial variation, with dry and wet districts side by side. Synchrony is therefore a tendency rather than a rule, and the measures above quantify how strong it is. Panel (b) turns the same fact into a cumulative curve. Within each destination and census cycle it ranks the years from the most to the least drought-affected and plots the cumulative

[14] The same pre-period migration table as Section 6.1.4; it covers 5,146 corridors and 344 destinations, carrying 95.6% of the urban flow.

[15] The Herfindahl–Hirschman index is the sum of squared shares, here the share of each year in the destination's induced inflow over its window. It equals $1/T_d$ when the inflow is spread evenly across the T_d years and one when it all arrives in a single year, so the measure is zero under a uniform spread and rises with concentration. Its reciprocal, $1/\text{HHI}_d$, is the effective number of years carrying the inflow.

Figure 1: The Spatio-Temporal Structure of Drought Shocks and of Drought-Attributable Migration



Notes: Panel (a) plots drought intensity $\text{drought}_{ot} = \max(0, -\text{rain_z}_{ot})$ for every origin district (rows, sorted within country by mean drought) and every year of the country's span (columns), with free scales because the windows differ in length. It is drawn from the rainfall data rather than the estimation sample, so it also covers years the migration windows leave out. White cells are genuine zeros, not missing data: the instrument is censored at zero, which is 33–70% of origin-years depending on the country. Panel (b) ranks, within each destination–census cycle, the years from the most to the least drought-affected and plots the cumulative share of the cycle total against the rank fraction, averaged by country, with the uniform diagonal as reference. The solid series is the first-stage prediction summed over a destination's origins with equal weights; the faint line uses pre-period flow weights, the classifier of Table 3, and correlates 0.906 with it. Since a Lorenz curve normalises by the within-group total, $\hat{\beta}$ cancels exactly: the curve is a transformation of the instrument, not evidence that the first stage drives its shape. The dashed grey line is observed migration from all causes, discussed in the text.

Table 2: How Synchronous Are the Shocks Reaching a Destination?

	Median	25th pct.	75th pct.	Destinations
Pairwise correlation of drought across a destination's origins	0.674	0.552	0.796	318
Variance ratio of induced inflow (= 1 if shocks independent)	3.38	2.30	4.98	318
Effective number of years carrying the induced inflow (1/HHI)	4.6	3.8	5.7	343
out of a window of	12	10	12	

Notes: Distribution across destinations of the three shock-structure measures defined in Section 3.5. All use predetermined weights: the origin shares of each destination's inflow in the three years preceding its recall window, census years −13 to −15. The pairwise correlation and the variance ratio are computed on origins with at least eight observed years, and the correlation additionally requires three such origins per destination, which is why it covers fewer of them. The effective number of years is $1/\text{HHI}_d$, computed by census cycle. Table 3 interacts the first stage with these measures.

share of the induced inflow against the rank fraction, with the uniform diagonal as the no-concentration benchmark. The induced inflow lies far above that diagonal: a median of 5.0 effective years out of twelve, with the most affected quarter of years carrying 70% of it. Panel (b) also plots total observed migration, whose curve sits much closer to the diagonal – 8.1 effective years against 5.0. This is dilution rather than evidence against the mechanism. The first stage implies that a one-standard-deviation drought raises the inflow by 6.8% of the mean, so observed flows are roughly 93% baseline migration and 7% drought-induced, and a concentration curve on that mixture is dominated by the baseline. Injecting the estimated induced component into the observed series moves its concentration from 9.56 to 9.54 effective years, 0.5% of the distance to the predicted curve. Total flows are mechanically incapable of revealing shock synchrony at this effect size, and we report the observed curve as a descriptive contrast only.

Drought reaches a destination through many origins at once. Table 2 reports descriptive statistics for the three indicators, showing an important synchrony. The median destination draws on origins whose droughts correlate at 0.67, and the variance of the shock it receives is 3.4 times what independence would imply. A permutation test fixes the magnitude without any tuning parameter: shuffling years independently within each origin destroys the synchrony and nothing else, and the standard deviation of a country's mean drought falls from 0.292 to 0.081 ($p < 0.001$). Synchrony alone accounts for 92% of the year-to-year variance of the average shock. It also compresses the induced inflow into a few years: because a destination's origins dry together, the migration they induce does not arrive as a steady trickle. Supplementary Appendix Figure S1 reports the same facts country by country, every country lying well above the independence benchmark on both measures.

Shock structure amplifies the first stage. Table 3 interacts the first stage with these three destination-level measures. Column (1) interacts the instrument with its temporal concentration, how unevenly the induced inflow is spread across the years of the window: where it is most concentrated, the first stage is 53% stronger. Column (2) uses the variance ratio, how far the shocks covary relative to the independence benchmark: +23%. Column (3) interacts with the average pairwise correlation of drought across the destination's origins and gives +10%, not distinguishable from zero ($p = 0.54$) – it is the noisiest of the three and covers the fewest destinations. The first two survive controlling for squared drought and for drought interacted with city size, so neither a non-linearity in drought nor the size of the destination is doing the work.

The amplification operates on volume alone. Interacting the same three measures with the housing outcomes leaves the composition margin untouched (+0.13 pp per standard deviation for temporal concentration, $p = 0.760$; −0.03 pp for the variance ratio, $p = 0.908$), though these interactions are under-powered rather than informative nulls: the minimum detectable effect is between 92% and 162% of the main propensity effect.

These results matter for climate projections. Projections for the Sahel and East Africa imply not only drier growing seasons on average but droughts that are more spatially extensive, hence more correlated across the origins feeding any given city (IPCC, 2021). Our estimates say that this second dimension matters on its own: for the same average intensity, a more spatially correlated shock regime delivers larger and more concentrated migration surges toward urban destinations. Projections built on mean intensity alone will therefore understate the pressure on receiving cities, whose capacity to absorb arrivals is a function

ofage.

Table 3: The Spatio-Temporal Structure of Shocks Amplifies the First Stage

	(1) Temporal concentration	(2) Variance ratio	(3) Pairwise correlation
Drought intensity	+0.00406***	+0.00336***	+0.00337***
	(0.00070)	(0.00066)	(0.00067)
Drought × shock structure	+0.00216***	+0.00077***	+0.00033
	(0.00070)	(0.00029)	(0.00053)
Amplification per SD, %	+53	+23	+10
Interaction coefficient under additional controls			
+ drought squared	+0.00213***	+0.00078***	+0.00035
	(0.00070)	(0.00029)	(0.00054)
+ drought × city-size class	+0.00209***	+0.00089***	+0.00033
	(0.00069)	(0.00031)	(0.00054)
Observations	235,922	216,990	216,990
Destinations	343	318	318
OD pairs	16,489	15,043	15,043

Notes: The dependent variable is the urban migration rate; the specification is the paper's main first stage interacted with the standardized shock-structure measure C_d of Table 2, with standard errors clustered at the origin district. The level of C_d is absorbed by the origin–destination fixed effects, so the interaction reads as the change in the first-stage coefficient per standard deviation of C_d, and the amplification row expresses it as a percentage of the main effect. The last two rows repeat the interaction adding squared drought, and drought interacted with the destination's Africapolis size class. Column (3) covers fewer destinations because the pairwise correlation requires at least three origins with eight observed years. Stars: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

3.6 Reduced Form

Table 4 and Table 21 report the reduced-form relationship between drought at origin and the scale and propensity outcomes, respectively, estimated as

$$Y_{odt} = \beta\, D_{ot} + \gamma_1\, |t - t_c| + \gamma_{od} + \delta_t + \varepsilon_{odt}, \tag{3}$$

where Y_{odt} is either the low-quality housing inflow rate S_{odt} (the scale outcome) or the share of migrants in low-quality housing P_{odt} (the propensity outcome) in corridor (o, d) at time t; D_{ot} is drought intensity at origin; γ_{od} and δ_t are origin-destination pair and year fixed effects; and $|t-t_c|$ is the distance-to-census control. The reduced form regresses the outcome directly on the instrument, without instrumenting migration; it captures the total effect of a drought shock at origin on the outcome and rests only on the exogeneity of rainfall variation at origin.

Residential status - Scale - Columns (1)-(4) use the contemporaneous drought instrument and columns (5)-(8) its cumulative (three-season mean) counterpart on the lag-restricted sample; within each group, the first column is our main specification and the remaining columns progressively saturate the fixed effects, up to destination-by-year effects that absorb all destination-year shocks. The table shows the stability of the estimate: under the contemporaneous instrument the coefficient is essentially identical across all four configurations, indicating that the relationship is driven by within-corridor variation in drought rather than by fixed differences across origins or destinations or by destination-level shocks. The main specification in column (1) implies that a one standard deviation increase in drought intensity at origin raises the low-quality housing inflow rate by 3.3% of its mean; under the cumulative instrument (column (5)), the same shock raises it by 6.7%, consistent with several consecutive poor seasons displacing more households than a single one.

Table 4: Reduced Form: Drought Intensity and Low-Quality Housing Inflows, Both Instruments

	Contemporaneous drought				*3-season mean drought (lagged)*			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Main spec.				Main spec.			
Drought intensity	0.00103***	0.00106***	0.000546	0.000546	0.00266***	0.00278***	0.00151	0.00150
	(0.000350)	(0.000354)	(0.000431)	(0.000437)	(0.000735)	(0.000751)	(0.000999)	(0.00101)
Effect of 1-SD drought (% mean)	+4.3%	+4.5%	+2.3%	+2.3%	+6.6%	+6.9%	+3.8%	+3.7%
Origin FE	No	Yes	No	No	No	Yes	No	No
Destination FE	No	Yes	No	No	No	Yes	No	No
Origin-Destination FE	Yes	No	Yes	Yes	Yes	No	Yes	Yes
Year FE	Yes	Yes	No	Yes	Yes	Yes	No	Yes
Country × Year FE	No	No	Yes	No	No	No	Yes	No
Destination × Year FE	No	No	No	Yes	No	No	No	Yes
Dist. to census control	Yes	Yes	No[a]	No[a]	Yes	Yes	No[a]	No[a]
Destination controls	No	No	No	No	No	No	No	No
Origin controls	No	No	No	No	No	No	No	No
R^2	0.370	0.088	0.372	0.393	0.383	0.091	0.386	0.407
Within R^2	0.003	0.002	0.000	0.000	0.003	0.002	0.000	0.000
Mean outcome	0.01196	0.01196	0.01196	0.01196	0.01189	0.01189	0.01189	0.01189
Observations	241,540	241,540	241,540	241,540	235,920	235,920	235,920	235,920
OD pairs	16,846	16,846	16,846	16,846	16,846	16,846	16,846	16,846

Notes: Reduced form: the low-quality-housing inflow rate, per 100 urban residents, regressed directly on drought intensity at origin. Columns (1)–(4) use the contemporaneous instrument; columns (5)–(8) its cumulative three-season counterpart on the lag-restricted sample. Within each group the first column is the paper's main specification and the remaining columns saturate the fixed effects. [a]The distance-to-census control is mechanically absorbed under country × year or destination × year fixed effects. Effect of 1-SD drought: coefficient × SD(instrument) ÷ mean outcome × 100, on each column's own sample. Sample: urban migrants, all destinations, 8 countries, 12-year window. SEs clustered at *admin_origin*, in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Residential status - Propensity - Beyond raising the number of arrivals, drought may shift *who* arrives toward more housing-vulnerable migrants. Table 21 in Appendix 10.3.2 re-

ports the reduced form for this propensity channel: the direct effect of drought at origin on the composition of the migrant inflow. Columns (1)–(2) use the contemporaneous drought instrument and columns (3)–(4) the cumulative (three-season mean) instrument on the lag-restricted sample. Odd columns report the extensive margin, based on the paper's main dependent variable: the share of migrants in corridor (o, d, t) living in low-quality housing, $P_{o,d,t}$. Even columns report the depth of deprivation: the mean number of deficient housing dimensions among the same migrants. With the contemporaneous instrument, a one standard deviation drought raises the share of arriving migrants in low-quality housing by 0.34 percentage points, off a 46% baseline, while the depth of deprivation moves little. The cumulative instrument delivers a stronger and broader pattern: the extensive-margin shift roughly doubles and the depth of deprivation also rises. Because the reduced form does not rely on the first stage, these lagged estimates remain interpretable despite the weak first stage of the cumulative instrument. These results indicate that persistent drought sends not only more migrants but also sends migrants that are more likely to live in low-quality housing.

Figure 15 in Appendix 10.3.3 plots binned migration and low-quality housing inflows against drought intensity at origin, after residualizing on the main specification's fixed effects and controls. Both the first-stage and reduced-form relationships are close to linear and hold across the full distribution of drought shocks, rather than being driven by extreme outliers.

Labor Status - Tables 22 and 23 report the reduced forms for the employment results, regressing the scale and propensity outcomes directly on origin drought for both instruments. On the scale margin, all six inflows rise significantly at the 1% level: a one-standard-deviation drought raises the employed, unemployed, and inactive inflows by 6.8%, 10.7%, and 4.9% of their means with the contemporaneous instrument, and by 8.2%, 21.3%, and 8.2% with the cumulative one. On the propensity margin, the same shock shifts the composition of arrivals by about half a percentage point from inactivity toward employment under both instruments (+0.43 and −0.51 pp contemporaneous; +0.42 and −0.56 pp lagged), while the unemployed share barely moves. Since the reduced form does not rely on the first stage, the lagged columns provide valid inference where the lagged first stage of Table 28 is weak, and confirm the same reallocation.

4 Main Results

4.1 Residential Integration

4.1.1 Scale

Table 5 reports the main estimates of the causal effect of drought-induced migration on the urban low-quality housing inflow (the scale outcome) at destination. Columns (1)-(3) report the OLS, first-stage and 2SLS estimates using the contemporaneous drought instrument - column (3) is the paper's main estimate - and columns (4)-(6) report the same estimates using the cumulative instrument, on the sample restricted to complete lag coverage. The sample is urban migrants to all destinations, with origin–destination pair and year fixed

Table 5: Drought-Induced Migration and Low-Quality Housing Inflows: Main Results

	Contemporaneous drought			*3-season mean drought (lagged)*		
	OLS	FS	2SLS	OLS	FS	2SLS
	(1)	(2)	(3)	(4)	(5)	(6)
Urban migration rate	0.46865***		0.28455***	0.46443***		0.28082***
	(0.02498)		(0.06323)	(0.02536)		(0.04813)
Drought intensity		0.00362***			0.00947***	
		(0.00066)			(0.00164)	
First-stage F			29.65			33.42
AR 95% CI			[0.132, 0.397]			[0.171, 0.370]
Effect of 1-SD drought (% mean)	+7.1%	+6.8%	+4.3%	+11.0%	+10.5%	+6.6%
Origin-Destination FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Dist. to census control	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.697	0.486	0.647	0.702	0.499	0.652
Within R^2	0.521	0.010	0.441	0.518	0.011	0.437
Mean outcome		0.011964			0.011892	
Mean migration rate		0.026820			0.026804	
SD(drought intensity)		0.502			0.297	
Observations		241,540			235,920	
OD pairs		16,846			16,846	

Notes: Columns (1)–(3): contemporaneous instrument; columns (4)–(6): 3-season mean instrument on the lag-restricted sample, reported as robustness. Column (3) is the paper's main estimate. OLS and 2SLS columns report the effect of the urban migration rate on the low-quality housing inflow rate; FS columns report the first stage of the migration rate on drought intensity, reproduced from Table 1 and its lagged counterpart. First-stage F is the cluster-robust Wald statistic on the instrument and AR 95% CI the cluster-robust Anderson–Rubin interval. Effect of 1-SD drought is the coefficient $\times$ SD(instrument) $\div$ mean outcome $\times$ 100, multiplied by the first-stage coefficient in the OLS and 2SLS columns. Sample: urban migrants, all destinations, 8 countries, 12-year window. SEs clustered at *admin_origin*, in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

effects and the distance-to-census control throughout.

The main scale result shows that drought-induced migration causally raises the inflow into low-quality housing at destination. The main scale result is the 2SLS coefficient in column (3): a 1 pp increase in the drought-induced urban migration rate raises the low-quality housing inflow rate at destination by 0.285 pp. Since both rates are expressed per hundred urban residents, the coefficient reads directly as a ratio of flows: for every additional hundred migrants that drought pushes into a destination, about 28 settle in low-quality housing (see Table 25 for more details on the calculation).

To interpret the magnitude of the 2SLS estimate, consider a representative origin-destination corridor with 10,000 urban residents at destination. At the baseline urban migration rate of 0.0268 migrants per 100 residents (the sample mean reported in Table 5), this corridor receives $0.0268 \times 100 = 2.68$ migrants in a year without drought, of whom $0.0120 \times 100 = 1.20$ settle in low-quality housing, the mean inflow rate reported in the same table. Now let drought intensity at origin rise by one standard deviation ($SD = 0.502$, as reported in the table). The first stage translates this shock into migration: the migration rate rises by $\beta_{FS} \times SD = 0.00362 \times 0.502 = 0.0018$, or 6.8% of its mean, where β_{FS} is the first-stage coefficient from column (2) of Table 5 - 0.18 additional migrants for the corridor, bringing the inflow to 2.86. Combining the two stages translates the same shock into housing: the low-quality housing inflow rate rises by $\beta_{2SLS} \times \beta_{FS} \times SD = 0.285 \times 0.00362 \times 0.502 = 0.00052$ - equivalently, the reduced-form coefficient of Table 4 times SD, since with a single instrument the reduced form equals the product of the two stages - or 0.052 additional residents in low-quality housing. The ratio of the two changes is, by construction, the 2SLS coefficient itself: because the low-quality housing inflow and the migration rate share the same denominator - destination urban population - $\beta_{2SLS} = 0.052/0.18 = 0.28$ reads directly as the share of the marginal inflow that settles in low-quality housing. *Of every hundred migrants that drought pushes into a city - migrants who would not have moved otherwise - roughly twenty-eight end up in low-quality housing.* This 28% is the central quantitative result of the paper; Appendix Table 25 reports the calculation step by step.

Column (6) reports the 2SLS estimate under the cumulative instrument: 0.281, significant at the 1% level, with a strong first stage ($F = 33.42$) and an Anderson-Rubin interval of $[0.17, 0.37]$. The interpretation is the same ratio reading as for the main estimate, applied to a different set of marginal migrants: of every hundred migrants displaced by persistent drought - three consecutive poor seasons rather than a single shock - about 28 settle in low-quality housing, the same proportion as for those displaced by a single-season shock. The two estimates are statistically indistinguishable and each lies well inside the other's Anderson-Rubin interval. The effect therefore does not appear to intensify with the persistence of the shock: households pushed out by repeated poor seasons are absorbed into low-quality housing at the same rate as those pushed out by a single one.

The OLS estimates (columns (1) and (4)) give the raw within-corridor correlation between migration and low-quality housing inflows, without a causal interpretation: corridors with more migration may differ systematically in ways correlated with housing quality. They are substantially larger than their 2SLS counterparts – 0.469 against 0.285 under the contemporaneous instrument, 0.464 against 0.281 under the cumulative one – so the correlation overstates what drought-induced migration does. Neither comparison answers whether the migrants that drought adds are individually more likely to end up in low-quality housing

than those who would have moved anyway, a question about the composition of the inflow that the propensity analysis addresses next.

Table 26 in Appendix 10.4 restricts the sample first to destinations that are main cities, and then to rural-to-main-city corridors. The effect is robust across both restrictions and is strongest along the rural-to-urban margin, consistent with the agricultural-push mechanism at the core of the paper: drought displaces rural households whose livelihoods depend most directly on rainfall, and these are the migrants who most raise urban low-quality housing inflows.

4.1.2 Propensity Shift

Table 6: Propensity for Low-Quality Housing: Main Results

	Contemporaneous drought				*3-season mean drought (lagged)*			
	Deprived Dummy - Main def.		Number of deprivations Continuous		Deprived Dummy - Main def.		Number of deprivations Continuous	
	OLS	2SLS	OLS	2SLS	OLS	2SLS	OLS	2SLS
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Urban migration rate	0.02983**	1.21591**	0.01818	0.94933	0.03382**	1.98110**	0.03086	3.19449*
	(0.01509)	(0.59790)	(0.03971)	(1.06501)	(0.01534)	(0.92526)	(0.03767)	(1.63623)
First-stage F		15.50		15.50		8.24		8.24
AR 95% CI		[0.17, 2.94]		[-1.27, 3.52]		[0.64, 6.47]		[0.60, 10.48]
Effect of 1-SD drought	+0.01 pp	+0.37 pp	+0.0001	+0.0029	+0.01 pp	+0.58 pp	+0.0001	+0.0093
R^2	0.392	0.295	0.450	0.436	0.396	0.141	0.455	0.303
Within R^2	0.007	-0.151	0.002	-0.023	0.007	-0.411	0.002	-0.278
Mean outcome (average migrant)	0.406	0.406	0.662	0.662	0.404	0.404	0.656	0.656
Mean migration rate	0.089	0.089	0.089	0.089	0.089	0.089	0.089	0.089
Implied marginal migrant		0.515		0.747		0.580		0.940
Observations	72,723	72,723	72,723	72,723	71,207	71,207	71,207	71,207
OD pairs	10,537	10,537	10,537	10,537	10,510	10,510	10,510	10,510

Notes: Shares among urban in-migrants under the main UN-Habitat definition – at least one deprived dimension among floor, wall, roof, sanitation and crowding, metal roofs durable – and the mean number of deprived dimensions. Water is excluded: the harmonized IPUMS water-supply codes in these censuses cannot reliably separate deficient from adequate access. 2SLS with migration instrumented by contemporaneous (cols (1)–(4)) or 3-season mean (cols (5)–(8), lag-restricted sample) drought. Weak lagged first stage ($F \approx 7$): lagged point magnitudes not interpretable – AR intervals only. Main specification (od-pair and year FE, `dist_census` control); SEs clustered at `admin_origin`. Effect of 1-SD drought in pp of the share / dimension units. 'Mean outcome (average migrant)' and 'Mean migration rate' are means of each column's own estimation sample. 'Implied marginal migrant' is $\overline{P} + \beta\overline{M}$, the propensity of the marginal migrant himself: since $S = P \times M$, the 2SLS coefficient equals $(p_{\text{marg}} - \overline{P})/\overline{M}$. Both means are taken on that same sample – never on the full panel, where the mean migration rate is three times smaller and would shrink the implied gap by the same factor. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 6 examines the propensity shift. The outcome is $P_{o,d,t}$, the share of the migrants arriving on corridor $o \to d$ in year t who live in low-quality housing under the paper's main definition, and the specification is that of the scale outcome with the inflow rate replaced by this share. The regression asks whether the share of migrants in slums rises when drought

pushes the number of arrivals up, within a corridor and net of year effects. A positive coefficient admits two readings: the compliers are more often badly housed than the infra-marginal migrants of the same cohort, or the additional arrivals crowd those infra-marginal migrants out of decent housing. The design cannot separate them, but the end of this section bounds their combined magnitude. The coefficient is a derivative with respect to a migration rate whose mean is 0.089, so its level is not read directly; the row reporting the effect of a one-standard-deviation drought converts it into percentage points of the share.

Columns (1)–(2) and (5)–(6) use the share itself, the extensive margin; columns (3)–(4) and (7)–(8) the mean number of deprived dimensions among the same migrants, the depth of deprivation. Within each pair the first column is OLS and the second the 2SLS estimate, which is the one we interpret. Columns (1)–(4) use the contemporaneous instrument and columns (5)–(8) the cumulative instrument on the lag-restricted sample. The OLS coefficient in column (1) is small, 0.030: across all migration, larger inflows barely change the deprived-housing share of arrivals, so ordinary migration comes at roughly the average housing mix. The 2SLS estimate is forty times larger.[16] Robustness across alternative deprivation definitions is reported in Appendix Table 31.

Drought at the origin sends larger cohorts, and those cohorts are differently composed. The main propensity result is the 2SLS estimate in column (2): a one-standard-deviation drought raises the share of arriving migrants living in low-quality housing by 0.37 percentage points, from 40.6% to 41.0%. The arithmetic runs as follows. Because the share is defined only on corridor-years with a positive urban flow, the relevant first stage is the one estimated on that restricted sample (Table 20, Panel B, column (1)): a coefficient of 0.00601, so a one-standard-deviation drought moves the migration rate by $0.00601 \times 0.503 = 0.0030$, or 3.4% of its mean. The propensity coefficient, 1.216, is a slope with respect to that rate, so applying it to that change gives $1.216 \times 0.0030 = 0.0037$, the 0.37 percentage points.

Under the cumulative instrument the effect appears on both margins: persistent drought raises the share of migrants in low-quality housing (column (6)) and the depth of their deprivation (column (8)).[17] The first stage on this sample (Table 20, Panel B, column (2)) gives a coefficient of 0.01007, so a one-standard-deviation cumulative drought moves the migration rate by 0.0029, or 3.3% of its mean. Applying the two coefficients to that change, the share of arriving migrants in low-quality housing rises by 0.58 percentage points off a baseline of 40.4% — about half again the contemporaneous shift — and the mean number of deprived dimensions rises by 0.009 off a baseline of 0.66. These patterns are consistent with the reduced-form estimates (Table 21), which do not depend on first-stage strength, though they cannot net out the migration channel. Persistent drought thus displaces a more housing-vulnerable population across a broader range of margins, consistent with repeated poor seasons eroding household resources before departure.

[16] The contrast runs opposite to the one at the scale margin, where OLS exceeds 2SLS (0.469 against 0.285). There the raw correlation is inflated, since the two rates share a denominator and corridor-years with more migration differ in other ways. Here it is not a matter of bias but of populations: OLS averages over every source of a migration surge, most of which leave the composition of the inflow unchanged, whereas the instrument isolates the one source that shifts it.

[17] Because the first stage is weak on this subsample ($F = 8.24$), inference rests on the Anderson–Rubin intervals, which remain valid under weak instruments and exclude zero on both margins ($[0.64, 6.47]$ and $[0.60, 10.48]$); the magnitudes below are therefore indicative.

4.1.3 Decomposing the Effect: Scale versus Propensity

Drought fills low-quality housing through two channels that can operate at once: it sends more migrants (the scale channel), and it may change how those migrants are housed relative to the ones a corridor sends in ordinary years (the propensity channel). On the sample where both margins are defined, a one-standard-deviation drought raises the number of migrants by 3.4% while raising their propensity for low-quality housing by 0.37 percentage points, from 40.6 to 41.0 percent of arrivals.

Volume accounts for 80% *of the response, composition for* 20%. - Because the scale outcome and the migration rate share the same urban-population denominator, the scale outcome is exactly the product of the two margins: $S_{odt} = P_{odt}M_{odt}$. Expanding this product around corridor-specific means yields

$$S_{odt} = \overline{P}_{od}M_{odt} + \overline{M}_{od}P_{odt} + \left(P_{odt} - \overline{P}_{od}\right)\left(M_{odt} - \overline{M}_{od}\right) + \text{const}_{od}, \tag{4}$$

where the corridor-specific constant is absorbed by the corridor fixed effect. Estimating Equation (2) with each of the three remaining terms as the outcome therefore provides an exact decomposition of the scale coefficient.

On the sample for which P_{odt} is defined, the scale coefficient is 0.320, compared with 0.285 in the full panel. Its three components are 0.299, 0.063, and -0.042, respectively. The first captures the additional low-quality-housing inflow mechanically generated when drought increases migration while holding the corridor's prevailing housing composition fixed. The second captures the contribution of changes in migrant composition. The third is the within-corridor covariance between the two deviations, and it belongs to the volume channel rather than to composition: grouped with the first term it comes to 0.257, which is 80% of the coefficient, against 0.063, or 20%, for composition; the two shares sum to the whole. Drought therefore increases the inflow into low-quality housing primarily by sending more migrants, and only secondarily by changing their housing composition.

4.1.4 Compliers *vs* Always takers

Drought-induced migrants are not distinguishable from the always-takers they arrive with. Section 3 established that β_1 in (2) is itself a propensity: the shared denominator cancels, so the coefficient counts how many of every hundred drought-induced migrants settle in low-quality housing. It is 0.285 on the full panel and 0.320 on the sample where $P_{o,d,t}$ is defined. The always-takers of the same corridors supply its benchmark. Had the additional arrivals been housed at their corridor's standing share $\overline{P}_{o,d}$, each would have raised the inflow by exactly $\overline{P}_{o,d}$, so the benchmark is the coefficient β_B obtained by replacing $S_{o,d,t}$ with $\overline{P}_{o,d}\,\text{MigRate}_{o,d,t}$ in (2), which is 0.299. We estimate the difference in a single regression rather than as the difference between two separately estimated coefficients, taking

$$G_{o,d,t} \equiv S_{o,d,t} - \overline{P}_{o,d}\,\text{MigRate}_{o,d,t}$$

as the outcome of the second stage:

$$G_{o,d,t} = \beta_G \widehat{\text{MigRate}}_{o,d,t} + \gamma_2|t - t_c| + \mu_{o,d} + \lambda_t + \nu_{o,d,t}, \tag{5}$$

with the first stage in Equation (1) unchanged. Because the 2Sits outcome, $\beta_G = \beta_1 - \beta_B$, and Equation (5) directlyprovides a cluster-robust standard error for this difference.

The estimate is $\beta_G = 0.021$ (0.085), with $p = 0.80$ and a 95% confidence interval of $[-0.145, 0.187]$. A positive β_G could arise either because drought-induced migrants are more likely than their corridor's existing migrant population to live in lowse their arrival displaces infra-marginal migrants into low-quality housing.Equation (5) cannot distinguish between these two mechanisms, but it bounds their combined effect: relative to housing the additional arrivals at the corridor's prevailing rate, the estimated deviation lies between -14.5 and $+18.7$ percentage points and iishable from zero.[18]

Section 7.1 tests crowding-out directly and encounters the same limitation: drought-induced migration leaves unchanged the housing outcomes of migrants who arrived along the same corridor in earlier years and therefore cannot be compliers with the current arrival cohort—where selection and crowding-out could both operate—thetest cannot distinguish between the two mechanisms either ($[-0.21, 3.53]$). The paper therefore bounds the two interpretations jointly and does not attribute the observed shift to either one.

One further interpretation remains possible. Evaluating the propensity coefficient against the panel-wide means rather than each corridor's own mean would place the low-quality-housing rate among drought-induced migrants at

$$\overline{P} + \beta_P \overline{M} = 0.406 + 1.216 \times 0.089 = 0.515,$$

a gap of 10.8 percentage points above the 40.6% rate among the average arrivals. This is not the point estimate from Equation (5). Expanding $S_{o,d,t} = P_{o,d,t}\,\text{MigRate}_{o,d,t}$ around the panel-wide means introduces a thirthe instrument with$(P_{o,d,t} - \overline{P})(\text{MigRate}_{o,d,t} - \overline{M})$. Its instrumented coefficient is -0.194. The calculation above discards this term, whereas Equation (5) retains it and returns an estimate of 0.021. The implied 10.8-percentage-point gap nevertheless lies within the confidence interval, so the data cannot distinguish it from zero. What the design establishes more precisely is the shift in the inflow itself, which Section 4.1.3 shows accounts for 20% of the aggregate response.

4.2 Labor-Market Integration

The housing results show that drought-induced migration raises the inflow into low-quality housing. In this section, we address a complement: whether and how drought-induced migrants - the compliers of our IV estimate - are absorbed into the urban labor market, in terms of employment status: employed, unemployed, and inactive. For all three, we use the same identification strategy, counting the number of migrants within the corridor (o, d, t) that are employed, unemployed, and inactive per 100 urban residents, and we measure both the scale and the propensity channel.

Table 7: Drought-Induced Migration and the Labor-Force Status of the Inflow: Scale

	Employed		*Unemployed*		*Inactive*	
	OLS	2SLS	OLS	2SLS	OLS	2SLS
	(1)	(2)	(3)	(4)	(5)	(6)
Urban migration rate	0.52978***	0.67630***	0.06600***	0.13033***	0.40057***	0.20025***
	(0.01562)	(0.05616)	(0.00741)	(0.02215)	(0.01511)	(0.05651)
First-stage F		28.84		28.84		28.84
AR 95% CI		[0.572, 0.808]		[0.092, 0.186]		[0.058, 0.299]
Effect of 1-SD drought (% mean)	+5.8%	+7.4%	+6.1%	+12.1%	+7.9%	+4.0%
R^2	0.828	0.801	0.374	0.277	0.687	0.593
Within R^2	0.672	0.621	0.142	0.009	0.550	0.414
Mean outcome	0.00972	0.00972	0.00114	0.00114	0.00537	0.00537
Observations	241,540	241,540	241,540	241,540	241,540	241,540

Notes: Outcomes: employed / unemployed / inactive in-migrants per 100 urban residents, unwinsorized. Population: in-migrants aged 15 or over *at arrival* (age minus years since migration). EMPSTAT is not asked below age 5 and records 5–14 year-olds as inactive in over 80% of cases, so on the unrestricted sample the inactive category pooled migrant children with adults out of the labour force; among those aged 15 or over at arrival, labour-force status is known for 99.8% against 94.6% overall. The housing tables are not restricted, housing being an attribute of the dwelling. The endogenous regressor is the migration rate of in-migrants aged 15 or over at arrival, so the first stage differs from that of the housing tables – its value for this table is reported above – and each coefficient reads per *adult* migrant. 2SLS: migration instrumented by contemporaneous drought. Main specification (od-pair and year FE, `dist_census` control); SEs clustered at `admin_origin`. Undetermined labor-force status excluded from the known-status denominator. Sample: urban migrants, all destinations; 8 countries, 12-year window. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

4.2.1 Scale

This section investigates how climate shocks translate into employment status - we measure the effect of climate-induced migration on the employment status of migrant. We restrict the population to adults, meaning individuals aged 15 or over in the year of their arrival, on both sides of the regression: the outcome counts and the migration rate itself count adult in-migrants only, the destination population in the denominator being unrestricted and cancelling between the two. The first stage is therefore specific to these tables ($F = 28.84$ against 29.65 for housing), and each coefficient reads per adult migrant. Table 7 addresses the scale margin, replacing the outcome of the main specification with the number of employed, unemployed, and inactive adult in-migrants per 100 urban residents. [19]The specification, instrument, and sample are otherwise identical to Table 5 , the endogenous migration rate apart.

Table 7 shows that climate shocks drives more individuals employed in uban areas. All three inflows increase significantly with migration, both in OLS and 2SLS. The 2SLS coefficient for the employed inflow is 0.676, significant at the 1% level: an increase in the adult migration rate of one adult in-migrant per 100 urban residents raises the number of employed adult in-migrants by 0.676 per 100 urban residents. The corresponding coefficients are 0.130 for the unemployed inflow and 0.200 for the inactive inflow, both also significant at the 1% level. Because each outcome shares the denominator of the migration rate, these coefficients read directly as the labor-force status of the marginal migrants themselves: of every 100 additional adult drought-induced migrants, about 68 arrive into employment, 13 into unemployment, and 20 into inactivity — the three statuses exhausting the inflow, since labour-force status is recorded for 99.8% of adult arrivals.

In relative terms, a one-standard-deviation drought raises the employed, unemployed, and inactive inflows by 7.4%, 12.1%, and 4.0% of their respective means. The larger relative rise in unemployment comes off a very small base - the unemployed inflow is an eighth of the employed one in levels - so in absolute terms employment dominates: the urban labor market absorbs a substantial part of the drought-induced inflow into work. These migrants find employment far more often than they join the ranks of the unemployed, even as nearly three in ten of all arrivals, children included, settle in low-quality housing.

In OLS the same total is allocated differently – 0.530 employed, 0.066 unemployed and 0.401 inactive, against 0.676, 0.130 and 0.200 in 2SLS – so that inactivity falls from 40% of ordinary migration to 20% of the drought-driven margin: the migration drought induces is markedly more attached to the labour force than migration in general. Whether drought-induced migrants indeed differ from the average arrival in their labor-force status is precisely what the propensity margin tests directly, holding the size of the inflow fixed; we turn to it next.

[18]Power is the binding constraint: the minimumdetectable gap is some twenty-four percentage points, limited by the first stage in this sample ($F = 15.50$) and by an outcome constructed from a median of thirty urban in-migrants per corridor-year. Since $\overline{P}_{o,d}$ is averaged over the corridor's entire sample window, it includes the drought years themselves, which attenuates the estimated difference toward zero.

[19]Migrants whose labor-force status is undetermined in the census are excluded from the counts, which removes only 0.2% of adult arrivals.

4.2.2 Propensity

Table 8: Drought-Induced Migration and the Labor-Force Status of the Inflow: Propensity

	Employed		*Unemployed*		*Inactive*	
	OLS	2SLS	OLS	2SLS	OLS	2SLS
	(1)	(2)	(3)	(4)	(5)	(6)
Urban migration rate	-0.07987***	0.94367	0.01948**	1.19160*	0.06039***	-2.13526*
	(0.02058)	(1.04871)	(0.00913)	(0.64380)	(0.01924)	(1.15674)
First-stage F		11.27		11.27		11.27
AR 95% CI		[-1.031, 4.107]		[0.151, 3.477]		[-6.357, -0.342]
Effect of 1-SD drought	−0.01 pp	+0.15 pp	+0.00 pp	+0.19 pp	+0.01 pp	−0.34 pp
R^2	0.348	0.322	0.265	0.122	0.349	0.218
Within R^2	0.009	-0.030	0.000	-0.196	0.011	-0.189
Mean outcome (average migrant)	0.6381	0.6381	0.0680	0.0680	0.2939	0.2939
Mean migration rate	0.061	0.061	0.061	0.061	0.061	0.061
Implied marginal migrant		0.695		0.140		0.164
Observations	64,554	64,554	64,554	64,554	64,554	64,554

Notes: Outcomes: employed / unemployed / inactive shares among known-status in-migrants; the three shares sum to one, so the 2SLS coefficients sum to zero by construction. 'Mean outcome (average migrant)' and 'Mean migration rate' are means of this table's own estimation sample – the corridor-years with known labor-force status, not the full panel. 'Implied marginal migrant' is $\overline{P} + \beta\overline{M}$; this reading truncates the exact identity $\beta_1 = \overline{P} + \overline{M}\beta_P + \text{cross}$, whose third term is larger in absolute value than the second, and is reported for comparability only. Population: in-migrants aged 15 or over *at arrival* (age minus years since migration). EMPSTAT is not asked below age 5 and records 5–14 year-olds as inactive in over 80% of cases, so on the unrestricted sample the inactive category pooled migrant children with adults out of the labour force; among those aged 15 or over at arrival, labour-force status is known for 99.8% against 94.6% overall. The housing tables are not restricted, housing being an attribute of the dwelling. The endogenous regressor is the migration rate of in-migrants aged 15 or over at arrival, so the first stage differs from that of the housing tables – its value for this table is reported above – and each coefficient reads per *adult* migrant. 2SLS: migration instrumented by contemporaneous drought. Main specification (od-pair and year FE, `dist_census` control); SEs clustered at `admin_origin`. Undetermined labor-force status excluded from the known-status denominator. Sample: urban migrants, all destinations; 8 countries, 12-year window. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 8 turns to the propensity margin, looking at how the drought shock shifts the distribution of the arriving cohort. The outcomes are now the shares of employed, unemployed, and inactive migrants among adult in-migrants with known labor-force status. As in the housing propensity analysis, the denominator is the number of migrants rather than the urban population, so the estimates hold the size of the inflow fixed and isolate changes in its composition. Because the three shares sum to one, the 2SLS coefficients sum to zero by construction: drought can only reallocate migrants across statuses. The specification and instrument are identical to the main table; on this smaller sample of corridors where adult labour-force status is observed ($N = 64{,}554$), the first stage is weaker, though still above the conventional threshold ($F = 11.27$).

Drought tilts the inflow toward the labour force: the inactive share falls by 2.135 and the unemployed share rises by 1.192, both at the 10% level (and with Anderson–Rubin intervals

excluding zero), while the employed share rises by 0.944 without being separately significant. A one-standard-deviation drought shifts the composition by −0.34 percentage points away from inactivity, +0.19 toward unemployment and +0.15 toward employment - mostly into the labor force.

OLS runs the other way — −0.080 on the employed share and +0.060 on the inactive, both at the 1% level — so the drought-responsive margin is not the average source of migration: it differs in the direction of labour-force participation. Whether the individual migrant differs is a question this design bounds rather than answers.

Table 28 in Appendix 10.4.3 re-estimates both margins with the cumulative drought instrument: the scale results are unchanged, and the shift of arrivals from inactivity toward employment persists.

4.2.3 Decomposing the Effect: Scale versus Propensity

Employment decomposes exactly as housing does. Because the employed inflow and the migration rate share the urban-population denominator, the inflow is all but exactly the product of the two margins, and expanding it around corridor means as in (4) partitions the scale coefficient. On the sample where the employed share is defined — where that coefficient is 0.803 against the 0.676 of the full panel — the three parts are 0.675, 0.156 and −0.034: what the corridor's standing employment rate mechanically produces once drought adds adult arrivals, the contribution of the composition shift, and a negligible remainder.[20] About four fifths of the additional employed inflow therefore operates through the number of adult migrants and one fifth through their composition — the same split as on housing, where the two channels divide 80 and 20.

4.2.4 Compliers *vs* Always takers: Employment

The same test applies to the three labour-force statuses, and because their shares sum to one the three gaps must offset — a check the estimates pass. The marginal adult migrant's employment rate is 0.803 against a corridor benchmark of 0.675, so $\beta_G = +0.127$ (0.117); unemployment gives +0.050 (0.035) and inactivity −0.166 (0.112), the three summing to +0.011 and the three benchmarks to exactly one. The pattern is the one the propensity estimates give: drought-induced adults come out of inactivity and into the labour force, divided between work and unemployment. No single gap is distinguishable from zero ($p =$ 0.28, 0.15 and 0.14), and the housing gap is +0.021 (0.085). On none of these margins, therefore, can the compliers be separated from the always-takers they join — a limit of power rather than a finding of equality.

Table 9: What Kind of Job Do Drought-Induced Migrants Find? Class of Worker

	All	Wage/salary	Self-employed	Unpaid/other
	(1)	(2)	(3)	(4)
Urban migration rate	0.74143***	0.44137***	0.30199***	-0.00194
	(0.09949)	(0.08234)	(0.05898)	(0.03031)
First-stage F	13.92	13.92	13.92	13.92
AR 95% CI	[0.574, 1.048]	[0.292, 0.676]	[0.205, 0.490]	[-0.090, 0.052]
Of 100 marginal migrants	74.1	44.1	30.2	-0.2
Share, marginal migrant (%)	100.0	59.5	40.7	-0.3
Share, average arrival (%)	100.0	59.3	33.4	7.4
Gap (pp)	–	+0.3	+7.4	-7.6*
AR 95% CI (gap)	–	[-15.3, +15.8]	[-4.7, +25.5]	[-20.7, -0.6]
Origin-Destination FE	Yes	Yes	Yes	Yes
Arrival-year FE	Yes	Yes	Yes	Yes
Dist. to census control	Yes	Yes	Yes	Yes
Mean outcome	0.008772	0.004531	0.003440	0.000802
Observations	224,668	224,668	224,668	224,668

Notes: In-migrants of each job type per 100 urban residents at destination, unwinsorized, among urban in-migrants with EMPSTAT = 1 who were aged 15 or over *at arrival*, as in Tables 7 and 8. The endogenous regressor is the migration rate of in-migrants aged 15 or over at arrival, the same population as the numerators, so each coefficient reads per *adult* migrant and the first stage differs from the housing tables. Main specification: 2SLS of the urban migration rate on contemporaneous origin drought, pair and arrival-year fixed effects, distance-to-census control, SEs clustered at `admin_origin`. Outcome and migration rate share a denominator, so each coefficient is the number of 100 marginal drought-induced migrants in that category and column (1) is the sum of (2)–(4). 'Share, marginal migrant' divides by that total; 'Share, average arrival' is the migrant-weighted share among all urban in-migrants of the sample; 'Gap' is their difference, estimated by 2SLS on the excess outcome and rescaled by the total, its AR interval scaled alike. IPUMS `CLASSWK`, carried by 10 of the 11 urban census samples (all but Kenya 2009). $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 10: What Kind of Job Do Drought-Induced Migrants Find? Sector

	All	Agriculture	Industry	Services
	(1)	(2)	(3)	(4)
Urban migration rate	0.59636***	0.14910***	0.04924	0.39803***
	(0.11682)	(0.05195)	(0.04411)	(0.10900)
First-stage F	8.43	8.43	8.43	8.43
AR 95% CI	[0.392, 1.062]	[0.044, 0.328]	[-0.094, 0.143]	[0.207, 0.829]
Of 100 marginal migrants	59.6	14.9	4.9	39.8
Share, marginal migrant (%)	100.0	25.0	8.3	66.7
Share, average arrival (%)	100.0	7.8	15.9	76.3
Gap (pp)	–	+17.2**	-7.6	-9.6
AR 95% CI (gap)	–	[-0.5, +43.7]	[-36.9, +6.3]	[-32.7, +20.1]
Origin-Destination FE	Yes	Yes	Yes	Yes
Arrival-year FE	Yes	Yes	Yes	Yes
Dist. to census control	Yes	Yes	Yes	Yes
Mean outcome	0.008146	0.001313	0.001293	0.005541
Observations	209,548	209,548	209,548	209,548

Notes: In-migrants of each job type per 100 urban residents at destination, unwinsorized, among urban in-migrants with EMPSTAT = 1 who were aged 15 or over *at arrival*, as in Tables 7 and 8. The endogenous regressor is the migration rate of in-migrants aged 15 or over at arrival, the same population as the numerators, so each coefficient reads per *adult* migrant and the first stage differs from the housing tables. Main specification: 2SLS of the urban migration rate on contemporaneous origin drought, pair and arrival-year fixed effects, distance-to-census control, SEs clustered at `admin_origin`. Outcome and migration rate share a denominator, so each coefficient is the number of 100 marginal drought-induced migrants in that category and column (1) is the sum of (2)–(4). 'Share, marginal migrant' divides by that total; 'Share, average arrival' is the migrant-weighted share among all urban in-migrants of the sample; 'Gap' is their difference, estimated by 2SLS on the excess outcome and rescaled by the total, its AR interval scaled alike. IPUMS `INDGEN`, carried by 9 of the 11 urban census samples (all but Kenya 2009 and Uganda 2014), three-way because the services subdivision is not comparable across censuses. The first stage is weak on this sample ($F = 8.43$), so inference rests on the Anderson–Rubin intervals, which remain valid under weak instruments, and the point magnitudes are indicative. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

4.2.5 Employment Status and Sector of Employment

The previous section shows that climate shocks raise the number of migrants in employment (Table 7) and shift the composition of the arriving cohort toward the labour force and away from inactivity (Table 8). This section investigates which type of jobs these climate-induced migrants find.

Climate shocks translate into paid work, whether salaried or self-employed, and not into unpaid work. Of every hundred additional adult drought-induced migrants, 74 take up employment whose class of worker is recorded: 44 in wage or salaried jobs and 30 self-employed (Table 9). None enter unpaid family labour — the coefficient is -0.002, with an Anderson–Rubin interval of $[-0.090, 0.052]$ — although such work absorbs 7.4% of urban in-migrants in the same censuses, and that gap of -7.6 percentage points is itself distinguishable from zero ($[-20.7, -0.6]$). This runs against the presumption that distress migration ends in dependent arrangements: the shock adds paid workers and no unpaid ones.

The paid jobs are mainly concentrated in services and urban agriculture, and we do not observe integration into industrial work. On the nine censuses recording sector, 60 of every hundred adult drought-induced migrants are working: 40 in services, 15 in urban agriculture, and 5 in industry, the last indistinguishable from zero ($[-0.094, 0.143]$; Table 10). Climate-driven urbanisation therefore adds service and agricultural employment and no industrial employment at all. The tension with Section **??** is direct: drought steers migration toward the cities with the largest industrial base, and sends none of those migrants into industry. This is urbanisation without structural transformation observed on the margin that produces it, alongside the urban agricultural employment shares of up to 40% documented for smaller African urban units (Henderson and Kriticos, 2018; McMillan et al., 2014). The first stage is weak on this sample ($F = 8.43$), so these magnitudes rest on the Anderson–Rubin intervals and are read as indicative: the agricultural share of the marginal migrant, 25.0% against 7.8% for the average arrival, is a suggestive rather than an established contrast, its interval reaching to zero ($[-0.5, +43.7]$). The two totals are not comparable, the sector sample being one census smaller.

5 The Aggregate Picture

Residential integration. - The corridor-level estimates can be aggregated to gauge the size of the phenomenon. Over 1981–2018, across eight countries and 351 destinations, the censuses record about 11.6 million urban in-migrants in the corridors we observe. Drought accounts for roughly 390,000 of these moves, about 3.3% of observed urban in-migration.[21] Applying

[20] The three terms sum to 0.798 against a coefficient of 0.803, a residual of 0.6%. The identity $S = P \times \text{MigRate}$ is exact only when the migration rate counts the same population as the denominator of the share. The employed share is taken over adult in-migrants with a recorded labour-force status, while the migration rate counts all adult in-migrants, as in Tables 7 and 8; status is recorded for 99.8% of adult arrivals, and that 0.2% is the residual.

[21] The number of drought-induced migrants is obtained by multiplying each corridor-year's realized drought intensity by the first-stage coefficient and the destination's urban population, and summing across corridors and years, which yields $\approx$ 387,700 migrants, or 3.3% of the 11.6 million observed. The counterfactual is the absence of drought everywhere. Applying the 2SLS estimate to this flow gives the additional residents of

the 2SLS estimate, under which 28 of every 100 marginal migrants settle in low-quality housing (Table 5), drought placed roughly 110,000 additional people in low-quality housing over the period. Drought degraded urban housing conditions mostly by adding people to the cities and secondarily by re-sorting a given flow into worse housing, in the proportions of Section 4.1.3. We do not split the 110,000 between the two channels: the decomposition is identified on the corridor-years where the propensity is defined, whereas this aggregate is computed on the full panel, and combining the two would mix samples.

Labour-market integration. - The same aggregation applies to the labour-market results, on the population they are estimated for: of the 11.6 million observed urban in-migrants, 8.3 million were aged 15 or over at arrival, and drought accounts for about 226,000 of those adult moves. Applying the coefficients of Table 7, roughly 153,000 arrived into employment, 30,000 into unemployment and 45,000 into inactivity. Drought therefore added workers to the cities on a scale comparable to the residents of low-quality housing it added, and the three statuses exhaust the adult inflow. The adult flow is smaller than the 390,000 of the previous paragraph because roughly two in five drought-induced migrants arrive as children, for whom labour-force status is not defined.

What kind of work. - Of the adult drought-induced migrants in the ten census samples that record class of worker, about 54,000 entered wage or salaried jobs and 37,000 self-employment, against essentially none in unpaid family labour (Table 9). On the nine samples recording sector, about 32,000 entered services, 12,000 urban agriculture and 4,000 industry (Table 10). These two counts rest on smaller samples and on their own first stages, so they cannot be read against the 226,000 of the previous paragraph; the sector figures in particular rest on a weak first stage ($F = 8.43$) and are indicative only.

These magnitudes are conservative. They cover only the corridors observed in the censuses, within the twelve-year windows preceding each census; they capture permanent, inter-regional moves and miss temporary or short-distance migration; and they hold destination conditions fixed, abstracting from any equilibrium response of housing supply.

Working Poor - Table 11 cross-tabulates the two dimensions on the same individuals, the four coefficients summing exactly to the coefficient on the total. Of every hundred adults drought displaces, 45 arrive employed and adequately housed, 22 employed and in low-quality housing, 29 not employed and adequately housed, and 4 in neither: deprived housing receives about a quarter of the inflow, not most of it. Read by housing, the table says what neither margin could alone. Among the drought-induced adults who end up in low-quality housing, 85% are employed; among those who do not, 61%. In the baseline inflow the two rates are indistinguishable, 63% and 65%. Drought does not send the less employable into worse housing — it sends the employed there. The cell that combines deprived housing with no work does not move at all (+0.038, Anderson–Rubin interval $[-0.069, 0.110]$), falling from 12.5% of the baseline inflow to 4% of the marginal one: climate migration adds working people to deprived housing and no inactive poor.

low-quality housing: $387{,}700 \times 0.285 \approx 110{,}300$. Both coefficients are taken from Table 5.

Table 11: Drought-Induced Migration across Employment and Housing Cells: The Working Poor

	In low-quality housing		*Not in low-quality housing*	
	Employed	Not employed	Employed	Not employed
	2SLS	2SLS	2SLS	2SLS
	(1)	(2)	(3)	(4)
Urban migration rate	0.22015***	0.03794	0.45505***	0.29052***
	(0.04108)	(0.04177)	(0.05618)	(0.04122)
First-stage F	28.84	28.84	28.84	28.84
AR 95% CI	[0.136, 0.307]	[-0.069, 0.110]	[0.351, 0.587]	[0.209, 0.381]
Mean outcome	0.004084	0.002918	0.005647	0.003601
Employed share of group (%)	85.3		61.0	
baseline inflow (%)	63.1		64.8	
Observations	241,540	241,540	241,540	241,540

Notes: Outcomes: in-migrants per 100 urban residents in the four employment × low-quality housing cells (main definition), measured on the same individuals – migrants with both labor-force status and housing observed (71.3% of urban migrants). No outcome is winsorized; the sample is restricted to destinations with at least 2,000 urban residents. Column (1) is the "working poor" cell. 2SLS: migration instrumented by contemporaneous drought. Main specification (od-pair and year FE, `dist_census` control); SEs clustered at `admin_origin`. The estimates describe the corridor-level composition of the marginal inflow, not an individual-level causal link between employment and housing. Column (2) is imprecise: the Anderson–Rubin interval includes zero. Sample: urban migrants, all destinations; 8 countries, 12-year window. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

How many. - Applied to the 226,000 drought-induced adult migrants, the four cells give about 103,000 employed and adequately housed, 66,000 not employed and adequately housed, 50,000 working poor and 9,000 in neither, the working-poor cell close to six times the last. Drought thus placed some 58,000 additional adults in low-quality housing; set against the 110,000 of Section 5, this implies that close to half of the people it placed there are migrant children — the worst-housed group in the inflow, and one for whom labour-force status is not defined.

6 Heterogeneity

6.1 Origin and Destination Characteristics

A destination can matter in two ways: by receiving more climate migration, or by housing its migrants differently. The first is a property of the first stage, the second of the second stage, and in our data they do not move together. This section tests destination heterogeneity along seven characteristics – city size, corridor distance, baseline slum shares (which could capture housing saturation), baseline share of labor force in industry, past built-up expansion, baseline migration share, and the pre-period corridor network. We also look at origin characteristics such as the share of area under harvested cropland. The organising fact is an asymmetry: these characteristics shift the volume of climate migration substantially, but its conversion into bad housing and employment is driven by the size of the city. The residential burden – the working poor, migrants who find work but not decent housing – concentrates in secondary and small cities.

6.1.1 Destination City Size

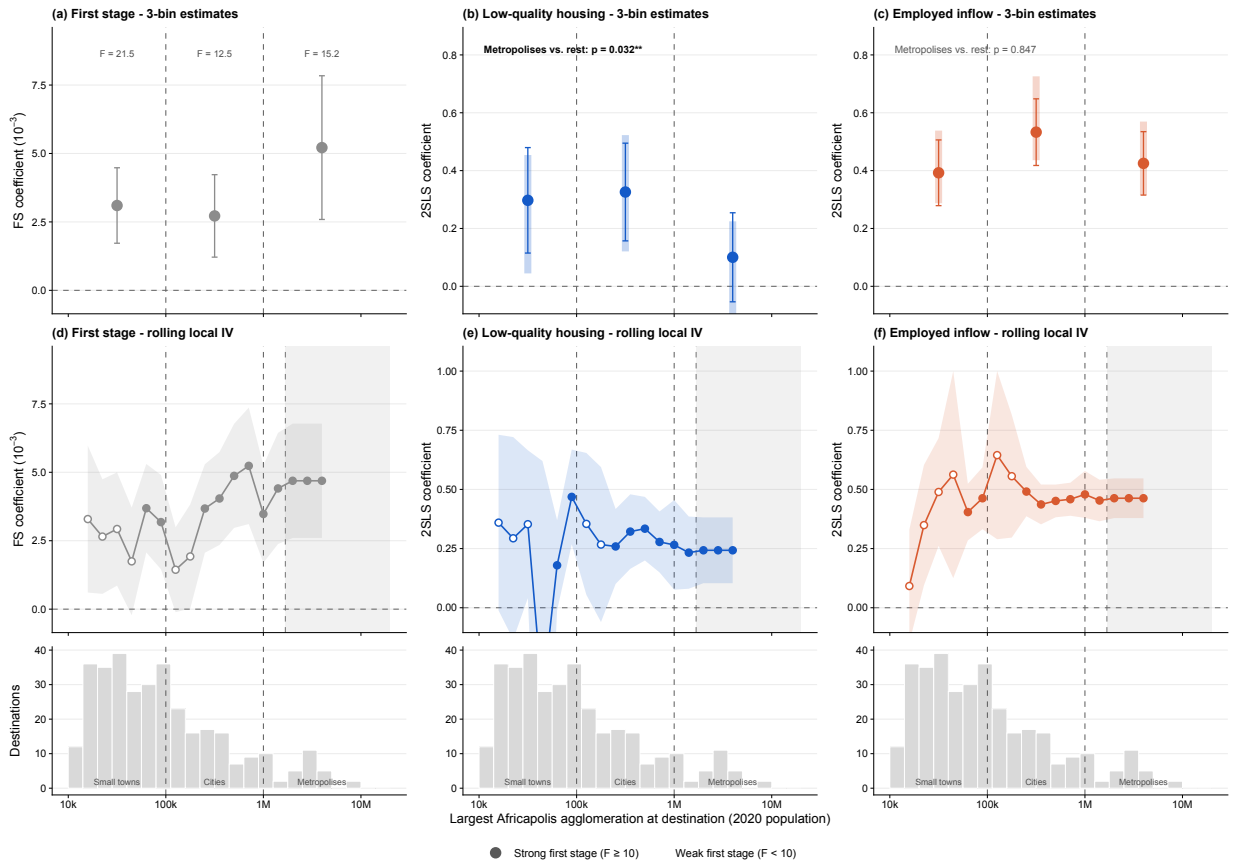
We examine whether the effect varies with the type of destination, classifying destinations by the population of the largest Africapolis agglomeration located in or within 10 km of the destination unit (OECD/SWAC, 2024): small towns (< 100k), secondary cities (100k–1M) and metropolises (≥ 1M). We use this size-based classification as our main cut for three reasons: it is external to the census data that produce our outcomes, it does not require choosing a threshold on a continuous and smoothly distributed urban share, and the agglomeration boundaries are those of the urban-settlement literature rather than of administrative units. Section S1.6 reports the same exercise on the census urban share as a complementary check. We estimate a single specification in which the migration rate is fully interacted with the destination-type indicators:

$$S_{o,d,t} \;=\; \sum_{k=1}^{3} \beta_k \big(\text{MigRate}_{o,d,t} \times \mathbf{1}\{d \in k\}\big) + \sum_{k=1}^{3} \delta_k \big(\text{dist}_{o,d,t} \times \mathbf{1}\{d \in k\}\big) + \alpha_{od} + \mu_{t \times k} + \varepsilon_{o,d,t}, \tag{6}$$

where each interacted migration term is instrumented by the corresponding interaction of drought intensity at origin, $Z_{o,t} \times \mathbf{1}\{d \in k\}$ (three endogenous variables, three instruments: just-identified), α_{od} are corridor fixed effects, and $\mu_{t \times k}$ arrival-year × bin fixed effects. No uninteracted migration term is included, so each β_k measures the absolute effect within

destinations of type k rather than a difference from a reference category. The system is numerically equivalent to estimating the main specification separately by destination type, while the pooling delivers the covariance across the β_k used to test for differences.[22] Figure 2 reports the binned estimates with the first stage by group (panels (a)–(c)), alongside a continuous rolling-window counterpart that re-estimates the main specification along the logarithm of destination city size (panels (d)–(f), indicative only) and the distribution of destinations.

Figure 2: Heterogeneity by Destination City Size (Africapolis): Low-Quality Housing and Employment



Notes: Destinations are classified by the 2020 population of the largest Africapolis (OECD/SWAC, 2024) agglomeration in or within 10 km of the unit: small towns < 100k, secondary cities 100k–1M, metropolises ≥ 1M ($N = 240{,}340$ corridor-years). The classification is independent of the census urban share. Panels (a)–(c): 3-bin saturated interacted IV (migration × bin instrumented by drought × bin, origin–destination and arrival-year × bin fixed effects, bin-specific distance-to-census control, SEs clustered at the origin district), plotted at zone midpoints; whiskers are 95% Wald intervals and shaded bars Anderson–Rubin 95% intervals. On housing, secondary cities exceed metropolises ($p = 0.035$) and metropolises differ from the other two zones pooled ($p = 0.032$); no employment contrast is significant. Panels (d)–(f): rolling-window local IV along log city size, indicative only; the shaded zone marks where the window no longer slides. Hollow points flag a Sanderson–Windmeijer conditional $F < 10$.
Source: Authors' computations using IPUMS census, CHIRPS climate, and Africapolis 2024 data.

[22]Saturation by bin is required for identification: with pooled year fixed effects or a continuous interaction, the interacted instruments do not separately identify the interacted endogenous variables (Sanderson-Windmeijer conditional F near zero).

The housing burden falls on secondary and small cities, not metropolises. - Of every 100 marginal migrants, about 33 enter low-quality housing in a secondary city of 100k–1M (0.326) and about 30 in a small town below 100k (0.297); the two are statistically indistinguishable ($p = 0.809$), so the effect is uniform across the non-primate tier. In metropolises above one million the coefficient falls to 0.100, with an Anderson–Rubin interval that includes zero. This attenuation at the top is the one difference the data establish: the metropolis coefficient lies significantly below the other zones pooled ($p = 0.032$) and below secondary cities pairwise ($p = 0.035$), and the rolling estimates in panel (e) trace the same profile continuously along city size.

Employment absorption, by contrast, does not vary with city size. - Drought migrants find work in destinations of every size – 0.393 in small towns, 0.533 in secondary cities, 0.425 in metropolises, each well away from zero – and metropolises do not differ from the rest ($p = 0.847$). The two margins therefore separate: urban labor markets absorb drought migrants everywhere, while the housing burden concentrates in secondary and small cities.

The gradient is a second-stage result, not a first-stage artifact. - Drought does not send disproportionately more migrants to smaller destinations; if anything the first stage is strongest in metropolises (0.00521 against 0.00272 in secondary cities, $p = 0.095$), precisely where the housing coefficient is smallest. A causal forest confirms that drought's effect on migration varies with corridor distance and origin rurality, not destination size (the destination urban share enters the best linear projection at $p = 0.123$). Because the volume of arrivals is not tilted toward smaller destinations, the size gradient in the housing effect reflects where migrants sort once they arrive, not where drought sends them. The two forces offset in the reduced form, which is flat across city sizes ($p = 0.328$); the contrast is identified by the 2SLS.

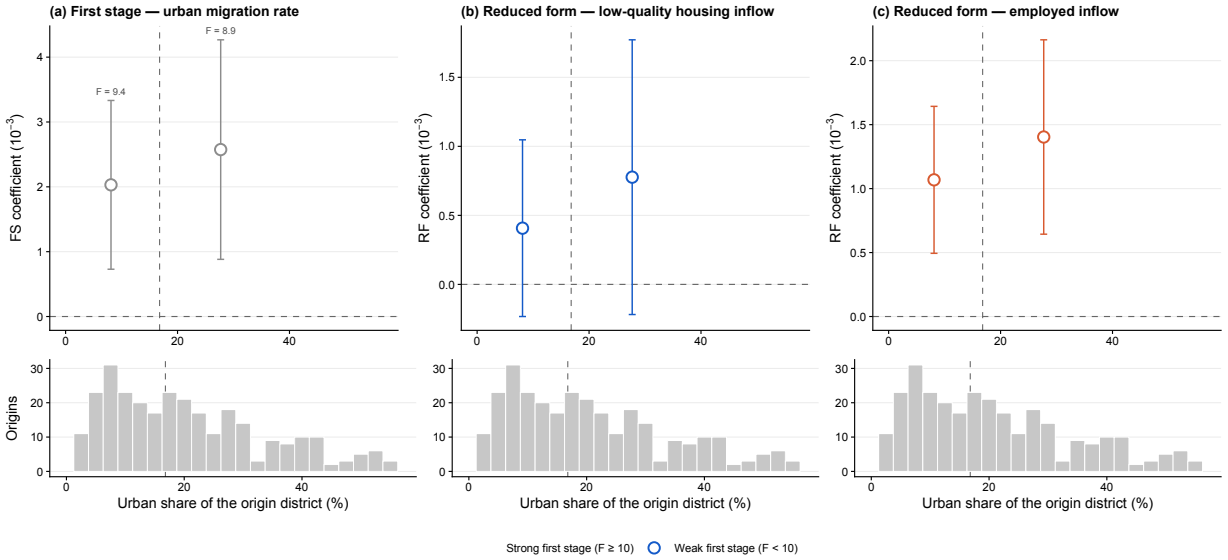
The inflow geography is consistent with the size result. - The share of each destination's in-migration attributed to drought declines with destination size, averaging 7.4% among destinations receiving fewer than 50,000 migrants over the panel against 3.3% among those above 200,000, and reaching 10 to 14% in secondary destinations such as Bushenyi, Kisii and Sikasso against 1.2% in Mombasa (Appendix Figure 19). Classifying destinations by census urban share yields the same direction – the coefficient declines as destinations urbanize – though no contrast along that gradient reaches significance; Section S1.6 reports that exercise.

6.1.2 Origin Characteristics

At origin, the rural world responds uniformly. - The counterpart of city size at origin is rurality, which the agricultural-push mechanism would predict to concentrate the effect. It does not: within the rural world the response is flat. Splitting the 315 rural origin districts (88% of the panel) at their median urban share, the first stage is statistically indistinguishable across halves ($p = 0.618$), as are both reduced forms and the 2SLS conversions. The direction is monotone – less rural origins respond slightly more in every split we construct – but never

separable from zero, and it vanishes within country. Drought displaces the rural world fairly uniformly, which justifies treating it as a single block; the only discontinuity is at the urban threshold, where the 36 majority-urban origins respond markedly more in levels.[23]

Figure 3: Heterogeneity Within the Rural World: Origin Districts Below the Urban Threshold



Notes: Origin districts whose census urban share is below 0.55 (315 origins, 14,739 corridors, $N = 212{,}481$), split at the pooled median of that share (16.8%, dashed line). The classifier is time-invariant, so origin–destination fixed effects stay nested within bins. Panel (a) is the first stage in each half, panels (b)–(c) the two reduced forms, from a saturated interacted specification (instrument × bin, origin–destination and arrival-year × bin fixed effects, bin-specific distance-to-census control, SEs clustered at the origin district); whiskers are 95% Wald intervals. Hollow points flag a conditional first stage below ten, which both halves are ($F = 9.4$ and 8.9). No contrast is significant: $p = 0.618$ on the first stage, 0.540 on low-quality housing and 0.491 on the employed inflow.

Cropland, not rurality. - The census urban share classifies origins by where people live; the agricultural-push mechanism needs to be established on where do they come from and whether this affects their outcomes at destination. For every origin district we compute the share of its area under harvested cropland from CROPGRIDS v1.08 (Tang et al., 2024), summing the harvested area of six staples – maize, cassava, wheat, sorghum, rice and soybean – over the district's 0.05° grid cells and dividing by its land area. [24] Along this classifier the first stage shows that climate-induces migrants are coming from agricultural areas. Split into terciles [25] it rises 0.99, 1.56 and 7.42×10^{-3}, a contrast of $+6.43 \times 10^{-3}$ ($p = 0.0006$)

[23]Splitting the rural world into terciles leaves conditional first stages between 3.7 and 10.8×10^{-3}, so these nulls reflect limited power as much as an absence of effect.

[24]The measure is available for 350 of our 354 origins, covering 99.5% of the panel, and spreads them widely.

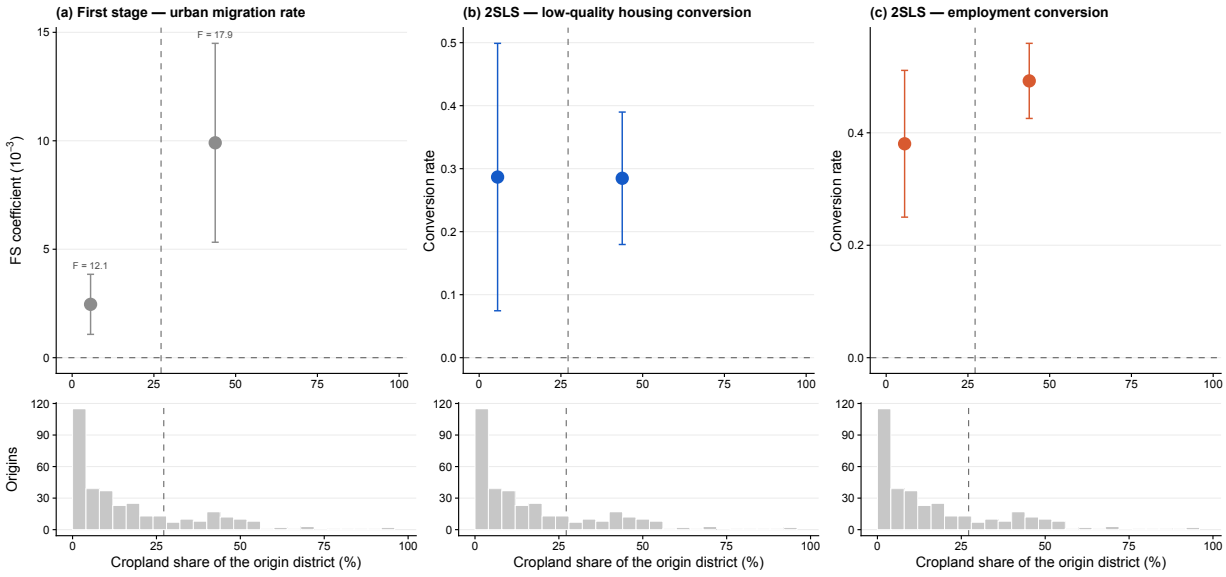
[25]The terciles cut at 4.3% and 18.7% of district area, with median cropland shares of 1.2%, 9.7% and 40.3%

and a ratio of roughly seven and a half. The drought-induced migration this paper studies originates almost entirely in farmland.

The two measures are therefore not substitutes: the census urban share captures settlement, which drought does not discriminate on, while cropland share captures exposure to rainfall, which it does.

The agricultural-push channel is corroborated on crop yields. - Drought should destroy the harvest before it drives migration, and crop statistics let us check this directly. Matching HarvestStat Africa (Lee et al., 2025) to our origin districts (2,535 district-years, 214 districts, six countries), a one-standard-deviation increase in drought lowers maize yield per hectare by 3.1% ($p = 0.0004$) and production by 3.0% ($p = 0.081$), while harvested area does not move (+0.1%). Drought destroys the harvest on the land that is farmed. Together with the cropland gradient above, this is the agricultural-push channel in two pieces – drought destroys the harvest at origin, and migration concentrates where that harvest is the livelihood. We characterise the channel this way rather than through a formal mediation of migration on yields, which the data (within-district yield reliability of 0.30, 69% panel coverage, effect driven by Kenya) cannot identify.

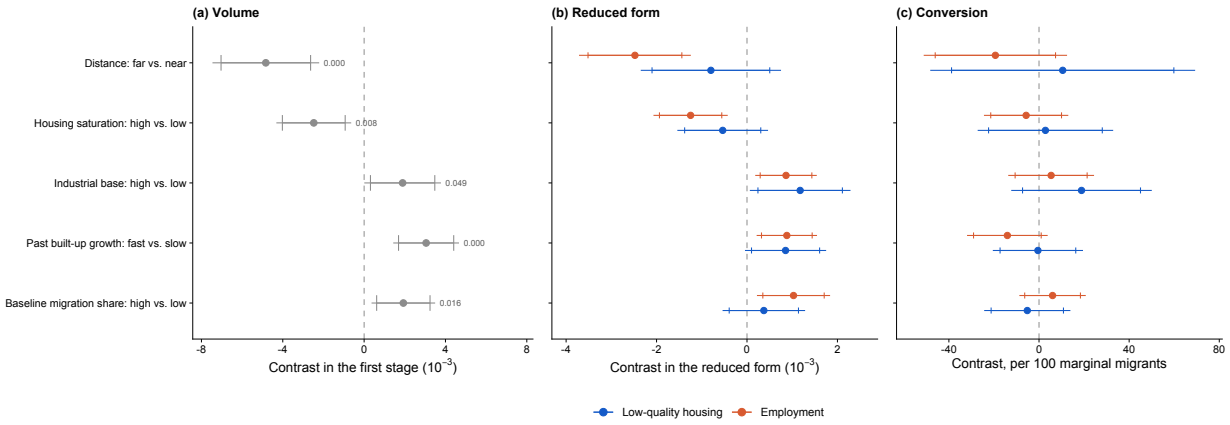
Figure 4: The Drought Response by Cropland Share of the Origin District



Notes: Cropland share is the harvested area of six staple crops (maize, cassava, wheat, sorghum, rice, soybean) over the origin district's 0.05° cells, divided by its land area, from CROPGRIDS v1.08 (Tang et al., 2024); crop absence is coded −1 in the source rasters and recoded to zero before the zonal sums. Available for 350 of 354 origins (99.5% of the panel) and split at the upper quartile (27.2%, dashed line): 262 origins and 180,334 corridor-years below, 88 and 60,006 above. The classifier is time-invariant, so origin–destination fixed effects stay nested within bins. Panel (a) is the first stage, panels (b)–(c) the two 2SLS conversions, from a saturated interacted specification (migration × bin instrumented by drought × bin, origin–destination and arrival-year × bin fixed effects, bin-specific distance-to-census control, SEs clustered at the origin district); whiskers are 95% Wald intervals. Both conditional first stages exceed ten ($F = 12.1$ and 17.9), which is why the quartile split is used rather than terciles. Contrasts: $p = 0.0023$ on the first stage, 0.987 on the housing conversion, 0.135 on the employment one.

6.1.3 Other Destination Dimensions

Figure 5: Destination Characteristics: the Volume of Climate Migration Varies, its Conversion into Low-Quality Housing Does Not



Notes: Each row is one destination characteristic, from a two-bin saturated interacted IV (instrument × bin, origin–destination and arrival-year × bin fixed effects, bin-specific distance-to-census control, SEs clustered at the origin district). Splits are at the within-country median, except city size, where the pre-specified contrast is metropolises (≥ 1M, Africapolis) against the other two zones pooled. Panel (a) reports the contrast in the first stage, panel (b) in the two reduced forms, panel (c) in the two 2SLS conversions, blue for low-quality housing and orange for the employed inflow. Each contrast spans its 95% Wald interval, with ticks at the bounds of the 90% interval: significant at 10% where the ticks fall on the same side of zero, at 5% where the whole line does. Of the twelve conversion contrasts, only one differs from zero – metropolises house drought-induced migrants far better than the rest (−20.6 per 100, $p = 0.032$) – while every other characteristic moves the volume by factors of 1.7 to 6 and leaves the conversion unchanged. Samples, top to bottom: $N = 240{,}340$; 157,556; 230,676; 215,556; 241,540; 241,540. Supplementary Appendix Table S6 reports the coefficients bin by bin.
Source: Authors' computations using IPUMS census, CHIRPS climate, Africapolis and GHSL data.

The city-size contrast of Section 6.1.2 is the paper's pre-specified primary hypothesis. This section asks, of a secondary family of destination characteristics, whether each changes the number of drought-induced migrants a city receives and what becomes of them. The answer is uniform: every characteristic moves the volume, none moves the conversion. Figure 5 plots the contrasts between bins, each characteristic being split at its within-country median. It shows where climate-induced migrants go: toward destinations that are close ($p = 0.0003$), had been building fast ($p = 0.0002$), were already migrant-intensive ($p = 0.016$) and have an industrial labour market ($p = 0.049$), but not toward those whose housing stock is already saturated ($p = 0.008$). Saturation here is the share of the destination's *resident* urban population — not of its migrants — living in low-quality housing at the last usable census, so the classifier is a property of the place, measured outside the flow this paper studies. It acts as a ceiling rather than a gradient: the first stage collapses to zero beyond roughly 75%, so destinations whose stock is already overwhelmingly informal stop receiving drought migrants altogether. What attracts is the composition of local employment — the industrial, formally employing part visible from a distance, not the overall employment rate ($p = 0.157$) — and proximity: drought migration is overwhelmingly short-distance, sixfold stronger below the within-country median distance than above. A Romano–Wolf correction over the six characteristics leaves two volume contrasts significant at 5% and one at 10%, and none of the twelve conversion contrasts (Appendix Table S3).

Past built-up expansion, built from the GHSL data described in Section 2.5, rules out the

natural supply-side reading: a supply constraint predicts the effect should concentrate where building was slow, yet the first stage is two and a half times larger where past expansion was fast ($p = 0.0002$). A binding constraint would deter arrivals, not attract them, so we read built-up growth as a proxy for urban dynamism rather than supply elasticity in the sense of Saiz (2010).

Across all five characteristics the 2SLS conversion of arrivals into low-quality housing is statistically indistinguishable (Appendix Figures S11–S16; $p = 0.853$, 0.236, 0.726, 0.962 and 0.594), as is employment. Housing saturation, industrial composition, distance, past built-up expansion and baseline migrant intensity thus determine how many drought migrants a city receives; city size determines what becomes of them. Where climate migration lands and how badly it is housed are separate questions, and only the second concerns absorption.

6.1.4 Where the Pressure Lands

Climate Opens New Corridors to Secondary Cities. The heterogeneity results establish one half of the answer. Secondary and small cities convert drought-induced migrants into low-quality housing at roughly three times the metropolitan rate (0.325 and 0.296 against 0.097), and city size is the only destination characteristic that moves the conversion at all: distance, housing saturation, industrial composition, past built-up expansion and baseline migrant intensity each shift the volume a city receives, none shifts what becomes of the arrivals. But a destination effect alone cannot explain why the burden falls on that tier, since it says nothing about how many migrants reach it. This section supplies the other half: the geography of the flows. Drought opens migration corridors that did not exist, and those new corridors lead to smaller, more deprived cities. The pressure concentrates there because climate sends people there and because those cities house them worse — two mechanisms that compound rather than compete.

Drought does not only intensify established flows; it opens new routes, which carry a small share of the inflow but a disproportionate share of its housing burden. For every corridor we measure a pre-period network: the urban migrants who travelled it in census years -13 to -15, outside the estimation window and therefore predetermined with respect to every drought realisation we exploit.[26] Of the 16,846 corridors, 4,812 carried migrants before the window and together move 95.6% of the urban flow; 5,855 carried none then and some during it, moving the remaining 4.4%. Panel A of Table 12 estimates the main specification separately on the two groups. On corridors with no prior network, drought raises the urban migration rate by 0.00174 ($p = 0.0006$, $F = 11.84$), against 0.00441 on established ones. In levels the established response is two and a half times larger, as one expects of corridors an order of magnitude bigger to begin with; relative to their own baseline the ranking reverses, a one-standard-deviation drought raising migration by 7.7% of the mean on corridors without a prior network against 3.6% on those with one. The extensive margin is thus the more climate-sensitive of the two, which is why it carries 15.2% of the low-quality-housing effect for 13.2% of the volume.

[26] Older recall is noisier, but since the variable only stratifies the sample, measurement error attenuates contrasts without biasing coefficients. Corridors with no prior network are kept in the estimation sample whether or not they later carry flow: restricting to those that do would condition on the outcome.

Table 12: Climate Opens New Corridors, and They Lead to Smaller, Worse-Housed Cities

Panel A. Extensive versus intensive margin (causal)

	No pre-period network (extensive)	Pre-period network (intensive)
First stage: urban migration rate	+0.00174***	+0.00441***
	(0.00051)	(0.00131)
First-stage F	11.84	11.36
Reduced form: low-quality housing	+0.00047*	+0.00102
	(0.00028)	(0.00066)
Reduced form: employed inflow	+0.00056***	+0.00245***
	(0.00018)	(0.00059)
Share of drought-induced migration	13.2%	86.8%
Share of the low-quality-housing effect	15.2%	84.8%
Share of the employment effect	8.1%	91.9%
Corridor-years	168,396	73,144
Corridors	12,034	4,812

Panel B. What the new corridors are (descriptive, flow-weighted)

	Existing	New	Difference	95% CI
Destination				
Metropolises (≥ 1M), %	65.6	8.9	-56.7***	[-60.9, -52.5]
Small towns (< 100k), %	11.0	55.4	+44.4***	[+40.8, +48.0]
Urban share of destination, %	69.9	25.0	-44.8***	[-48.0, -41.6]
Low-quality housing at destination, %	45.7	58.9	+13.3***	[+10.3, +16.3]
Built-up growth 1975–1990, %	69.6	101.5	+31.8***	[+26.1, +37.5]
Corridor and origin				
Distance, km	200	336	+136***	[+111, +161]
Urban share of origin, %	34.8	29.4	-5.5*	[-11.7, +0.8]
Mean drought intensity at origin	0.365	0.375	+0.010	[-0.007, +0.026]

Notes: A corridor has *no pre-period network* if no urban migrant travelled it in the three years preceding its 12-year recall window (census years −13 to −15), measured from the same IPUMS retrospective coding as the estimation panel but strictly outside it, for all eight countries. Panel A: main specification by group (origin–destination and arrival-year fixed effects, `dist_census` control, SEs clustered at the origin district). The causal object is the first stage on *all* corridors without a pre-period network; we do not estimate on *new* corridors alone, since being new is defined by a positive flow and conditioning on it would select on the outcome. Shares of the aggregate effect weight each group's coefficient by its realised drought exposure and destination population. Panel B compares the 5,855 corridors that opened during the window (no pre-period network, positive flow) with the 4,812 that already existed, excluding the 6,179 that never carried any flow; means are weighted by corridor flow and differences come from a regression on a new-corridor indicator with SEs clustered at the origin district. Panel B is descriptive: it characterises where the new corridors lead, not a causal effect of novelty on any outcome. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

New corridors reach smaller, poorer, more distant destinations than established ones, yet drought-induced migrants arriving along them are housed at the same rate as everyone else — the extra deprivation they face is the local rate, not a penalty for arriving by a new route. Panel B characterises the new corridors against the established ones, weighting by flow. They lead elsewhere: 8.9% of the new flow reaches metropolises of at least a million against 65.6% of the established flow, while small towns below 100,000 take 55.4% against 11.0%. Their destinations are markedly worse housed — 58.9% of the resident urban population in low-quality housing against 45.7%, a gap of 13.3 points — lie further away (336 km against 200), and differ in the formality of their labour markets rather than in employment as such: wage and salaried work accounts for 14.1% of the urban population against 23.4%, agriculture for 11.3% against 5.0%. They are not declining places, having built faster over 1975–1990 (101.5% against 69.6%). Panel C then asks whether a migrant arriving along one of these routes fares differently, estimating both conversions jointly on the same panel so that their difference is testable. On housing, nothing: 27.1 of every 100 drought-induced migrants enter low-quality housing on new corridors against 23.0 on established ones ($p = 0.766$), even though the stock at those destinations is 13.3 points more deprived. The extra deprivation migrants face there is the local rate, not a penalty attached to arriving by a new route. On employment there is a gap — 0.321 against 0.555 ($p = 0.026$) — consistent with destinations offering half as much wage work, though we read it as suggestive: it is a step rather than a gradient, and having no prior network is strongly correlated with being a small corridor.

6.2 Heterogeneity by Migrant: Gender, Age and Education

Appendix Table 30 profiles the drought-induced migrants themselves, comparing arrivals in high- versus no-drought years with the complier characterization of Abadie (2003). Both readings agree: drought-induced migrants are observationally similar to other migrants - the same gender balance and comparable or slightly higher education - but younger, with an over-representation of children under fifteen (42% versus 29%), consistent with whole-household displacement rather than the selective migration of young workers.

The effect itself is equally unselective on gender and age. Appendix Figures 17 and 18 re-estimate the main specification with subgroup-specific inflows and migration rates: drought displaces women and men, children and adults, proportionally alike (panel (a): relative first stages of +6 to +7% of each group's mean migration rate), and the housing and employment responses are correspondingly uniform across both dimensions.

Education is the exception, and it cuts in opposite directions on the two stages. Drought does not displace the least-educated adults: their migration response is indistinguishable from zero and significantly below that of the secondary-educated (−8.6 percentage points of relative response, $p = 0.001$) - a trapped-population pattern. Among the adults it does displace, the migration response tilts if anything toward the more educated (+8.8% against +5.6%, $p = 0.064$), yet the housing effect runs the other way - twice as strong for the primary-educated ($p = 0.036$, both groups well identified with conditional first stages above 14) - while employment absorption is education-blind ($p = 0.787$). Education thus eases displacement but shields from low-quality housing: the least educated stay behind, the secondary-educated escape bad housing, and the lower-educated among the displaced bear the residential burden. We verify that this selection operates along a single, additive dimen-

sion: a fully saturated model crossing gender, age at arrival, and education into twelve profile cells rejects equality of the first-stage responses (joint $p = 0.024$) but finds no interaction beyond the additive margins (joint $p = 0.59$) - drought selects migrants along a monotone educational gradient, from near-zero mobility among the least educated to the strongest response among the secondary-educated, without any distinct combined profile. The separate margins of Figure 17 are therefore a sufficient statistic for who drought moves (Supplementary Appendix Table S11). Attainment is measured at census and may post-date arrival, so the two displacement-side gradients remain suggestive. Full estimates are in Supplementary Appendix Tables S7–S10. Together with the profile evidence, the pattern points to barriers rather than composition: similar people, absorbed uniformly into work, sorted selectively into housing.

7 Robustness

This section presents a series of robustness checks and alternative specifications designed to assess the stability of the main results and the validity of the identification strategy. Several of these checks are reported in the appendix - the fixed-effects saturation in Table 4, the definition sweep, and the wet-side placebo of Figure 13 - and the remaining estimates are in progress.

Alternative specifications. - We consider four alternative specifications. First, we estimate a manually specified two-instrument design combining $\text{Drought}^g * p10$ and $N^g * dry$ to examine whether the results are robust to multi-instrument identification despite a weaker joint first stage. Second, we replace drought intensity with extreme precipitation measures based on high-percentile rainfall shocks to test whether migration responses are symmetric across the rainfall distribution or primarily driven by adverse drought events. Third, following Hultgren et al. (2025), we perform an out-of-sample validation to evaluate the stability of the post-LASSO instrument selection across held-out countries. Fourth, we replace the census-based destination population denominator with the annually reconstructed population series, $\text{Pop}_{d,t}$, to assess the sensitivity of the estimated migration rates to the choice of normalization.

Alternative samples. - To assess the external validity of the main findings, we extend the analysis to all fifteen countries in the dataset by exploiting five-year migration intervals rather than restricting attention to the eight countries with annual migration information. This exercise evaluates whether the estimated scale effect generalizes beyond the baseline sample.

Sensitivity analyses. - We conduct two additional sensitivity analyses. First, we replace the binary low-quality housing indicator with a continuous measure of housing deprivation defined as the number of deficient UN-Habitat dimensions (ranging from 0 to 5), allowing us to test whether drought-induced migration affects the entire distribution of housing quality rather than only the extensive margin of low-quality housing residence. Second, we perform

a leave-one-country-out exercise to verify that the estimated effects are not driven by any single country in the sample.

Leave-one-country-out. - Because the eight countries differ in size, census vintage and housing levels, we re-estimate the main specification eight times, each time dropping one country (Appendix Figure 20). The estimate stays positive in every subsample and ranges from 0.197 to 0.340 against 0.285 on the full sample, with each subsample estimate inside the full-sample confidence interval. Precision falls where the first stage weakens: dropping Benin or Kenya lowers the first-stage F to about 13 or 15, and in those two subsamples the Anderson–Rubin interval includes zero, a loss of power rather than a change in the estimate.

Level of clustering. - All estimates in the paper cluster standard errors at the origin district, which allows arbitrary correlation across the destinations and years a given origin feeds. Appendix Table S4 reports the four main results under three alternatives. Because the clustering level affects only the variance, **the coefficients are identical across columns**; what changes is precision. The two scale results survive at $p < 0.0001$ under every level. The housing propensity is the margin that moves: its p-value goes from 0.043 in the main specification to 0.058 when clustering at the origin $\times$ year cell and to 0.092 under two-way clustering, which is consistent with it being the paper's least robust result and reinforces the reading of Section 4.1.3 that the aggregate effect is a scale phenomenon. The employment propensity holds at the 10% level throughout. The demanding check is two-way clustering on origin and year, since it additionally permits arbitrary correlation across all origins within a year – precisely the spatial synchrony of shocks documented in Section 3.5, where the weighted pairwise correlation of drought across a destination's origins has a median of 0.67. All four results survive it. The first-stage F falls to 8.08 in that column, but this reflects the year dimension supplying only 38 clusters rather than any weakening of the instrument: the first-stage coefficient is unchanged, and the F is 29.65 and 24.66 under the two one-way levels on the same sample. We retain origin-level clustering as the main specification.

7.1 Exclusion Restriction

The instrument is drought at the origin of a corridor, and the exclusion restriction requires that it affect destination housing only through the migration it induces. The natural threat is geographic. Origins lie close to the destinations they feed, so origin drought may be correlated with shocks *at destination* – its own rainfall, conflict, agricultural prices – that could move destination housing and employment directly, bypassing migration altogether. This subsection tests that possibility on three channels, in each case by adding the destination shock as a control and by asking whether it moves the outcome on its own.

How much of the threat survives the fixed effects. - Less than one might fear. The raw correlation between origin drought and destination drought is 0.55 – not the 0.95 that the shift-share exposure index displays, because that index averages over a whole portfolio of nearby origins and is therefore far more collinear with local weather than any single origin is. Residualising both on corridor and arrival-year fixed effects brings the correlation down

to 0.35, so only about 12% of the instrument's variance is even potentially contaminated, and the instrument retains 81% of its raw dispersion. The design absorbs most of the problem before any control is added, because it compares years within a corridor rather than destinations with each other.

Controlling for destination shocks. - Table 13 reports the two scale results with and without each destination control. Because the three controls have different coverage, each forms a two-column group estimated on *its own* subsample, with the no-control column reported alongside: the comparison within a group is a pure control effect, and differences across groups are differences of coverage. This matters most for conflict, where ACLED's 1997 start restricts the panel to 70% of corridor-years, so the relevant benchmark is the 0.256 estimated on that subsample rather than the 0.285 of the full panel.

No control moves either result. The housing coefficient goes from 0.285 to 0.308 when destination drought is added, from 0.256 to 0.255 under conflict, and from 0.278 to 0.266 under crop prices; the employed inflow is unchanged in all three. The first-stage F is reported for both columns of each group and is essentially unaffected – 18.99 against 19.38 for conflict, 29.38 against 27.83 for prices – which is what one expects of controls measured at the destination: they do not touch the origin-to-migration relationship. Destination drought is the one that costs something, 29.65 to 23.73, for the spatial-correlation reason above. Entering all three jointly is a more demanding exercise and is reported separately in Supplementary Appendix Table S2; the housing result survives there too, but with the instrument visibly weakened.

Placebos. - The control test asks whether the destination shock changes the estimate; the placebo asks the prior question of whether it moves the outcome at all. Table 14 reports reduced forms in which no migration rate appears. Neither destination drought nor destination conflict has any detectable effect on the low-quality-housing inflow ($p = 0.50$ and $p = 0.98$), while origin drought retains its effect in the same regressions. There is simply no destination-weather or destination-conflict channel into housing to worry about.

The one placebo that bites is crop-price exposure, and it bites in the wrong direction to be a threat: the coefficient is *negative* and precisely estimated (-0.0104, $p < 0.001$). A favourable world price in a cropping destination *reduces* the inflow into low-quality housing, which is what one would expect if agricultural income retains people in the rural part of the unit rather than pushing them into its urban part. A channel of that sign cannot manufacture the positive housing effect we estimate; if anything it works against it.

The effect is specific to the arrival cohort. - A direct destination shock would affect everyone living there, not only that year's arrivals. It does not. Adding leads and lags of origin drought to the housing reduced form, the contemporaneous term is $+0.00109$ ($p = 0.025$) while drought three years *ahead* is -0.00004 ($p = 0.935$): a future shock at origin tells us nothing about the housing of the cohort arriving today. The small negative lag (-0.00062, $p = 0.096$) is consistent with modest intertemporal substitution – a past drought has already displaced some of the people who would otherwise have moved – rather than with a destination-wide shock.

What this leaves unresolved. - Two things, stated plainly. First, these tests cover the three destination-level confounders we can measure; they cannot speak to one we have not thought of. Second, the propensity margin is more fragile than the scale margin here as elsewhere: Supplementary Appendix Table S1 shows that within each control group it barely moves, but that the no-control estimates differ substantially *across* groups – from 1.216 on the full panel to 1.831 on the ACLED subsample – so its apparent sensitivity is largely a matter of coverage, on a sample where the first stage is in any case weaker. The scale results, which carry the paper's conclusions, are not threatened by destination-level confounding through any of the three channels we can test. A shift-share design on destination outcomes would be the natural way to go further; Section 3.5 explains in a footnote why the spatial synchrony of the shocks and the absence of an earlier census for five of our eight countries make it infeasible here.

Exclusion restriction. - The identifying assumption may be violated if drought at origin affects destination outcomes through channels other than migration, such as food price transmission, remittance flows, or government infrastructure responses. Origin–destination pair fixed effects absorb time-invariant correlations between origin characteristics and destination outcomes, while arrival-year fixed effects capture aggregate shocks common to all migration corridors. To further assess the plausibility of the exclusion restriction, we are investigating whether origin-specific drought shocks predict destination food prices after conditioning on the baseline specification. Evidence from this exercise will be reported in future versions of the paper.

Crowding-out. - The scale outcome counts every migrant of the arrival cohort living in low-quality housing, not only those whom drought displaced. A positive coefficient is therefore consistent with two mechanisms: drought-induced migrants settling in low-quality housing themselves, or drought-induced migrants crowding other arrivals out of decent housing into it. The aggregate margins cannot separate them, because on the sample where housing is observed the decent-housing response is the exact arithmetic complement of the low-quality one.

Two checks bound the concern. First, drought-induced migration does not degrade the housing of migrants who arrived on the same corridor in *earlier* years, who cannot be compliers of the current shock: the weak-instrument-robust interval on that outcome excludes any effect above 0.138 per marginal migrant, and a placebo on their total number is flat (Appendix Table 32). Crowding-out, if it operates, does not spill onto previously settled migrants. Second, the estimated effect (0.285) lies *below* the benchmark that no crowding-out and average sorting would imply — the baseline low-quality-housing share among migrants, 0.359 — whereas crowding-out would push it above.

Neither check identifies crowding-out *within* the arrival cohort, which is the channel that would bias the estimate. A direct test on the one migrant group whose own inflow does not respond to drought — adults with less than primary education, and therefore almost entirely always-takers — is inconclusive: their low-quality-housing propensity rises with drought-induced migration by 1.25, of the same order as the aggregate propensity effect, but the Anderson–Rubin interval, $[-0.21, +3.53]$, contains both zero and substantial crowding-

Table 13: Exclusion Restriction: Destination Shocks and the Scale Results

	Destination drought		Destination conflict		Destination crop prices	
	(1)	(2)	(3)	(4)	(5)	(6)
	No control	With control	No control	With control	No control	With control
Panel A. Low-quality housing inflow per 100 urban residents						
Urban migration rate	0.28455***	0.30748***	0.25556***	0.25529***	0.27763***	0.26575***
	(0.06323)	(0.07332)	(0.05460)	(0.05391)	(0.06453)	(0.06707)
First-stage F	29.65	23.73	18.99	19.38	29.38	27.83
AR 95% CI	[0.13, 0.40]	[0.12, 0.44]	[0.12, 0.36]	[0.12, 0.36]	[0.12, 0.39]	[0.10, 0.38]
Observations	241,540	241,540	169,920	169,920	240,340	240,340
Panel B. Employed inflow per 100 urban residents						
Urban migration rate	0.42760***	0.40278***	0.53594***	0.53583***	0.42959***	0.43142***
	(0.04330)	(0.04918)	(0.05234)	(0.05190)	(0.04440)	(0.04605)
First-stage F	29.65	23.73	18.99	19.38	29.38	27.83
AR 95% CI	[0.35, 0.53]	[0.32, 0.53]	[0.44, 0.68]	[0.44, 0.68]	[0.35, 0.54]	[0.35, 0.54]
Observations	241,540	241,540	169,920	169,920	240,340	240,340

Notes: 2SLS of the urban migration rate instrumented by contemporaneous drought intensity at origin, under the paper's main specification (origin–destination-pair and arrival-year fixed effects, distance-to-census control), with standard errors clustered at the origin district and cluster-robust Anderson–Rubin 95% intervals. **The instrument is origin drought in every column; every control is measured at the destination.** Each control forms a two-column group estimated on *its own* subsample, with the odd column reporting that subsample without the control: the comparison within a group is therefore a pure control effect, while differences across groups are differences of coverage. Destination drought is the instrument's construction applied to the destination unit (100% of the panel). Destination conflict is an indicator for any battle, act of violence against civilians or remote-violence event, plus log(1+fatalities), from ACLED matched to IPUMS units by point-in-polygon; ACLED begins in 1997 and reaches 70% of the panel. Crop prices are $\sum_c \text{share}_{cd} \log(P_{ct})$ and its first difference, with CROPGRIDS harvested-area shares for nine traded crops and World Bank Pink Sheet world prices (99.5% of the panel). **No control moves either scale result.** Housing goes from 0.285 to 0.307 under destination drought, from 0.256 to 0.255 under conflict and from 0.278 to 0.266 under crop prices; employment is unchanged throughout. The first-stage F is reported for both columns of each group and is essentially unaffected by the controls – 18.99 against 19.38 for conflict, 29.38 against 27.83 for prices – which is the point: these are destination characteristics and they do not touch the origin-to-migration first stage. Destination drought is the one that costs something (29.65 to 23.73), because rainfall is spatially correlated between an origin and its destinations. Supplementary Appendix Table S2 reports the three controls entered jointly. Stars: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 14: Placebo: Do Destination Shocks Move Destination Outcomes Directly?

	(1) Low-quality housing inflow	(2) Employed inflow	(3) Share in low-quality housing
Panel A. One shock at a time, each on its own maximal coverage			
Destination drought	+0.00020	+0.00084***	+0.00428
	(0.00029)	(0.00022)	(0.00366)
Destination conflict (any violent)	+0.00001	+0.00060*	+0.00597
	(0.00022)	(0.00032)	(0.00487)
Destination log(1 + fatalities)	+0.00019	+0.00044**	+0.00078
	(0.00015)	(0.00020)	(0.00225)
Destination crop-price exposure	-0.01038***	-0.00772***	-0.01534
	(0.00195)	(0.00160)	(0.03940)
Destination crop-price shock	-0.00079	+0.00331	+0.05995
	(0.00278)	(0.00295)	(0.06026)
Observations	241,540	241,540	72,723
Panel B. All shocks jointly, common subsample			
Destination drought	+0.00099***	+0.00131***	-0.00179
	(0.00028)	(0.00027)	(0.00361)
Destination conflict (any violent)	-0.00011	+0.00045	+0.00680
	(0.00022)	(0.00032)	(0.00491)
Destination log(1 + fatalities)	+0.00023	+0.00048**	+0.00061
	(0.00015)	(0.00020)	(0.00226)
Destination crop-price exposure	-0.00776***	-0.00744***	-0.02104
	(0.00189)	(0.00159)	(0.03859)
Destination crop-price shock	-0.00241	+0.00430	+0.02952
	(0.00280)	(0.00322)	(0.06308)
Origin drought (the instrument)	+0.00041	+0.00137***	+0.01041**
	(0.00028)	(0.00033)	(0.00415)
Observations	168,820	168,820	52,593

Notes: Reduced-form regressions: each destination outcome on destination shocks and origin drought jointly, with origin–destination-pair and arrival-year fixed effects, the distance-to-census control, and standard errors clustered at the origin district. No migration rate enters these regressions, so they ask whether a destination shock moves destination outcomes through *any* channel. Panel A estimates each shock on its own maximal coverage (destination drought 100% of the panel, conflict 70%, crop prices 99.5%); Panel B puts all of them in one regression on the subsample where all are observed. Column (1) is the outcome the paper's main result is built on. **Neither destination drought nor destination conflict has a direct effect on the low-quality-housing inflow**, while origin drought retains its effect in every specification. Crop-price exposure does enter column (1), but with the *opposite* sign to the paper's effect: a positive world price shock in a cropping destination *reduces* the inflow, consistent with farm income retaining people rather than pushing them into the city, so it cannot generate the estimated effect through a direct channel. Stars: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

out (Appendix Table 33). We therefore read the coefficient as a total effect on the corridor — each hundred additional drought-induced migrants is associated with twenty-eight additional migrants living in low-quality housing — rather than as a claim about which individuals occupy which dwellings.

Comparison with other migration drivers. - Two additional exercises will compare drought-induced migration with migration generated by alternative shocks. The first instruments migration using conflict events at origin (ACLED data); the second uses exogenous variation in mine openings at the admin-unit level. These comparisons sharpen the interpretation of the LATE estimated in the main analysis by contrasting push driven (drought, conflict) and pull-driven (mining) migration mechanisms and their respective urban absorption patterns.

8 A Conceptual Framework: Two Margins of Urban Integration

8.1 Purpose and Scope

The preceding sections establish three facts. Drought at origin raises rural-to-urban migration. The additional migrants raise the inflow into low-quality housing, overwhelmingly through the number of arrivals rather than through who they are. And the same migrants are absorbed into urban employment wherever they land. The framework below organises these facts into a single statement about where the friction lies. It is an interpretive device: it is not estimated, no parameter is calibrated, and no welfare number is derived from it. Its role is to make precise what the reduced-form estimates do and do not identify, and to name the object a policy would have to move.

We are explicit about why it stops there. The one thing a structural apparatus would buy over the reduced form is a welfare metric, and that requires separating an access friction from a price – a deposit that excludes from a rent that reflects genuine scarcity. Census microdata contain neither prices nor incomes, so the separation is unavailable here. A structural extension on a single country where housing prices exist is the natural home for that question, and we flag it as future work rather than attempt it with data that cannot support it.

8.2 Two Margins and Roy Sorting

Consider migrants who have already arrived in a destination city; the decision to move is taken as given, since that is what the instrument identifies. Each migrant integrates along two margins, employment (E) and housing (R), and on each chooses between a high-integration state H – formal or wage work; adequate housing – and a low-integration state L – informal or agricultural work; low-quality housing. On each margin the migrant draws a pair of state-specific capacities (s_{iL}, s_{iH}) indexing how well she fares in each state, and the draws are margin-specific: (s^E_{iL}, s^E_{iH}) and (s^R_{iL}, s^R_{iH}). Formal housing is valued for the bundle of services it delivers rather than for a return to labour, which keeps the two margins separable and consistent with the decoupling the data show. Writing w_H and w_L for the

value per unit of capacity in each state – on the housing margin, inclusive of that bundle – a migrant absent frictions chooses the high state whenever $w_H s_{iH} > w_L s_{iL}$, that is, above the ray OC of slope w_L/w_H (Appendix Figure 21).

A proportional barrier τ on access to the high state – deposits, guarantors, documentation, the absence of local networks – reduces its effective value to $(1-\tau)w_H$ and rotates the cutoff to the steeper ray OE of slope $w_L/[(1-\tau)w_H]$. The two rays partition the space into four regions. Region I, above OE, holds migrants whose comparative advantage is strong enough that they take the high state despite the barrier. Region IV, below OC, holds those who would choose the low state in any case. Between them lies the object of interest: Regions II and III, the migrants who would take the high state absent the barrier but do not take it in its presence, whether their comparative advantage lies in the high state (II, above the 45° line) or in the low one (III). This trapped fan is the misallocation a barrier creates, and its width is what a policy that lowered τ would close.

8.3 Where the Barrier Lies: Three Candidates, One Survivor

The framework is empty until the data say which dimension indexes τ. Three candidates present themselves, and the paper's estimates adjudicate among them. Two are rejected.

Climate status: tested and rejected. - The natural first candidate – and the one an earlier version of this framework imposed – is that drought-induced migrants face a barrier other arrivals do not. This is testable, and the test is the paper's propensity estimate, not an auxiliary exercise. Instrumenting with origin drought compares the migrants the shock adds at the margin with the infra-marginal migrants arriving on the same corridor in the same year; since drought-induced migrants are not selected on education (Section 6.2), the comparison holds skills approximately fixed, which is precisely the condition under which a difference in outcomes can be read as a difference in barriers. The estimate is small and fragile: a one-standard-deviation drought raises the share of arrivals in low-quality housing by 0.37 percentage points (Table 6), and the cleanest version of the complier comparison cannot distinguish the marginal migrant from the average one. We therefore do not index τ by climate status. Climate enters the framework through the mass of arrivals M_c, which the first stage identifies, and not through the sorting problem at all.

Education: tested and eliminated. - The second candidate is that the barrier falls on the less-educated. It does not survive either, and the reason is instructive. Descriptively, education moves both margins by the same amount: between the lowest and highest education groups the log-odds gap is +0.87 on housing and +0.87 on employment (Appendix Table 16). Causally, the same symmetry holds. Estimating the conversion of drought-induced migration separately by education, the housing conversion falls from 0.39 among the primary-educated to 0.22 among those with secondary schooling or more, a gap of +0.18 ($p = 0.092$); the employment conversion rises from 0.79 to 0.91 over the same groups, a gap of +0.12 in the same direction of disadvantage ($p = 0.175$).[27] The two gaps are of the same sign and compa-

[27]The least-educated group cannot be read: its conditional first stage is $F = 1.95$, and its Anderson–Rubin interval on the housing conversion, $[-1.80, +3.38]$, contains zero. Point estimates below the paper's $F \geq 8$

rable size, so their difference – which is what an education-specific *residential* barrier would require – is nowhere near distinguishable from zero. Education degrades both margins together, which is the signature of a human-capital effect operating through the capacity draws rather than of a friction operating on access to one state. We therefore treat education as a dimension the data *eliminate*, not one we omit.

The destination city: the surviving asymmetry. - The third candidate survives, and it is the only one that does. Classifying destinations by agglomeration size, the housing conversion of drought-induced migration is 0.30 in small towns, 0.33 in secondary cities and 0.10 in metropolises; the contrast between metropolises and the rest is -0.21 ($p = 0.023$), and all three conditional first stages exceed ten. The employment conversion over the same three groups is 0.39, 0.53 and 0.43, and the corresponding contrast is -0.04 with a standard error of 0.06 – a 95% interval of $[-0.16, +0.08]$ that excludes a gap of the size observed on housing. The asymmetry the framework needs is therefore present, and present only here: the residential margin varies threefold across the urban hierarchy while the occupational margin does not vary at all. In the language of the diagram, τ^R is indexed by the destination city and τ^E is not indexed by anything – panel (a) of Appendix Figure 21 has no trapped region, panel (b) has one, and its width is a property of the city a migrant reaches rather than of the migrant.

8.4 The Working Poor as a Decoupling

The term *working poor* denotes a configuration, not a level of disadvantage: integrated on employment and trapped on housing. The distinction matters for reading the results above. Secondary cities display exactly this configuration – of every hundred drought-induced migrants they receive, 53 arrive into work and 33 into low-quality housing, the highest value on both margins in the urban hierarchy – while metropolises do not, converting 43 into work and only 10 into low-quality housing. The less-educated, by contrast, are not working poor in this sense: they are worse on both margins at once (0.39 into low-quality housing against 0.22, but also 0.79 into work against 0.91), which is a uniform shortfall rather than a decoupling. Table 11 confirms the configuration on the same individuals: 63% of the drought-induced migrants entering low-quality housing arrive employed, against 44% of the baseline inflow. The decoupling the framework describes is therefore geographic. It happens in the secondary tier, and it happens there to everyone who arrives.

8.5 What the Framework Implies

Because the barrier is a property of the destination, the arithmetic of the aggregate burden follows from two objects the instrument already delivers: the induced volume M_c by city type, from the first stage, and the conversion S_c, from the two-stage estimates. Metropolises receive 59% of the drought-induced migration volume but carry only 31% of the induced low-quality-housing inflow; the secondary and small tier receives 41% of the volume and carries 69% of the burden, and per urban resident the induced burden there is 2.0 times

gate are reported for completeness only.

the metropolitan one – a ranking that would reverse were the conversion rate uniform. The counterfactual follows directly and requires nothing beyond these estimates: holding migration flows unchanged and bringing the conversion rate of small and secondary cities down to the metropolitan rate of 0.10 would avert 47% of the induced low-quality-housing inflow, and 27% if the policy reached secondary cities alone.

The policy reading is the paper's title in another form. The lever is the destination, not the migrant and not the flow: a climate shock determines how many people arrive, and the city they reach determines what becomes of them. Since the barrier is indexed neither by climate status nor by education, a policy need not identify climate migrants to reach them – improving residential absorption in the secondary tier reaches them by reaching everyone who arrives there, which is also what makes it administratively feasible.

8.6 Limits

Two limits bound what the framework can claim, and we state both rather than leave them to the reader.

The first is identification of the barrier itself. A rotation of the cutoff is one explanation of a lower formal-housing share; two others produce it without any friction. Migrants may sort on prices or preferences, choosing low-quality housing to economise on rent, and a city where formal housing is genuinely scarcer will show a lower share for that reason alone. Alternatively the composition of capacity draws may differ across cities, shifting the distribution across a fixed cutoff rather than rotating the cutoff itself. Observational data cannot separate these: a proportional barrier rotates the cutoff while a fixed cost such as a deposit shifts it, so the two are distinguishable along the income distribution, but only with price and income data. We therefore use τ^R as a name for the residential wedge the estimates reveal, not as a claim that the wedge is a deadweight friction, and we do not attach a welfare number to closing it.

The second is statistical. The city-size contrast on the housing conversion is significant at conventional levels on its own ($p = 0.023$) but does not survive the Romano–Wolf correction applied to the paper's exploratory family of heterogeneity tests ($p_{RW} = 0.282$). We defend it as a pre-specified primary hypothesis rather than as a survivor of multiple-testing correction, and the framework rests on it to that extent and no further. The employment null, by contrast, is tight rather than merely imprecise, and it is the conjunction of the two – a residential gradient alongside an occupational flat – that the framework formalises.

9 Conclusion

This paper provides causal evidence on the contribution of drought-induced migration to the growth of low-quality housing in Sub-Saharan Africa. Using harmonized census microdata for eight countries, censuses 1990–2019 with migration windows covering 1981–2018, it combines a cohort-bilateral panel of migration flows with high-resolution rainfall data to construct exogenous variation in drought intensity at origin. A post-IV LASSO procedure selects drought intensity as the instrument for bilateral migration, and the 2SLS estimates identify

a Local Average Treatment Effect on drought-induced migrants, precisely the population climate policy concerns.

Drought-induced migration raises urban low-quality housing inflows at destination: of every hundred migrants that drought pushes into a city, about twenty-eight settle in low-quality housing, and over 1981–2018 drought placed roughly 110,000 additional people in low-quality housing across our corridors. The effect operates mainly through the scale of arrivals, about four fifths of it, and only secondarily through their composition: a one-standard-deviation drought raises the share of arrivals in low-quality housing from 40.6 to 41.0 percent. Yet these migrants are absorbed into urban employment across destinations, and 85% of those who end up in low-quality housing are employed against 61% of those who do not: the margin of exclusion is residential, not occupational. The housing effect originates in farmland rather than in rurality as such — the first stage is some seven times stronger out of the most cultivated third of origin districts, while within the rural world the response is flat — which is the agricultural-push mechanism. The burden of climage-induced migrants being working poors concentrates in secondary and small, rapidly urbanizing cities. third of that level in metropolises above 1 million, a gap significant at the 5% level.

These findings have direct implications for urban and climate policy in Sub-Saharan Africa. Because the effect runs mainly through the scale of arrivals rather than through their composition, the priority is expanding urban residential absorption. That the effect concentrates in secondary and small urbanizing cities, and fades in the largest metropolises, suggests that housing policies focused on primate metropolitan areas may overlook precisely the destinations where climate-induced migration weighs most heavily on housing. If low-quality housing itself exposes residents to climate hazards (Marx et al., 2019), climate-induced growth in low-quality housing may amplify rather than mitigate the costs of climate variability. A structural model of migrant integration, jointly modeling housing and labor-market choices, would quantify the welfare cost borne by these migrants and allow the evaluation of housing support in the secondary tier.

Because the residential barrier is indexed by the destination rather than by the migrant, a policy need not identify climate migrants to reach them: improving residential absorption in the fast-growing secondary cities reaches them by reaching everyone who arrives there.

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10 Appendix

10.1 Descriptive Statistics

Table 15 reports descriptive statistics for all variables used in the analysis: the corridor-year estimation panel (Panel A), the migrant propensity outcomes (Panel B), and destination characteristics (Panel C).

Table 15: Descriptive Statistics

Variable	Mean	SD	P25	Median	P75	N
Panel A: Corridor-year panel (main estimation sample)						
Urban migration rate (per 100 urban residents)	0.0268	0.1003	0.0000	0.0000	0.0099	241,540
Low-quality housing inflow rate (per 100)	0.0120	0.0589	0.0000	0.0000	0.0000	241,540
Employed inflow rate (per 100)	0.0106	0.0405	0.0000	0.0000	0.0000	241,540
Unemployed inflow rate (per 100)	0.0013	0.0095	0.0000	0.0000	0.0000	241,540
Inactive inflow rate (per 100)	0.0118	0.0522	0.0000	0.0000	0.0000	241,540
Drought intensity at origin	0.350	0.502	0.000	0.000	0.625	241,540
Cumulative 3-season drought at origin	0.366	0.297	0.106	0.347	0.551	235,920
Distance to census $\lvert t - t_c \rvert$ (years)	6.08	3.29	3.00	6.00	9.00	241,540
Urban population at destination ('000)	123.8	315.6	16.0	41.4	95.9	241,540
Panel B: Migrant propensity outcomes (corridors with positive denominators)						
Share of migrants in low-quality housing	0.406	0.388	0.000	0.333	0.750	72,723
Mean number of deprived dimensions (0–5)	0.662	0.818	0.000	0.429	1.000	72,723
Share of migrants employed	0.505	0.362	0.200	0.500	0.842	71,503
Share of migrants unemployed	0.058	0.155	0.000	0.000	0.036	71,503
Share of migrants inactive	0.437	0.358	0.000	0.429	0.667	71,503
Panel C: Destination characteristics (one row per destination, most recent census)						
Urban population share of the destination	0.256	0.241	0.085	0.188	0.351	351

Notes: Descriptive statistics for the variables used in the analysis. Panel A: the main corridor-year estimation panel (8 countries, Egypt excluded; Mali 1998 included; 12-year capped windows with the census year excluded). The panel is the full origin–destination rectangle and retains pairs that never record an urban migrant: 37% of the 16,846 origin–destination pairs register no urban migrant at all over the window, and 70% of corridor-years record zero urban migrant arrivals, which accounts for the zero lower quartiles of the rate variables. Zero-flow corridor-years are kept because dropping them would condition the sample on the endogenous regressor. Rates are counts per 100 urban residents at destination; no outcome is winsorized, and the sample is restricted to destinations with at least 2,000 urban residents (the conventional minimum size of an urban locality). Drought intensity is the standardized negative seasonal rainfall anomaly at origin, $\max(0, -\tilde{z})$, equal to zero by construction in wetter-than-average seasons; the cumulative version averages the current and two preceding seasons (defined where lags exist). The inactive inflow counts migrants out of the labor force with determined status. Panel B: shares among migrants (housing: denominator = migrants; employment: denominator = migrants with known labor-force status); defined only for corridor-years with a positive denominator. Panel C: census urban population share, one observation per destination at its most recent census.

Table 16 breaks the two outcome shares down by migrant education and by destination city size. It is purely descriptive, and its role is to give the baseline against which the causal estimates are read. Three features of that baseline matter. First, the housing gradient in education is steep: 49.5% of urban in-migrants with less than primary schooling live in low-quality housing, against 35.0% of the primary-educated and 29.1% of those with secondary schooling or more. Second, the same gradient is present on the employment margin and is of the same magnitude – in log-odds, +0.87 on housing and +0.87 on employment between the top and bottom education groups – so education is not a dimension along which the two margins separate. Third, the low-quality-housing share falls with city size, from 44.1% in small towns to 37.1% in secondary cities and 33.8% in metropolises, while the employment share is nearly flat across the three (49.2, 48.6 and 53.1%). It is the destination, not the education of the migrant, that separates the residential from the occupational margin, and it does so in the raw data before any instrument is applied.

We stress what this table does not establish. A difference in shares across education levels mixes any access friction with the composition of skills, since education shifts both, and no observational comparison can separate them. The paper's causal statements rest instead on the instrumented estimates, which compare drought-induced migrants with the infra-marginal migrants arriving on the same corridor – a comparison that holds skills fixed because, as Section 6.2 shows, drought-induced migrants are not selected on education. The purpose of the table is to show that the gradients above are features of the migrant population as a whole, so that the finding that drought-induced migrants settle at their destination's local rate means the climate shock adds arrivals to this landscape rather than altering it.

Table 17 documents the panel census by census. For each of the eight countries whose censuses record a year of arrival, it reports the range of arrival years that census can describe, the number of origins, destinations and corridors it contributes, and the urban in-migrants it observes. The eleven census samples together yield 241,540 corridor-years, 16,846 corridors and 11.6 million urban in-migrants, of whom 36% live in low-quality housing and 51% are employed. Both figures are shares of the urban in-migrants themselves, not of the destination's resident population, and they vary widely across census samples – from 15% to 55% on housing and from 31% to 65% on employment – which is why the estimating equations absorb destination and arrival-year effects throughout. Coverage is uneven by construction: Benin 2013 alone supplies 28% of the corridor-years, since its 77 origins and 75 destinations generate 5,700 corridors over a full twelve-year window, while Malawi 2008 supplies 3%. Arrival windows are shorter than twelve years in three cases because a preceding census in the same country already describes the earlier years – Kenya 2019, Mali 2009 and Zambia 2000 – and in one further case, Zambia 1990, because the CHIRPS rainfall record begins in 1981 and earlier arrival years cannot be matched to a shock.

10.1.1 Descriptive Maps

10.1.2 Housing-Quality Definition

Figure 11 shows how the index behaves in the data. For each country it reports the share of households counted deprived on each dimension, separately for all urban residents and for the urban in-migrants on whom the paper's outcome is built. Three features are worth noting.

Table 16: Housing and Employment of Urban In-Migrants, by Education and Destination City Size

	Share in low-quality housing (%)	Share employed (%)	Migrants
Panel A. By education (migrants aged 25 and over)			
Less than primary	49.5	60.3	900,280
Primary completed	35.0	69.2	2,380,194
Secondary or more	29.1	78.4	2,647,353
All aged 25 and over	34.6	72.0	5,927,827
Panel B. By destination city size (all migrants)			
Small town (<100k)	44.1	49.2	1,500,424
Secondary city (100k–1M)	37.1	48.6	2,754,880
Metropolis (≥1M)	33.8	53.1	7,285,082
Panel C. Education × destination city size (aged 25 and over)			
Small town (<100k)			
Less than primary	62.9	62.1	158,170
Primary completed	42.7	72.9	268,554
Secondary or more	31.1	81.0	289,399
Secondary city (100k–1M)			
Less than primary	53.3	60.2	237,723
Primary completed	37.2	70.4	520,449
Secondary or more	29.0	79.9	535,977
Metropolis (≥1M)			
Less than primary	43.4	59.8	500,238
Primary completed	32.9	68.2	1,581,991
Secondary or more	28.9	77.6	1,807,966
All urban in-migrants	35.9	51.5	11,603,276

Notes: Observed shares among urban in-migrants in the estimation panel, pooled over corridor-years and weighted by migrant counts, so each cell is an individual-level share rather than an average of corridor shares. Low-quality housing follows the paper's main definition, with migrants whose housing is observed as the denominator; employment is the share employed among those whose labour-force status is observed, so the two columns differ slightly in denominator and the 'Migrants' column reports the housing one. Education is observed only from age 25, so Panels A and C cover that group (5,927,827 migrants, 51% of the panel) while Panel B and the final row cover all migrants: this age restriction, not the education split, is why employment is higher in Panels A and C, and the subtotal row makes the two comparable. City size is the Africapolis class used throughout, available for 99% of the panel.

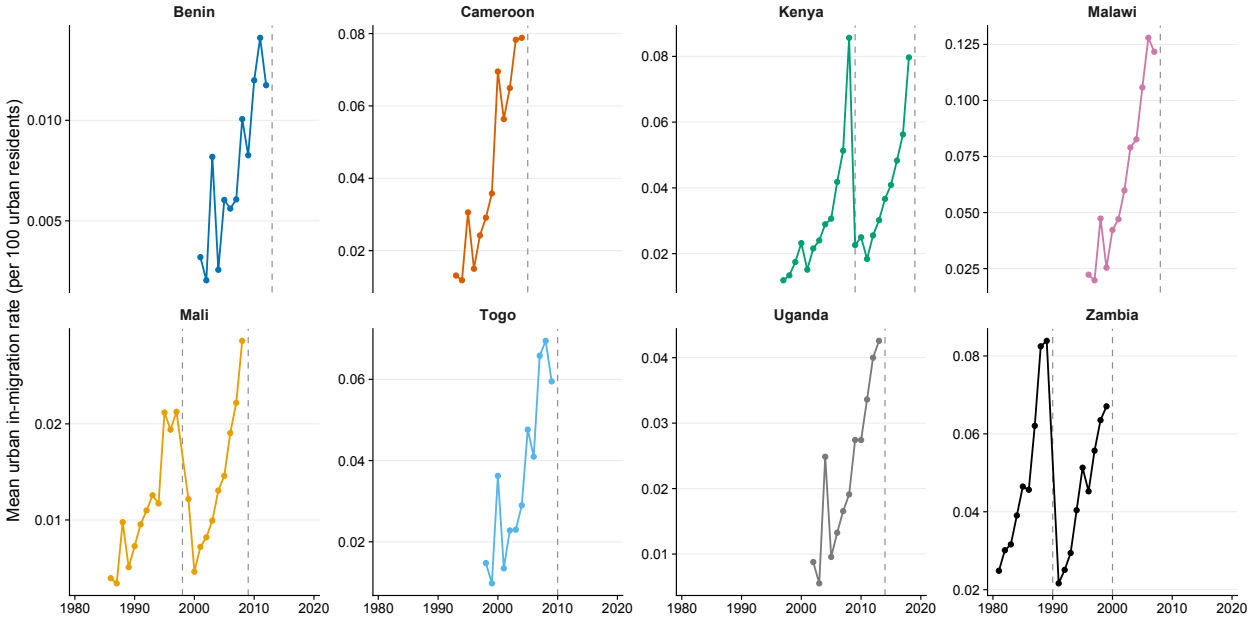
The table is descriptive and carries no causal content. The education gradient in housing in particular should not be read as a barrier facing the less-educated: a difference in shares across education levels mixes any access friction with the composition of skills, since education shifts both, and observational data cannot separate them. The paper's causal statements rest on the instrumented estimates (Tables 5 and 6), which compare drought-induced migrants with the infra-marginal arrivals on the same corridor and so hold skills fixed. What this table supplies is the baseline those estimates are read against: the gradients below hold for the migrant population as a whole, so drought-induced migrants settling at their destination's local rate means the shock adds arrivals to this landscape rather than altering it.

Table 17: Coverage of the Estimation Panel, by Country and Census

Census	Arrival window	Years	Origins	Destin-ations	Corridors	Corridor-years	Urban in-migrants	Share of urban in-migrants (%)	
								In low-quality housing	Employed
Benin									
2013	2001–2012	12	77	75	5,700	68,400	470,870	38	46
Cameroon									
2005	1993–2004	12	39	39	1,482	17,784	1,914,190	19	31
Kenya									
2009	1997–2008	12	38	38	1,406	16,872	2,595,370	33	64
2019	2009–2018	10	38	38	1,406	14,060	2,954,250	49	55
Malawi									
2008	1996–2007	12	26	26	650	7,800	385,220	54	46
Mali									
1998	1986–1997	12	47	47	2,162	25,714	190,090	55	43
2009	1999–2008	10	47	25	1,150	11,500	310,060	44	48
Togo									
2010	1998–2009	12	36	35	1,225	14,700	447,140	36	54
Uganda									
2014	2002–2013	12	36	36	1,260	15,120	1,140,030	15	65
Zambia									
1990	1981–1989	9	53	54	2,810	25,290	552,210	46	40
2000	1991–1999	9	55	50	2,700	24,300	641,590	39	38
All	1981–2018	38	354	351	16,846	241,540	11,601,020	36	51

Notes: The eight countries whose censuses record a year of arrival, so that bilateral yearly flows can be constructed; each row is one census sample, and censuses listed in Section 2 but absent here lack that question. Origins and destinations are IPUMS units, GEOLEVEL2 except in Malawi and Uganda, which enter at GEOLEVEL1. The arrival window is twelve years, excludes the census year, and is truncated where a preceding census in the same country already describes the earlier years (Kenya 2019, Mali 2009, Zambia 2000) or where it would reach before 1981, when the CHIRPS record begins (Zambia 1990). The last two columns are shares of the urban in-migrants themselves, not of the destination's resident population: the share in low-quality housing among those whose housing is observed, and the share employed among those whose labour-force status is observed, so the two denominators differ slightly from each other and from the 'Urban in-migrants' column, which reports the housing one. The panel holds 241,540 corridor-years of a constructed table of 300,431. One sample is constructed but cannot enter – Zambia 2010, whose harmonized urban classification is missing on every record – a loss of measurement rather than of design, its window not overlapping the preceding census.

Figure 6: Average Urban In-Migration Rates Over Time, by Country



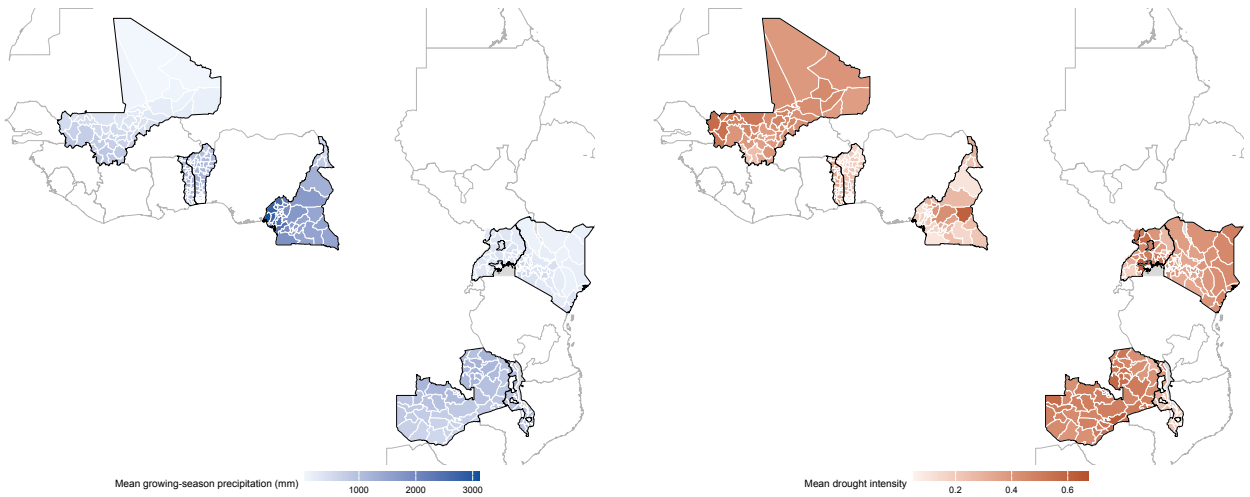
Notes: Mean urban in-migration rate (in-migrant arrivals per 100 urban residents at destination) across corridor-year observations of the yearly estimation panel, by calendar year and country. Dashed vertical lines mark census years; each census contributes a capped retrospective window of yearly migration cohorts. Within each window, rates rise mechanically toward the census year, reflecting recall and onward-migration attrition of earlier cohorts; the distance-to-census control absorbs this gradient in estimation. Y-axis scales differ across panels.

Source: Authors' computations from IPUMS census microdata.

Figure 7: Descriptive Maps: Climate at Origin

(a) Mean growing-season precipitation

(b) Mean drought intensity



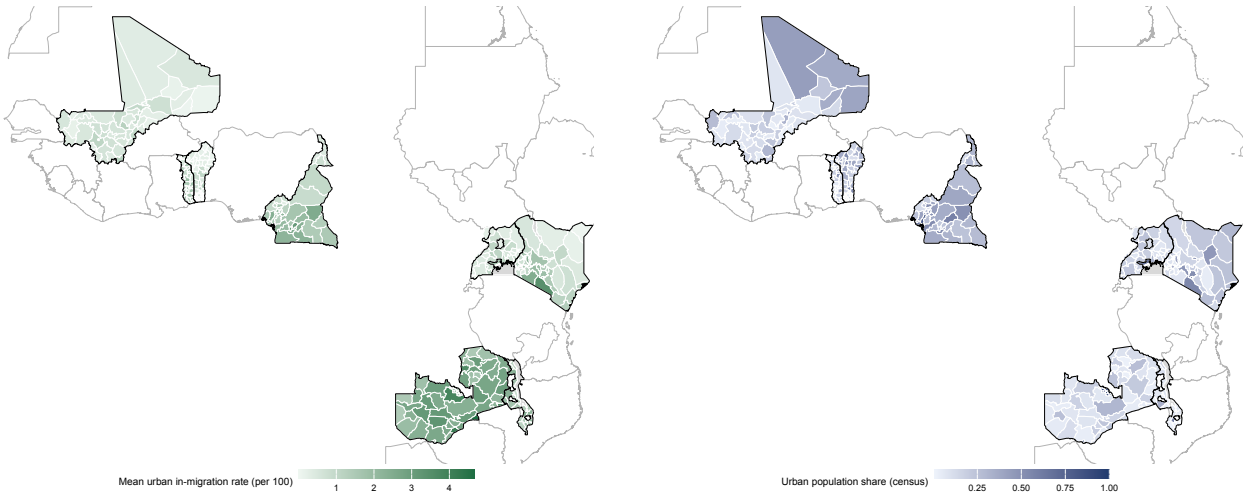
Notes: The eight countries of the estimation sample are outlined in black, with their administrative units (second level; first level for Malawi and Uganda); other African countries covered by IPUMS are outlined in grey. Panel (a): mean cumulative precipitation over the growing season (mm), by origin unit – the Precip^{g}_{cum} measure underlying the instrument. Panel (b): mean of the standardized drought intensity $\max(0, -\tilde{z})$ over the panel years, by origin unit.

Source: Authors' computations from CHIRPS data and IPUMS harmonized shapefiles.

Figure 8: Descriptive Maps: Migration and Urbanization at Destination

(a) Mean urban in-migration rate

(b) Urban population share



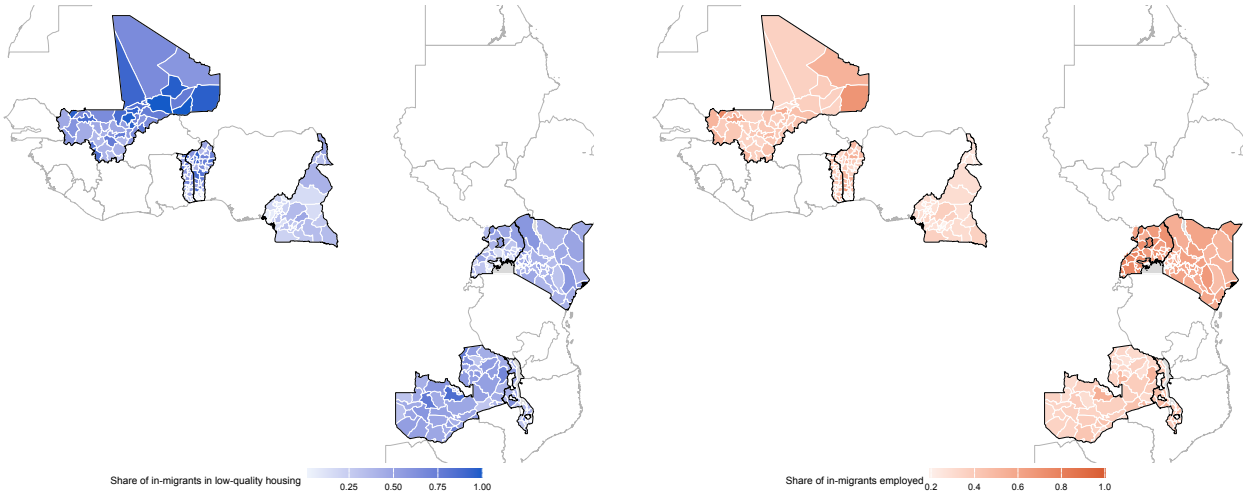
Notes: Panel (a): mean yearly urban in-migrant arrivals per 100 urban residents, by destination unit, aggregated over all origins. Panel (b): urban population share of the destination (urban over total census population, most recent census). Grey units: not observed in the estimation panel.

Source: Authors' computations from IPUMS census microdata.

Figure 9: Descriptive Maps: Migrant Outcomes at Destination

(a) Share of in-migrants in low-quality housing

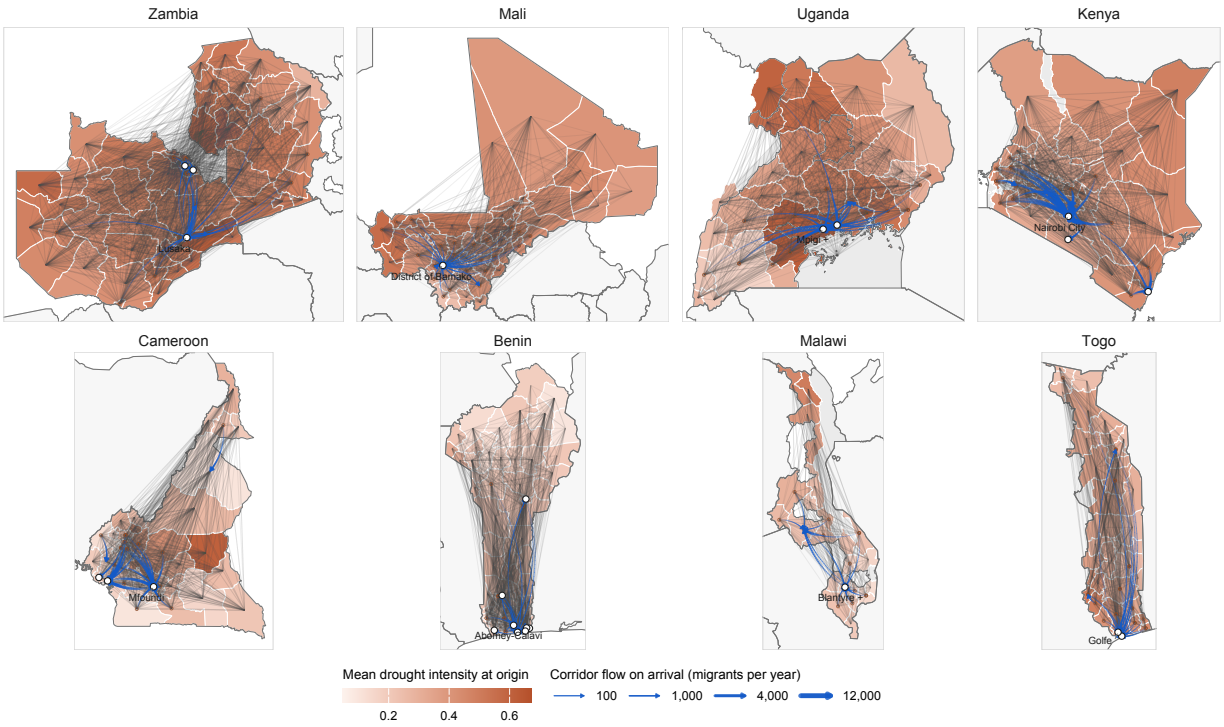
(b) Share of in-migrants employed



Notes: The propensity outcomes of the paper, aggregated by destination over the panel: among urban in-migrants, the share residing in low-quality housing (panel (a), main UN-Habitat definition; denominator: migrants) and the share employed (panel (b); denominator: migrants with known labor-force status). Grey units: no in-migrants observed.

Source: Authors' computations from IPUMS census microdata.

Figure 10: Observed Urban In-Migration Corridors, by Country



Notes: Observed bilateral migration corridors, one panel per country (241,540 corridor-years). Units are shaded by mean drought intensity at origin, on the scale of Figure 7(b); grey units are never observed as an origin. The faint web shows all 10,399 corridors with a positive urban flow; highlighted arcs are the 189 largest (top 25 per country), 44% of urban in-migration. Arcs run from origin (open dot) to destination (arrowhead), with width proportional to the square root of the mean ANNUAL flow – totals are not comparable across countries, since Kenya, Mali and Zambia contribute two censuses. White circles mark main-city destinations (census urban share ≥ 0.55). Flows count urban in-migrants, the numerator of the endogenous variable. Centroids are interior points of administrative polygons, not city locations.

Source: Authors' computations from IPUMS census microdata and IPUMS harmonized shapefiles.

Table 18: Coding of the Housing-Quality Dimensions

Dimension	Classified as deficient
Panel A: Structural quality of the dwelling	
Floor	Unfinished natural floor – earth, sand, or dung – or salvaged wood (codes 100–120, 235)
Walls	No walls, or rudimentary, organic, or salvaged materials – cane, palm, logs, grass, scrap (codes 100–299, 405, 407) – and non-permanent earth materials: mud, adobe, clay, unburnt brick, pole and mud (codes 513, 515, 527, 533, 535, 539, 540, 541). Cement-stabilised earth, burnt or stabilised brick, and wood are treated as permanent
Roof	Wood, thatch, or palm; mud or earth; cardboard, plastic, or salvaged materials; other (codes 40–53, 60–61, 70–73, 90)
Panel B: Access to basic services and living conditions	
Sanitation	No toilet facility (code 10)
Crowding	More than three persons per habitable room

Notes: Coding of the five housing-quality dimensions, in harmonized IPUMS categories (codes in parentheses). The coding is identical across countries; the availability of each dimension varies by census (Egypt records no floor, wall, or roof variable and is excluded from the housing analysis; sanitation is unavailable in Kenya 2009, and crowding in Cameroon 2005 and Uganda 2014, whose censuses record only rooms used for sleeping). Water access is not part of the index: the UN-Habitat criterion counts boreholes and protected wells as improved sources, but the harmonized IPUMS water-supply variable in these censuses distinguishes types of piped supply rather than improved from unimproved sources, so it cannot reliably separate deficient from adequate access in this setting. Sanitation follows the UN-Habitat standard, under which a pit latrine with a slab counts as improved, so only the complete absence of a facility is coded deficient. The crowding denominator is the habitable room; for Zambia, where the census records living rooms and bedrooms separately, it is their sum. Sheet-metal roofing (codes 30–39) is classified as durable in the main definition and as deficient in the robustness definition. A household resides in low-quality housing if at least one available dimension is deficient (main definition); the robustness definition requires at least two. The continuous deprivation count sums the deficient dimensions (0–5).

First, in-migrants are less deprived than the resident urban population in every country and on almost every dimension, by 6 to 19 percentage points on the summary measure: recent arrivals are not, in the raw data, the worst-housed group in the city. This is not in tension with the paper's estimates, which compare the marginal drought-induced migrant with the average migrant arriving on the same corridor, holding the corridor fixed, and never compare migrants with residents. If anything it works against a mechanical reading of the results, since the unconditional correlation between migration and housing deprivation runs the other way. Second, crowding is the one dimension on which the two populations coincide: migrants occupy dwellings of comparable or better materials but at similar densities, which locates the pressure of in-migration on living space rather than on the built structure. Third, the dimensions contribute very unequally. Floors, walls and crowding do most of the work; roofs fire rarely once sheet metal is treated as adequate; and sanitation fires only where latrines are absent rather than unimproved. Water is absent from the index altogether, for the reason given in Section 2: these censuses distinguish only types of piped supply, while UN-Habitat counts boreholes and protected dug wells as improved sources, so the variable cannot separate deficient from adequate access. The index should therefore be read as a measure of housing durability and living space, not as the full four-pillar UN-Habitat definition, and its levels are not comparable to published slum headcounts. The category composition behind each dimension is reported in Supplementary Appendix Figures S17–S20.

10.1.3 Urbanization Data

Table 19 and Figure 12 validate the census-based urbanization classification against an external measure of city size, the population of the largest Africapolis agglomeration at destination. The two measures line up where it matters: across the three bins, the median agglomeration size rises from 47,000 to 86,000 to 880,000, the share of destinations hosting a city of at least 100,000 inhabitants rises from 28 to 43 to 70 percent, and the main-cities bin contains the region's actual metropolises (Nairobi, Kampala, Yaoundé, Bamako). The destination-level rank correlation is positive but moderate (Spearman 0.33), as expected: the census share captures the urban composition of the destination's population, while Africapolis measures the physical size of the nearest agglomeration, and coarser administrative units in Malawi and Uganda mechanically dilute urban shares (panel (b)). Appendix Figure S3 reports the country-specific cumulative distributions of both measures.

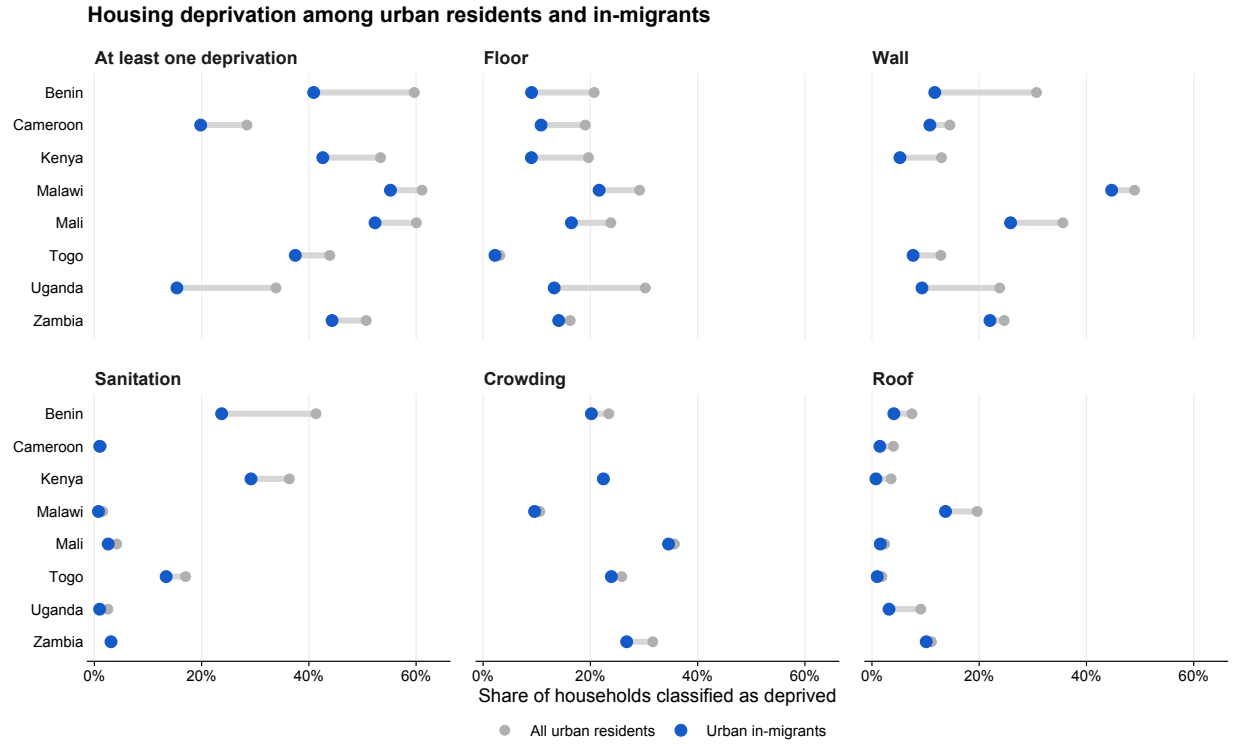
10.2 Identification Strategy

10.2.1 Post-IV Lasso Selection

Figure 13 reports a data-driven validation of the instrument choice. We build a pool of candidate instruments - drought intensity, drought dummies at alternative thresholds, precipitation and temperature polynomials, degree-day and evapotranspiration measures, and their lags and 3-season means - and apply the post-IV-LASSO selection of Belloni et al. (2014), alongside the single-candidate first-stage F. We split the exercise between drought shocks (panels (a)-(b)) and their extreme-wet counterparts (panels (c)-(d)).

The drought-intensity instrument is the strongest single candidate on both samples ($F =$

Figure 11: Housing Deprivation among Urban Residents and Urban In-Migrants



Notes: Share of households counted deprived on each dimension of the housing index, by country. Grey points are all urban residents, blue points urban in-migrants, and the segment joins the two for the same country and dimension. An in-migrant is a person whose unit of origin differs from the unit of enumeration and whose arrival year falls in that census's retrospective window; both populations are restricted to urban areas and person-weighted. The first panel applies the paper's threshold, at least one deprived dimension. Water deprivation is virtually absent in all samples and is therefore omitted from the figure. Crowding cannot be constructed for Cameroon 2005 and Uganda 2014, whose censuses record only rooms used for sleeping, so those two countries are absent from the crowding panel and mechanically lower in the summary panel. Sheet-metal roofs count as adequate.

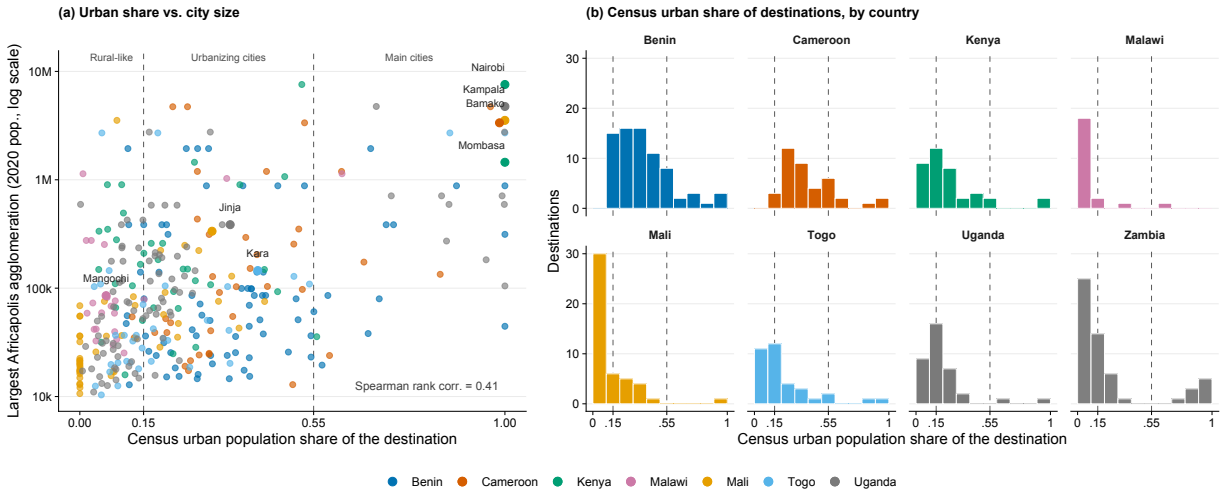
Source: Authors' computations from IPUMS census microdata.

Table 19: Validation of the Urbanization Classification: Census Urban Share versus Africapolis City Size

	Destinations	Corridor-years	Largest agglomeration		Example cities
Bin (urban share)			Median ('000)	$\geq$ 100k (%)	
Rural-like (< 0.15)	142	91,019	36	23%	Naivasha, Arua, Molo, Kitui
Urbanizing cities ($[0.15, 0.55)$)	169	122,821	90	43%	Nakuru, Lilongwe, Kisumu, Fort Portal
Main cities (≥ 0.55)	36	27,700	652	78%	Nairobi, Kampala, Yaounde, Bamako

Notes: Correspondence between the census urbanization bins and Africapolis city sizes. Bins: census urban population share of the destination. Agglomeration populations and names: largest Africapolis (2024) agglomeration in or within 10 km of the destination unit (2020 population; recovered boundary file, Nairobi and 15 further Kenya units included). Example cities are assigned to the bin of their own unit (highest urban share among the units they map to). Four rows carrying conurbation-cluster totals (Kisumu, Embu, Mbale, Kikima) are corrected using each agglomeration's pre-merge decadal growth (e.g., Kisumu 15.5M $\rightarrow$ 901k). All eight countries enter; corridor-years from the estimation sample ($N = 241{,}540$).

Figure 12: Census Urban Share versus Africapolis City Size at Destination



Notes: Panel (a) plots one observation per destination, colored by country. The horizontal axis reports the destination's urban population share in the most recent census. The vertical axis reports the 2020 population of the largest agglomeration in the Africapolis 2024 database located within, or within 10 km of, the destination unit, on a logarithmic scale. Panel (b) shows the distribution of census urban shares across destinations, separately by country. Dashed vertical lines indicate the classification thresholds of 0.15 and 0.55. Destinations without a matched Africapolis agglomeration of at least 10,000 inhabitants are excluded from panel (a). Four Africapolis agglomeration records reporting conurbation-cluster totals are corrected as described in the notes to Table 19. The Spearman rank correlation between the two measures is 0.33. Appendix Figure S3 reports their country-specific cumulative distributions.

Source: Authors' computations using IPUMS census data and the Africapolis 2024 database.

23.9 on the scale sample, $F = 13.0$ on the propensity sample) and is selected by the LASSO in both; adding lagged candidates, its 3-season mean dominates ($F = 27.4$), consistent with the cumulative instrument used throughout. At the default penalty the LASSO retains a small set of dry-side candidates rather than a single winner (drought intensity, dry days, and extreme degree days on the scale sample); we nevertheless use the single strongest candidate, both for transparency and to keep the model just-identified, so that Anderson-Rubin inference remains exact and the complier population is not mixed across instruments. he robustness section reports the main results under the LASSO-selected instrument sets, showing that our findings are not tied to this choice.

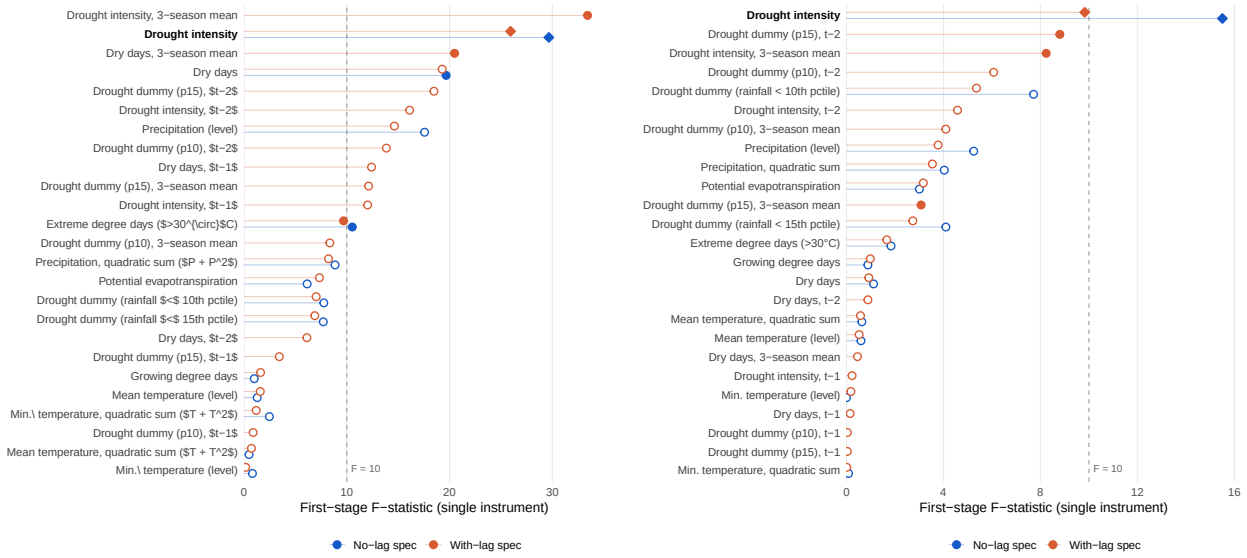
By contrast, Figure 14 shows the results for the excess-rainfall intensity - the exact positive mirror of the drought instrument - has essentially no predictive power ($F = 4.0$ and $F = 0.03$), and the flood dummies are similarly weak. The instrument is thus data-selected, and its power comes specifically from the dry tail of the rainfall distribution.

10.3 First Stage and Reduced Form

10.3.1 First Stage

Table 20 assesses the robustness of the first stage to sample restrictions and to the drought exposure window, reporting the paper's main specification for each sample and instrument; specification robustness itself is established on the full sample in Table 1. Columns (1)-(2) cover the main sample of urban migrations to all destinations, columns (3)-(4) restrict

Figure 13: Post-IV-LASSO Instrument Selection: Drought Candidates

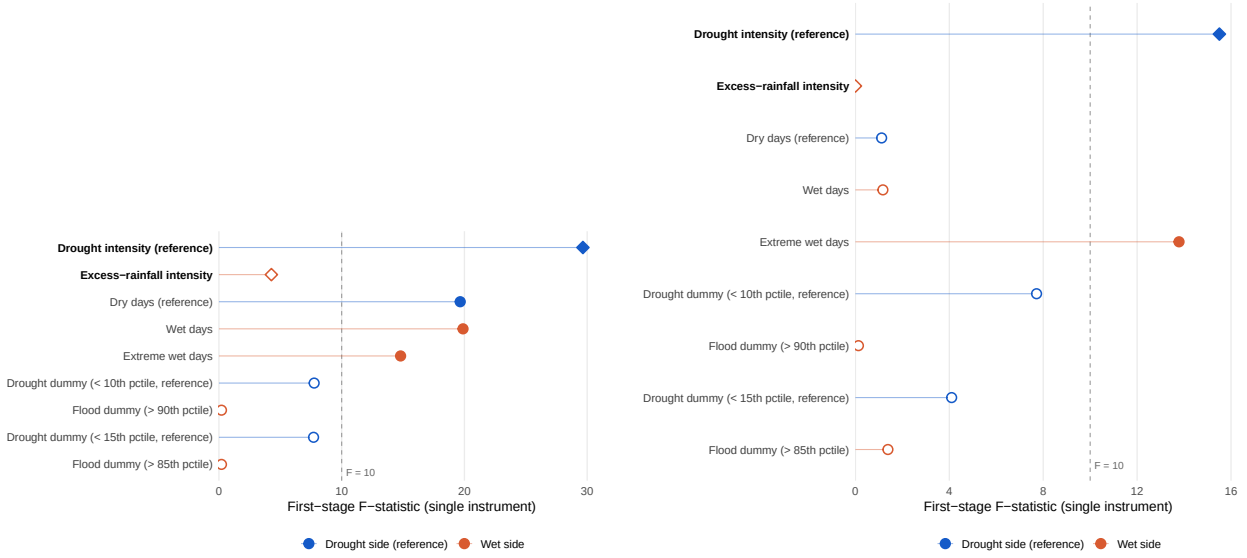


(a) Scale sample (b) Propensity sample

Notes: Each point reports the first-stage F-statistic of a single candidate instrument for the urban migration rate under the main specification, including origin–destination and year fixed effects and the distance-to-census control. Standard errors are clustered at the origin-district level. The panels compare candidate sets without and with lagged instruments for the scale and propensity samples, respectively. Filled markers denote candidates selected by post-IV-LASSO.

Source: Authors' computations using IPUMS census and CHIRPS climate data.

Figure 14: Placebo Instrument Selection: Wet-Side Mirror Candidates



(a) Scale sample (b) Propensity sample

Notes: This figure replaces each drought candidate with its wet-side mirror. Excess-rainfall intensity is defined as the positive mirror of drought intensity. Each point reports the single-instrument first-stage F-statistic under the main specification. The weaker first stages indicate that instrument strength is specific to the dry tail of the rainfall distribution.

Source: Authors' computations using IPUMS census and CHIRPS climate data.

destinations to main cities,
footnoteMain cities are defined as destinations with an urban population share weakly above the threshold, $\text{share}_d^{\text{urb}} \geq 0.55$, measured at the country's most recent census wave and held fixed across waves. Rural origins are defined as the complement: locations with $\text{share}_o^{\text{urb}} < 0.55$ or with no measured urban share. Robustness checks vary this threshold and use alternative definitions and datasets. and columns (5)-(6) further restrict to moves from rural origins to main cities. Within each pair, the first column uses the contemporaneous drought instrument and the second its cumulative (three-season mean) counterpart on the lag-restricted sample.

The contemporaneous instrument is a strong and significant predictor of migration in every sample, including moves from rural origins to major cities. On the full sample, the cumulative instrument delivers an even stronger first stage ($F = 27.4$ against 23.9), but its strength collapses once the sample is restricted to city destinations (F below conventional thresholds in columns (4) and (6)). This asymmetry, together with exclusion-restriction considerations-a multi-year accumulation of drought is more likely to affect destination outcomes through channels other than the timing of migration, such as persistent income losses at origin-leads us to retain the contemporaneous instrument as the paper's main one. The cumulative instrument is nonetheless reported alongside the main results throughout, and delivers consistent conclusions wherever its first stage is strong.

10.3.2 Reduced Form

Residential Status - This section reports the reduced-form counterparts of the paper's main estimates: each outcome is regressed directly on the drought instrument at origin, under the main specification, so that no result depends on the strength of the first stage. Table 21 reports the reduced form for the propensity margin. Columns (1)-(2) use the contemporaneous instrument and columns (3)-(4) the cumulative instrument on the lag-restricted sample; within each pair, the first column reports the share of in-migrants living in low-quality housing under the main definition (dummy) and the second the mean number of deprived housing dimensions among the same migrants (continuous). Its significance pattern mirrors the Anderson-Rubin intervals of Table 6, as the two tests coincide.

Labor-Force Status - Tables 22 and 23 report the reduced forms for the employment results, regressing the scale and propensity outcomes directly on origin drought for both instruments. On the scale margin, all six inflows rise significantly at the 1% level. On the propensity margin, drought shifts the composition of arrivals from inactivity toward employment under both instruments; since the reduced form does not rely on the first stage, the lagged columns provide valid inference where the lagged first stage of Table 28 is weak, and confirm the same reallocation.

10.3.3 Identifying Variation

Figures 15 and 16 provide a graphical representation of the relationships underlying the identification strategy. Each panel plots mean outcomes within forty equal-sized bins of drought intensity at origin, after residualizing all variables on origin-destination pair and

Table 20: First Stage: Restricted Samples and Lagged Instrument (Main Specification)

	All destinations		*Main cities*		*Rural → main cities*	
	Contemp.	Lagged	Contemp.	Lagged	Contemp.	Lagged
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. Scale sample: all corridor-years						
Drought intensity	0.00362***	0.00947***	0.00312***	0.00386	0.00250***	0.00369*
	(0.00066)	(0.00164)	(0.00092)	(0.00317)	(0.00065)	(0.00196)
First-stage F	29.65	33.42	11.61	1.48	14.54	3.55
Effect of 1-SD drought (% mean)	+6.8%	+10.5%	+4.8%	+3.4%	+5.1%	+4.3%
Mean migration rate	0.02682	0.02680	0.03317	0.03350	0.02487	0.02509
Observations	241,540	235,920	27,700	26,764	23,968	23,176
OD pairs	16,846	16,846	1,948	1,948	1,678	1,678
Panel B. Propensity sample: corridor-years with a positive urban flow						
Drought intensity	0.00601***	0.01007***	0.00317**	0.00429	0.00256***	0.00473**
	(0.00153)	(0.00351)	(0.00129)	(0.00430)	(0.00081)	(0.00239)
First-stage F	15.50	8.24	6.07	1.00	9.90	3.93
Effect of 1-SD drought (% mean)	+3.4%	+3.3%	+3.1%	+2.4%	+3.2%	+3.4%
Mean migration rate	0.08908	0.08881	0.05319	0.05393	0.04086	0.04134
Observations	72,723	71,207	17,273	16,624	14,587	14,066
OD pairs	10,537	10,510	1,633	1,632	1,395	1,394

Notes: First stage (DV: urban migration rate) at the paper's main specification – origin–destination pair and arrival-year fixed effects, distance-to-census control – in each sample and for each instrument. 'Contemp.' is $\max(0, -\tilde{z})$ at origin, 'Lagged' its 3-season mean over $\{t, t-1, t-2\}$ on the lag-restricted sample. Main cities are destinations with a census urban share of at least 0.55; the rural columns additionally restrict to origins below that threshold. Panel A is the sample on which the scale results are estimated; Panel B restricts to the corridor-years where the share of migrants in low-quality housing is defined, which is the sample of the propensity estimates. The first-stage coefficient is larger in Panel B, but so is the mean migration rate, which is why the same shock represents a smaller percentage of the mean there. Effect of 1-SD drought: coefficient × SD(instrument) ÷ mean migration rate × 100. 8 countries, 12-year window; SEs clustered at *admin_origin*, in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 21: Reduced Form: Drought Intensity and the Propensity for Low-Quality Housing

	Contemporaneous drought		*3-season mean (lagged)*	
	Deprived Dummy - Main def.	Number of deprivations Continuous	Deprived Dummy - Main def.	Number of deprivations Continuous
	(1)	(2)	(3)	(4)
Drought intensity	0.00731**	0.00571	0.01995***	0.03217**
	(0.00325)	(0.00637)	(0.00687)	(0.01360)
Effect of 1-SD drought	+0.37 pp	+0.0029	+0.58 pp	+0.0093
R^2	0.392	0.450	0.396	0.455
Within R^2	0.007	0.002	0.007	0.002
Mean outcome	0.406	0.662	0.404	0.656
Observations	72,723	72,723	71,207	71,207
OD pairs	10,537	10,537	10,510	10,510

Notes: Reduced form: propensity outcomes (main UN-Habitat share – floor, wall, roof, sanitation, crowding, water excluded as unmeasurable in these censuses; mean nb. of deprived dimensions) regressed directly on origin drought – no first-stage reliance; RF t-test = AR test of zero in the 2SLS table. Cols (3)–(4): lagged instrument, lag-restricted sample, directly interpretable. Main specification; SEs clustered at `admin_origin`. 1-SD effects in pp / dimension units. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 22: Reduced Form: Drought Intensity and the Labor-Force Status of the Inflow: Scale

	Contemporaneous drought			*3-season mean (lagged)*		
	Employed	Unemployed	Inactive	Employed	Unemployed	Inactive
	(1)	(2)	(3)	(4)	(5)	(6)
Drought intensity	0.00155***	0.00027***	0.00128***	0.00315***	0.00089***	0.00343***
	(0.00027)	(0.00005)	(0.00033)	(0.00063)	(0.00013)	(0.00078)
Effect of 1-SD drought	+7.3%	+10.1%	+5.5%	+8.8%	+19.5%	+8.7%
R^2	0.474	0.298	0.377	0.486	0.303	0.387
Within R^2	0.006	0.001	0.007	0.006	0.002	0.007
Mean outcome	0.0106	0.00134	0.0118	0.0106	0.00136	0.0118
Observations	241,540	241,540	241,540	235,920	235,920	235,920
OD pairs	16,846	16,846	16,846	16,846	16,846	16,846

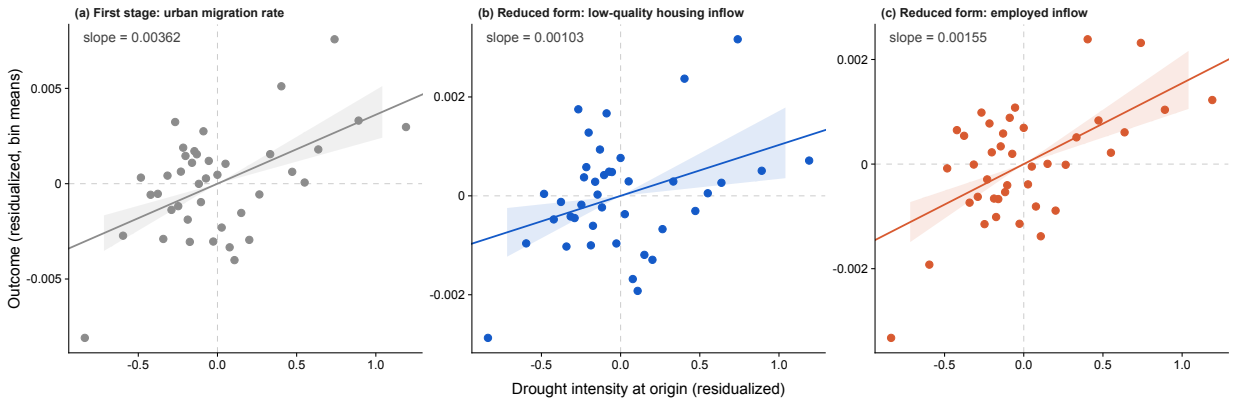
Notes: Reduced form: employed / unemployed / inactive in-migrants per 100 urban residents (unwinsorized) regressed directly on origin drought – no first-stage reliance; the RF t-test = AR test of zero in the 2SLS table. Cols (4)–(6): lagged instrument, lag-restricted sample. Effect of 1-SD drought in % of the mean outcome. Main specification (od-pair and year FE, `dist_census` control); SEs clustered at `admin_origin`. Undetermined labor-force status excluded. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 23: Reduced Form: Drought Intensity and the Labor-Force Status of the Inflow: Propensity

	Contemporaneous drought			*3-season mean (lagged)*		
	Employed	Unemployed	Inactive	Employed	Unemployed	Inactive
	(1)	(2)	(3)	(4)	(5)	(6)
Drought intensity	0.00839***	0.00175	-0.01014***	0.01403**	0.00500*	-0.01904***
	(0.00303)	(0.00137)	(0.00293)	(0.00599)	(0.00270)	(0.00576)
Effect of 1-SD drought	+0.42 pp	+0.09 pp	−0.51 pp	+0.41 pp	+0.14 pp	−0.55 pp
R^2	0.329	0.243	0.324	0.335	0.245	0.329
Within R^2	0.003	0.000	0.005	0.003	0.000	0.005
Mean outcome	0.505	0.0578	0.437	0.507	0.0586	0.435
Observations	71,503	71,503	71,503	70,036	70,036	70,036
OD pairs	10,452	10,452	10,452	10,420	10,420	10,420

Notes: Reduced form: employed / unemployed / inactive shares among known-status in-migrants regressed directly on origin drought – no first-stage reliance; the RF t-test = AR test of zero in the 2SLS table, the honest inference for the weak lagged first stage there. The three shares sum to one, so coefficients sum to zero by construction within each instrument. Cols (4)–(6): lagged instrument, lag-restricted sample. Effect of 1-SD drought in percentage points. Main specification (od-pair and year FE, `dist_census` control); SEs clustered at `admin_origin`. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

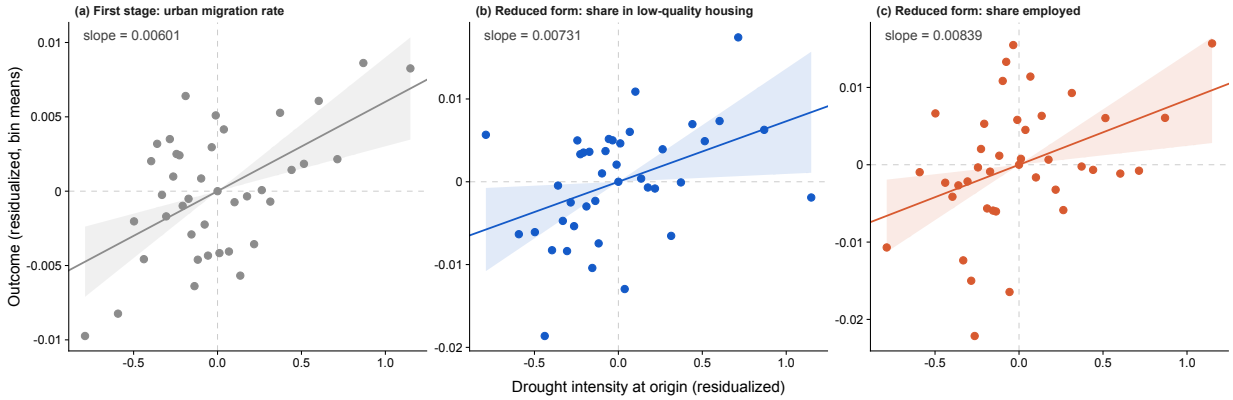
Figure 15: First Stage and Reduced Form: Binned Scatterplots under the Main Specification



Notes: Binned scatterplots of the first-stage (panel a) and reduced-form (panel b) relationships under the main specification. All variables are residualized on origin–destination pair and arrival-year fixed effects and the distance-to-census control (Frisch–Waugh); each dot is the mean within one of forty equal-sized bins of residualized drought intensity at origin. Panel (a) plots the urban migration rate; panel (b) the low-quality-housing inflow rate (main slum definition, unwinsorized). The fitted line is estimated on the underlying corridor-year data and passes through the origin by construction; its slope equals the corresponding regression coefficient exactly (0.00362 in panel a, 0.00103 in panel b). The shaded area is the 95% confidence band based on standard errors clustered at the origin administrative-unit level. Sample: urban migrants, all destinations; 8 countries; 241,540 corridor-years.
Source: Authors' computations from IPUMS census and CHIRPS climate data.

arrival-year fixed effects and the distance-to-census control; the fitted line is estimated on the underlying corridor-year data, so its slope equals the corresponding regression coefficient exactly, and the shaded area is the 95% confidence band based on standard errors clustered at the origin level. All relationships are positive, close to linear, and supported across the entire distribution of drought shocks rather than driven by a few extreme bins. Corridors experiencing unusually dry growing seasons send more migrants to urban areas, more migrants into low-quality housing, and more employed migrants (Figure 15), the visual counterparts of the first-stage and reduced-form estimates of Tables 1 and 4. The same holds for the propensity outcomes (Figure 16): among migrants, the shares in low-quality housing and in employment both rise with origin drought, the reduced-form counterpart of the sorting results.

Figure 16: First Stage and Reduced Form for the Propensity Outcomes: Binned Scatterplots



Notes: Binned scatterplots of the first-stage (panel a) and propensity reduced-form (panels b–c) relationships under the main specification. All variables are residualized on origin–destination pair and arrival-year fixed effects and the distance-to-census control (Frisch–Waugh); each dot is the mean within one of forty equal-sized bins of residualized drought intensity at origin. Panel (a) plots the urban migration rate; panel (b) the share of migrants in low-quality housing (main definition, denominator: migrants); panel (c) the share of migrants employed (denominator: known-status migrants). Shares are unwinsorized, as are all outcomes in the paper. The fitted line is estimated on the underlying corridor-year data and passes through the origin by construction; its slope equals the corresponding regression coefficient exactly (0.00628 in panel a, 0.00679 in panel b, 0.00850 in panel c). The shaded area is the 95% confidence band based on standard errors clustered at the origin administrative-unit level. Sample: urban migrants, all destinations; 8 countries; 72,821 corridor-years in panels (a)–(b) (corridors with at least one migrant) and 71,587 in panel (c) (corridors with at least one known-status migrant).

Source: Authors' computations from IPUMS census and CHIRPS climate data.

Drought and the harvest at origin. - Table 24 reports the reduced form of origin drought on the harvest itself, the first link of the agricultural-push channel discussed in Section 6.1.2. Crop statistics come from HarvestStat Africa v1.0 (Lee et al., 2025), matched to our origin districts through its areal crosswalk to IPUMS boundaries. Because production is the product of harvested area and yield, the three outcomes are an exact decomposition and their coefficients add up: the loss is entirely in yield per hectare, with harvested area unmoved. The sample is necessarily narrower than the estimation panel – 2,535 district-years, 214 districts and six countries – and the limits of the series are set out in the table notes.

Table 24: Drought and the Harvest at Origin: The First Link of the Agricultural Channel

	log Production		log Harvested area		log Yield	
	(1)	(2)	(3)	(4)	(5)	(6)
Drought intensity at origin	−0.064*	−0.059	0.002	−0.016	−0.065***	−0.044**
	(0.036)	(0.042)	(0.030)	(0.036)	(0.018)	(0.022)
Effect of 1-SD drought (%)	−3.03	−2.81	+0.09	−0.74	−3.11	−2.07
District FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	—	Yes	—	Yes	—
Country × year FE	—	Yes	—	Yes	—	Yes
Mean of dependent variable (level)	30,916 t		20,244 ha		1.43 t/ha	
Districts	214					
Countries	6					
District-years	2,535					

Notes: Crop statistics are from HarvestStat Africa v1.0 (Lee et al., 2025), aggregated onto the paper's origin districts through the areal crosswalk between its FEWS NET units and IPUMS GEOLEV boundaries, and restricted to maize, the one staple with broad coverage in every country. Yield is recomputed as production over harvested area after that aggregation, since yields cannot be averaged across units; the top and bottom percentiles of yield within each country are trimmed. The sample is 2,535 district-years, 214 districts and 6 countries (Benin, Cameroon, Kenya, Malawi, Mali, Zambia), over 1981–2018. Drought intensity at origin is the instrument of the main specification, $\max(0, -z)$ of the seasonal rainfall anomaly, standardized within district. Standard errors, in parentheses, are clustered at the origin district. 'Effect of 1-SD drought' is $b \times SD(drought_int_z) = 0.476$ on this sample, in percent of the dependent variable, matching the convention of the paper's other tables. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Because $\log \text{Production} = \log \text{Harvested area} + \log \text{Yield}$, the three outcomes are an exact decomposition and their coefficients add up within each fixed-effect structure. In the reference specification a one-standard-deviation increase in drought intensity lowers maize yield per hectare by 3.1% and total production by 3.0%, while harvested area does not move (+0.09%). Yield is the precisely estimated margin; production, being the sum of yield and area, inherits the area's noise and is significant only at the 10% level. The harvested-area columns matter for interpretation: HarvestStat records area harvested rather than area planted, so a drought that led farmers to abandon their worst plots would remove those plots from the denominator and could raise measured yield even as income collapsed. It does not: the loss is in the harvest, not in abandoned land.

This table is the first link of the agricultural-push channel and should be read with Figure 4, which is the second: drought destroys the harvest at origin, and the migration response concentrates in the districts where that harvest is the livelihood – the first stage is roughly four times stronger out of the most cultivated quarter of origins. We characterize the channel through these two pieces rather than through a formal year-by-year mediation of drought on migration through yields, which is not identified here: the two independent yield indices we can construct for the same district-year correlate at only 0.30 once district and year fixed effects are removed, the crop series covers 69% of the estimation panel, and the pooled estimate rests largely on Kenya, the only country significant on its own. Columns (2), (4) and (6) restrict the comparison to districts of the same country in the same year, which is demanding on six countries: the yield effect survives it, the production effect does not. Mali enters HarvestStat at the first administrative level while it enters our panel at the second, so its drought intensity is averaged over the origin districts sharing one crop observation rather than repeated against it; absent that step the same harvest would be matched to several different shocks, which would attenuate the estimate and overstate the sample by roughly a fifth.

10.4 Main Results

10.4.1 Residential Integration

Scale- Table 25 details the calculation behind the interpretation of the main estimate: for a representative corridor with 10,000 urban residents, it translates a one-standard-deviation drought at origin into additional migrants and additional residents in low-quality housing, whose ratio equals, by construction, the 2SLS coefficient of Table 5. Table 26 replicates the main table under the two sample restrictions used in the paper - destinations limited to main cities, and rural origins moving to main cities - with the contemporaneous instrument, whose first stage remains strong in both samples.

Table 25: Interpreting the Main Estimate: A Representative Corridor

Panel A. Step-by-step calculation

	Formula	Value
Baseline (no drought)		
Migrants	$\overline{\text{MigRate}}/100 \times 10{,}000$	$0.02682 \times 100 = 2.68$
Migrants in low-quality housing	$\overline{S}/100 \times 10{,}000$	$0.01196 \times 100 = 1.20$
Effect of a 1-SD drought shock		
Δ Migration rate	$\beta_{FS} \times SD(\text{drought})$	$0.00362 \times 0.502 = 0.00182$
Δ Migrants	$\Delta\text{MigRate}/100 \times 10{,}000$	+0.18
Δ Low-quality housing inflow (S)	$\beta_{2SLS} \times \beta_{FS} \times SD(\text{drought})$	$0.285 \times 0.00362 \times 0.502 = 0.00052$
Δ Migrants in low-quality housing	$\Delta S/100 \times 10{,}000$	+0.052
Share of new migrants in low-quality housing		
Δ Low-quality / Δ Migrants	β_{2SLS}	$0.052/0.18 = 28\%$

Panel B. Summary

	Without drought	With 1-SD drought	Change	Share of new migrants in low-quality housing
Number of migrants	2.68	2.86	+0.18 (+6.8%)	–
Migrants in low-quality housing	1.20	1.25	+0.052 (+4.3%)	28%

Notes: Illustration for a single origin–destination corridor with 10,000 urban residents at destination (all-destinations sample, main definition of low-quality housing). Notation as in the text: MigRate is the urban migration rate and S the low-quality housing inflow rate, both per 100 urban residents; bars denote sample means. Because S and MigRate share the same denominator (destination urban population), β_{2SLS} reads directly as the share of marginal, drought-induced migrants who settle in low-quality housing – about 28 of every 100. $\beta_{FS} = 0.00362$, $\beta_{2SLS} = 0.28455$, $SD(\text{drought}) = 0.502$; all values from Table 5.

10.4.2 Propensity

Propensity - Table 27 re-estimates the propensity outcomes - the share of in-migrants in low-quality housing under the main definition and the mean number of deprived dimensions - in the two restricted samples, with the contemporaneous instrument. The propensity

Table 26: Drought-Induced Migration and Low-Quality Housing Inflows: Main Cities and Rural-Origin Samples

	Main cities			*Rural origins → main cities*		
	OLS	FS	2SLS	OLS	FS	2SLS
	(1)	(2)	(3)	(4)	(5)	(6)
Urban migration rate	0.34155***		0.13052*	0.32717***		0.13922*
	(0.02545)		(0.07793)	(0.02831)		(0.07379)
Drought intensity		0.00312***			0.00254***	
		(0.00092)			(0.00065)	
First-stage F			11.61			15.16
AR 95% CI			[-0.123, 0.267]			[-0.066, 0.273]
Effect of 1-SD drought (% mean)	+4.6%	+4.8%	+1.8%	+4.7%	+5.2%	+2.0%
Origin-Destination FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Dist. to census control	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.879	0.770	0.814	0.856	0.757	0.804
Within R^2	0.589	0.025	0.365	0.528	0.042	0.355
Mean outcome		0.011634			0.008890	
Mean migration rate		0.033169			0.024660	
SD(drought intensity)		0.507			0.505	
Observations		27,700			24,280	
OD pairs		1,948			1,704	

Notes: Contemporaneous instrument throughout; the cumulative instrument is omitted, as its first stage collapses in these samples (see the first-stage appendix table). Columns (1)–(3): destinations restricted to main cities, with a census urban share of at least 0.55. Columns (4)–(6): additionally, rural origins only. OLS and 2SLS columns report the effect of the urban migration rate on the low-quality housing inflow rate; FS columns the sample-specific first stage. First-stage F is the cluster-robust Wald statistic and AR 95% CI the cluster-robust Anderson–Rubin interval. Effect of 1-SD drought is the coefficient × SD(instrument) ÷ mean outcome × 100, multiplied by the first-stage coefficient in the OLS and 2SLS columns. 8 countries, 12-year window. SEs clustered at *admin_origin*, in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

subsamples are considerably smaller than for the scale outcome, since they require corridors with observed housing characteristics, and both first stages are moderate ($F = 6.1$ in the main-cities sample, $F = 10.4$ in the rural-to-cities sample), so the Anderson-Rubin intervals carry the inference. The 2SLS point estimates on the extensive margin (2.22 and 2.41) are larger than the full-sample estimate of 1.15, but both Anderson-Rubin intervals are very wide and include zero. These subsamples therefore neither confirm nor contradict the main propensity result; they are too imprecise to be informative on their own.

Table 27: Propensity for Low-Quality Housing: Main Cities and Rural-Origin Samples

	Main cities				*Rural origins → main cities*			
	Deprived Dummy - Main def.		Number of deprivations Continuous		Deprived Dummy - Main def.		Number of deprivations Continuous	
	OLS	2SLS	OLS	2SLS	OLS	2SLS	OLS	2SLS
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Urban migration rate	0.07472***	2.21915	0.10170**	0.66641	0.14962**	2.40963	0.22845**	-0.75675
	(0.02551)	(1.80309)	(0.04359)	(2.61755)	(0.05970)	(2.23782)	(0.10708)	(3.49205)
First-stage F		6.07		6.07		10.44		10.44
AR 95% CI		[-0.85, 12.82]		[-6.93, 9.90]		[-1.83, 9.44]		[-9.88, 7.20]
Effect of 1-SD drought	+0.01 pp	+0.36 pp	+0.0002	+0.0011	+0.02 pp	+0.33 pp	+0.0003	−0.0010
R^2	0.361	0.229	0.384	0.381	0.364	0.305	0.383	0.379
Within R^2	0.011	-0.193	0.004	-0.001	0.011	-0.080	0.004	-0.002
Mean outcome	0.345	0.345	0.472	0.472	0.346	0.346	0.470	0.470
Observations	17,273	17,273	17,273	17,273	14,720	14,720	14,720	14,720
OD pairs	1,633	1,633	1,633	1,633	1,414	1,414	1,414	1,414

Notes: Propensity (composition) outcomes in the restricted samples, contemporaneous instrument only (the cumulative instrument has no first stage in these samples; see the first-stage appendix table). Columns (1)–(4): destinations restricted to main cities (*share_urban_known_dest* ≥ 0.55); columns (5)–(8): additionally, rural origins only. Within each sample, the first pair of columns reports the share of in-migrants living in low-quality housing under the main definition (dummy) and the second pair the mean number of deprived housing dimensions (continuous); denominator: migrants, unweighted, not winsorized. Main specification (origin–destination pair and arrival-year FE, distance-to-census control); SEs clustered at *admin_origin*, in parentheses. First-stage F: cluster-robust Wald on the instrument, on each column's estimation sample. AR 95% CI: cluster-robust Anderson–Rubin confidence interval. Effect of 1-SD drought: coefficient × first-stage coefficient × SD(instrument), in pp of the share (dummy) or dimension units (continuous). 8 countries, 12-year window. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

10.4.3 Labor Market Integration

Table 28 re-estimates the employment results of Tables 7 and 8 with the 3-season cumulative drought instrument. On the scale margin the first stage is strong ($F = 27.4$) and the employed, unemployed, and inactive inflows all rise significantly. On the propensity margin the first stage is weak ($F = 7.1$), so inference relies on the Anderson-Rubin intervals, which confirm the contemporaneous pattern: the employed share of arrivals rises, the inactive share falls, and the unemployed share is unchanged.

Table 28: Drought-Induced Migration and the Labor-Force Status of the Inflow: Lagged Instrument

	Scale			*Propensity*		
	Inflows per 100 urban residents			Shares of known-status in-migrants		
	Employed 2SLS	Unemployed 2SLS	Inactive 2SLS	Employed 2SLS	Unemployed 2SLS	Inactive 2SLS
	(1)	(2)	(3)	(4)	(5)	(6)
Urban migration rate	0.52274***	0.16607***	0.31710***	-0.30566	1.90688**	-1.60122
	(0.04253)	(0.02191)	(0.03569)	(0.98895)	(0.84324)	(1.05507)
First-stage F	32.93	32.93	32.93	7.46	7.46	7.46
AR 95% CI	[0.433, 0.609]	[0.130, 0.222]	[0.237, 0.384]	[-3.223, 2.282]	[0.769, 6.755]	[-6.683, 0.104]
Effect of 1-SD drought	+9.2%	+24.6%	+10.2%	−0.06 pp	+0.34 pp	−0.29 pp
R^2	0.831	0.161	0.676	0.353	-0.098	0.280
Within R^2	0.672	-0.157	0.525	0.008	-0.498	-0.105
Mean outcome	0.00977	0.00116	0.00536	0.639	0.0689	0.292
Observations	235,920	235,920	235,920	63,295	63,295	63,295
OD pairs	16,846	16,846	16,846	9,945	9,945	9,945

Notes: 2SLS estimates with the 3-season cumulative drought instrument on the lag-restricted sample. Columns (1)–(3): employed / unemployed / inactive in-migrants per 100 urban residents, unwinsorized; the effect of a 1-SD drought is in % of the mean. Columns (4)–(6): shares among known-status in-migrants (summing to one, so coefficients sum to zero by construction); the effect of a 1-SD drought is in percentage points. The propensity-side first stage is weak ($F \approx 7$): the lagged point magnitudes in columns (4)–(6) are not interpretable and inference should rely on the Anderson-Rubin intervals. Main specification (od-pair and year FE, `dist_census` control); SEs clustered at `admin_origin`. Undetermined labor-force status excluded. Sample: urban migrants, all destinations; 8 countries, 12-year window. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

10.4.4 Interpreting the effect

Decomposing the Effect for housing - Table 29 details the decomposition of the additional low-quality housing inflow from a one-standard-deviation drought into its scale and propensity components, reporting the formula, inputs, and implied contribution of each channel - roughly 90% through the number of migrants and 10% through their propensity to live in low-quality housing.

10.5 Individual Heterogeneity: Migrant Profiles

This section asks two distinct questions about the migrants behind the main result: who the drought-induced migrants are (Table 30), and whether the effect varies with their profile - for low-quality housing (Figure 17) and for employment (Figure 18).

Who Are the Drought-Induced Migrants? - Table 30 profiles the drought-induced migrants on gender, age at arrival, and education, comparing the naive read - migrants arriving in high- versus no-drought years - with the complier characterization of Abadie (2003), which describes the marginal migrants identified by the instrument, the same population as the main result. The two readings agree: drought-induced migrants are observationally similar to other migrants - the same gender balance and comparable or slightly higher education - but younger, with an over-representation of children under fifteen, consistent with whole-household displacement rather than the selective migration of young workers. Their higher propensity for low-quality housing is therefore not accounted for by observably lower human capital.

Does the Effect Vary by Profile? - Figure 17 estimates the effect separately by subgroup, on both margins, with subgroup-specific inflows and migration rates under the main specification. Effects and propensity shifts are broadly uniform across gender and age - every identified estimate sits close to the aggregate anchor, and no within-dimension contrast is significant. The only gradient is educational: on the scale margin, the effect is twice as strong for primary-educated adults as for those with secondary education or above ($p = 0.036$), suggestive since education is measured at census. Full estimates and contrasts are in Supplementary Appendix Tables S7 and S8.

Figure 18 repeats the exercise for the employment margins. The mirror image completes the picture: employment absorption is uniform where housing is not - there is no educational gradient in the employed inflow ($p = 0.787$), while the housing effect is twice as strong for the primary-educated. Full estimates are in Supplementary Appendix Tables S9 and S10.

10.6 Geography of the Drought-Induced Inflows

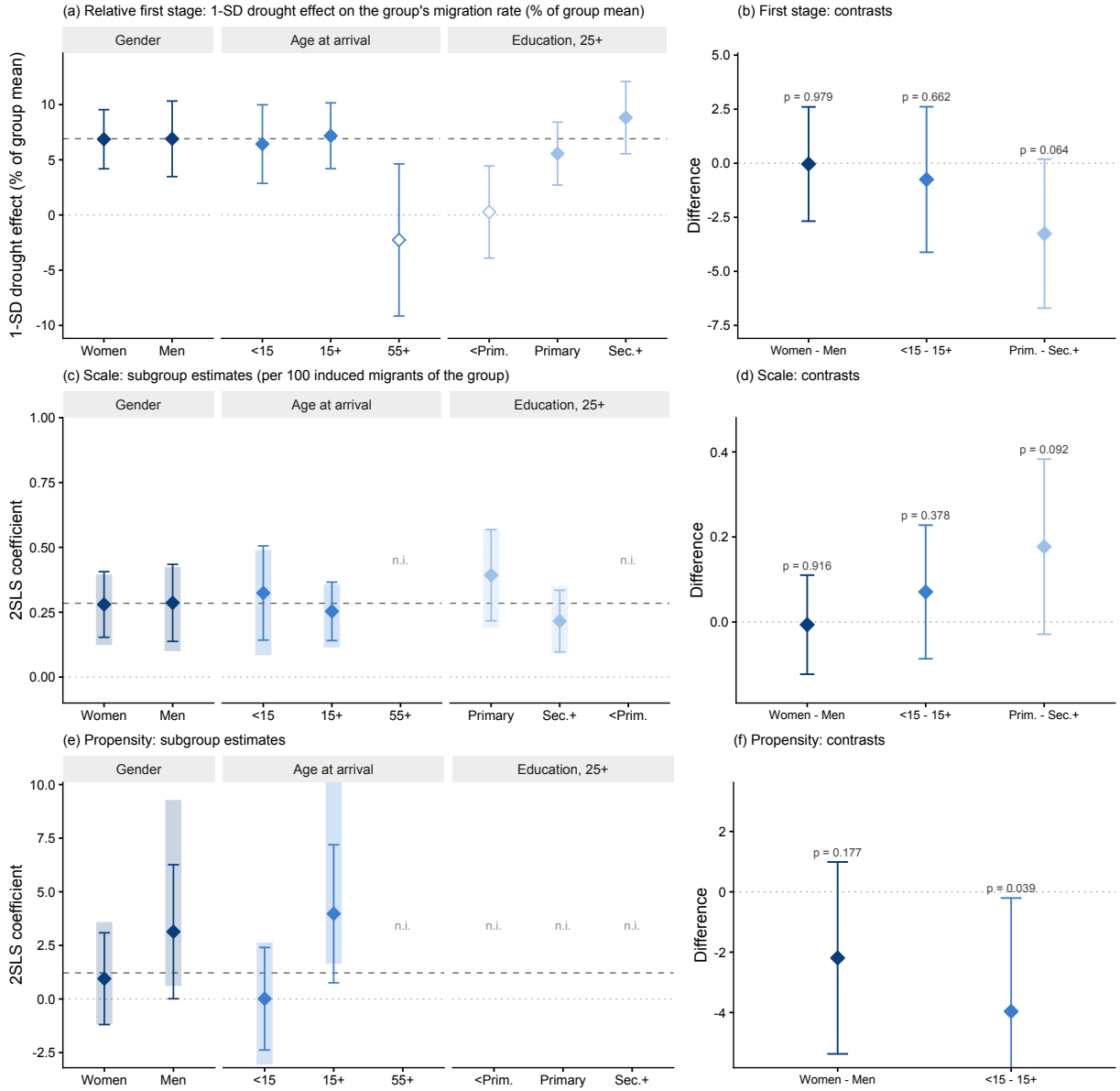
Figure 19 maps the destination side of the first stage: for each destination, the share of its observed urban in-migration that the first stage attributes to drought. It illustrates the destination heterogeneity of Section 6.1.2 geographically – the share is highest in intermediate destinations and lowest in the largest metropolises – and is an illustration of where the estimated inflows land, not an identified result.

Table 29: Decomposition of the Drought Effect on Low-Quality Housing Inflows: Scale versus Propensity

	Low-quality housing inflow rate (scale) (1)	Share of migrants in low-quality housing (propensity) (2)
Panel A: Both channels on the common (positive-flow) sample		
Urban migration rate	0.32022***	1.21591**
	(0.10891)	(0.59790)
First-stage F	15.50	15.50
AR 95% CI	[0.032, 0.525]	[0.170, 2.940]
Mean outcome	0.03974	0.4064
Observations	72,723	72,723
Panel B: Decomposition of a one-standard-deviation drought		
	Calculation	Share
Scale: $\Delta\text{MigRate} \times \overline{P}$ *(more migrants, same propensity)*	$0.00302 \times 0.4064 = 0.00123$	79%
Propensity: $\overline{\text{MigRate}} \times \Delta P$ *(same migrants, higher propensity)*	$0.08908 \times 0.00368 = 0.00033$	21%
Total	0.00156	100%

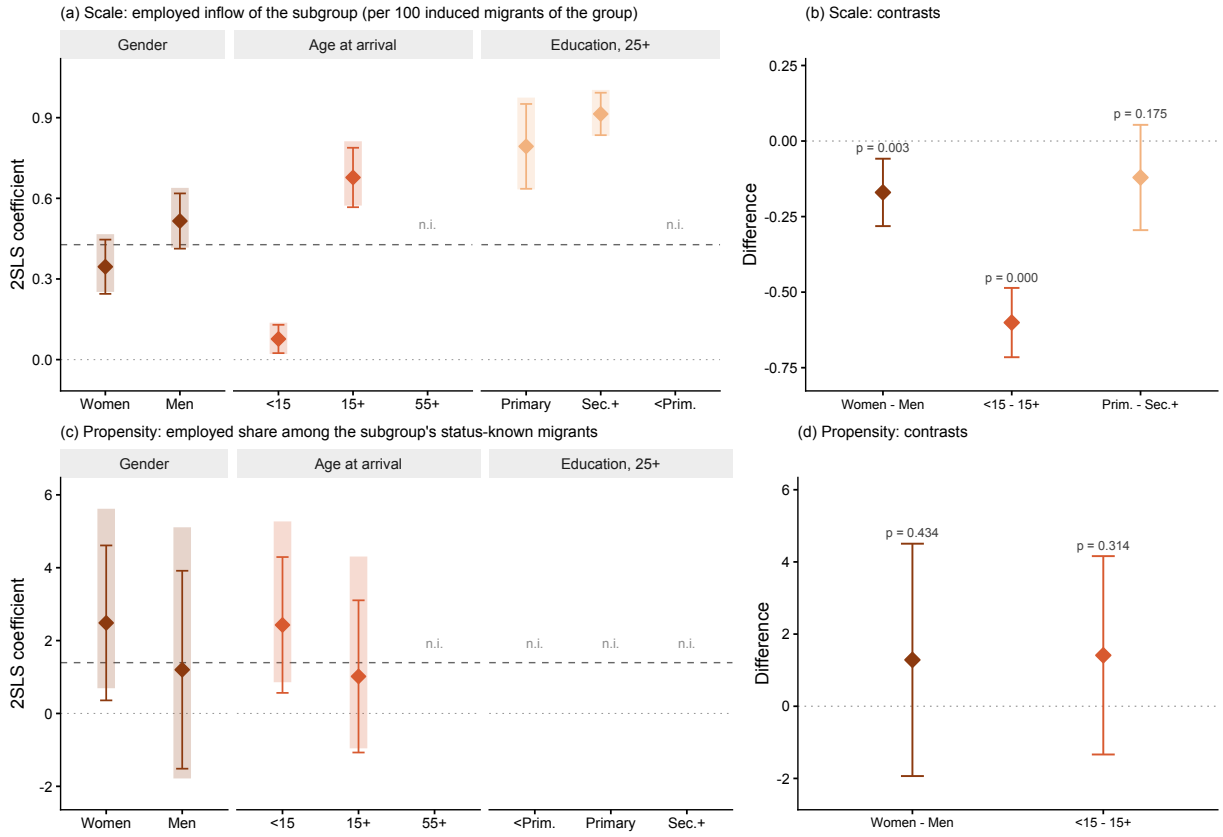
Notes: Decomposition of the additional low-quality housing inflow caused by a one-standard-deviation drought, $\Delta S \approx \Delta\text{MigRate} \times \overline{P} + \overline{\text{MigRate}} \times \Delta P$, per 100 urban residents at destination. The identity is an accounting split only if all four inputs come from the same sample, so both channels are estimated on the sub-sample where the propensity is defined: corridor-years with a positive urban migrant flow (N = 72,723). Panel A reports those two 2SLS estimates; column (1) is therefore *not* the paper's headline scale estimate, which is 0.28455 on the full panel (N = 241,540, Table 5). Restricting to positive-flow corridors conditions on the endogenous outcome and is not a defensible main specification; it is used here only so the two channels share a denominator. Panel B inputs, all from that sub-sample: $\Delta\text{MigRate} = \beta_{FS} \times SD(\text{drought}) = 0.00601 \times 0.503 = 0.00302$; $\overline{P} = 40.6\%$, the baseline share of in-migrants in low-quality housing; $\overline{\text{MigRate}} = 0.08908$; and $\Delta P = +0.37$ percentage points, the effect of the same shock on that share. The implied total (0.00156 per 100 residents) exceeds the direct estimate of 0.00097 on the same sample because the identity evaluates the scale channel at the *average* migrant's propensity (40.6%) rather than the *marginal* migrant's, which is the scale coefficient itself (32.0%). The decomposition therefore measures the relative weight of the two channels, not the level of the effect. Main specification throughout: origin–destination pair and arrival-year fixed effects, distance-to-census control, migration instrumented by drought intensity at origin, standard errors clustered at the origin district. AR: cluster-robust Anderson–Rubin confidence interval. Housing measure: definition F (five UN-Habitat dimensions, water excluded), unwinsorized; sample restricted to destinations with at least 2,000 urban residents. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Figure 17: Heterogeneous Effects by Migrant Profile



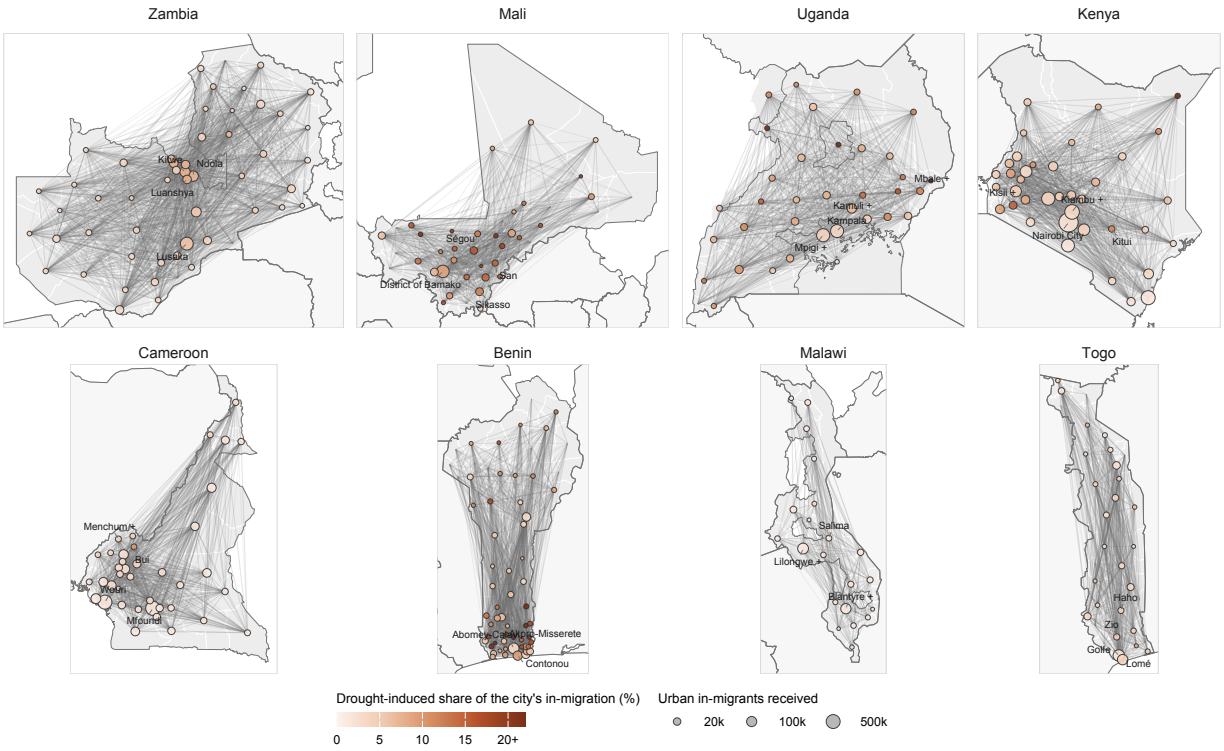
Notes: Panels (a)–(b): relative first stage - the effect of a 1-SD drought on the group's migration rate, as a % of the group's own mean (OLS, so valid regardless of instrument strength; dashed line: the aggregate +6.9%). Panels (c)–(f): subgroup-specific 2SLS under the main specification - the subgroup's low-quality-housing inflow per 100 urban residents (panel (c), unwinsorized; coefficients read per 100 induced migrants of the group) and the share of its migrants in low-quality housing (panel (e), positive-flow corridor-years), with the subgroup's migration rate instrumented by drought at origin. Whiskers: 95% Wald intervals; shaded bars: Anderson–Rubin 95%; dashed lines: the aggregate anchors (0.285, 1.216); shades of blue index the dimension. In the 2SLS panels, subgroups with conditional first-stage $F < 8$ appear as empty positions (n.i.) and are reported AR-only in Supplementary Appendix Tables S7–S8; hollow diamonds: $F < 10$; propensity first stages are weak throughout (F 8–13), so inference rests on the AR intervals. Panels (b), (d), (f): within-dimension contrasts from the saturated stacked interaction. Education is measured at census and may post-date arrival.
Source: Authors' computations using IPUMS census and CHIRPS climate data.

Figure 18: Heterogeneous Effects by Migrant Profile: Employment



Notes: As Figure 17, for the employment margins (the first stage, common to both figures, is in Figure 17(a)): the subgroup's employed inflow per 100 urban residents (panel (a)) and the employed share among its status-known migrants (panel (c)); dashed lines: the aggregate anchors (0.428, 1.395); shades of orange index the dimension; same display rule (empty n.i. positions: $F < 8$; hollow: F between 8 and 10; weak propensity first stages, F 9–13, AR inference). The age contrast in panel (b) is mechanical - children are rarely employed - and the under-15 column of panel (c) is child employment, conceptually thin. Full estimates: Supplementary Appendix Tables S9–S10.
Source: Authors' computations using IPUMS census and CHIRPS climate data.

Figure 19: Drought-Induced Share of Urban In-Migration by Destination, by Country



Notes: Circles mark destinations receiving at least 1,000 urban in-migrants (299 of 352); area is proportional to total in-migration, shade to the drought-induced share of it – the first-stage-implied inflow $\sum_{o,t} \hat{\beta}_{k(o,d)}\, \text{drought}_{o,t}\, \text{pop}_d^{urb}/100$ divided by the observed one. The share is bounded at destination level (p5 1.3%, median 4.5%, p95 18.7%) and capped at 22% on the scale; the corridor-level analogue is not usable. Grey lines show all 10,399 corridors with a positive urban flow, the darker arcs the 189 largest (top 25 per country). The first stage is estimated by within-country distance tercile rather than pooled, since one coefficient would give short and long corridors the same per-capita response: it falls from 0.00706 on the nearest third of corridors to 0.00323 and 0.00076. The mapped pattern is robust to this (rank correlation 0.935 with the pooled version), and the corrected allocation totals 430,494 drought-induced migrants against 400,139 pooled. The figure illustrates the geography of estimated inflows; the identified result is the bin-level 2SLS analysis of Figure S4. Centroids are interior points of administrative polygons, not city locations.

Source: Authors' computations from IPUMS census microdata, CHIRPS data and IPUMS harmonized shapefiles.

Table 30: Who Are the Drought-Induced Migrants? Selection on Observables

	Arrival-year drought (descriptive)			*Complier characterization (2SLS)*		
	No drought	High drought	Diff.	Drought-induced	All migrants	Diff.
	(1)	(2)	(3)	(4)	(5)	(6)
Share female (%)	50.5	50.9	+0.4	51.8	50.4	+1.4
Mean age at arrival (years)	21.0	20.3	-0.7***	16.4	21.1	-4.8***
Share below 15 at arrival (%)	29.1	30.3	+1.2	41.6	28.5	+13.1***
Share 15–29 at arrival (%)	52.5	52.2	-0.3	41.9	53.1	-11.2**
Share below primary, 25+ (%)	15.7	14.7	-1.0	11.2	15.2	-4.0
Share secondary or above, 25+ (%)	43.9	45.1	+1.2	47.0	44.7	+2.3
Migrants represented (millions)	5.72	1.29		11.60		

Notes: Characteristics of urban in-migrants, canonical estimation panel (8 countries, Egypt excluded; $N =$ 241,540 corridor-years; first-stage $F = 29.65$). Columns (1)–(3), descriptive: migrant-weighted means for arrivals in no-drought ($z = 0$) versus high-drought (top quartile of positive z) corridor-years; the difference is a migrant-weighted WLS contrast. Columns (4)–(6), complier characterization (Abadie, 2003): 2SLS of each characteristic-specific inflow rate (count per 100 urban residents, unwinsorized as in the main tables) on the migration rate instrumented by drought; column (4) is the ratio of coefficients and describes the IV's marginal drought-induced migrants – the same LATE population as the main result; the difference (6) is tested by 2SLS on the excess outcome. Age at arrival = census age minus years since migration. Education is measured at census and may post-date arrival; adults 25+ with known attainment (∼40–55% of migrants). SEs clustered at `admin_origin`. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

10.7 Robustness

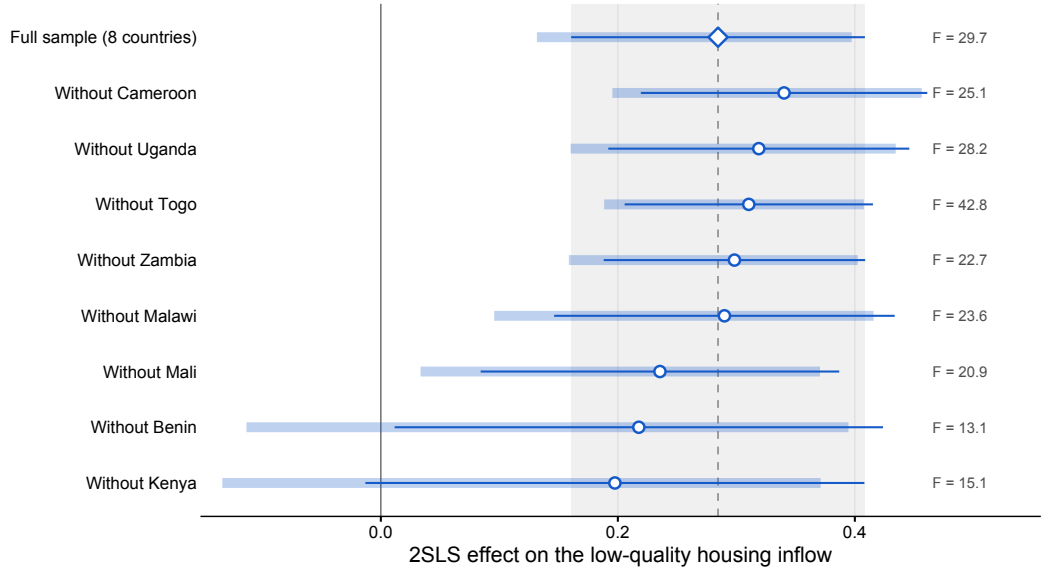
10.7.1 Leave-One-Country-Out

10.7.2 Housing Quality Gradient

10.8 Model

Figure 21 illustrates the two-state housing integration framework of Section 8.

Figure 20: Leave-One-Country-Out Robustness of the Main Effect



Notes: Each row re-estimates the main specification on the estimation sample with one country removed; the diamond is the full eight-country estimate and the shaded band its 95% confidence interval. Thin whiskers are cluster-robust Wald intervals, thick shaded bars the weak-instrument-robust Anderson–Rubin 95% intervals, and the annotation gives the first-stage F of each subsample. The outcome is the low-quality-housing inflow per 100 urban residents (main UN-Habitat definition, unwinsorized); migration is instrumented by drought intensity at origin under the main specification (origin–destination-pair and arrival-year fixed effects, distance-to-census control, standard errors clustered at `admin_origin`). The estimate ranges from 0.197 to 0.340 against 0.285 on the full sample; it stays positive in every subsample and every one of the eight falls inside the full-sample confidence interval, so no single country moves the magnitude. Precision, by contrast, depends on the countries that carry the first stage: dropping Benin or Kenya lowers the first-stage F to about 11 and 12 respectively, and in Benin, Kenya and Mali the Anderson–Rubin interval then includes zero. That reflects lost power rather than a change in the estimate, since the point estimates in those subsamples (0.094, 0.092, 0.134) sit inside the full-sample confidence interval.

Source: Authors' computations from IPUMS census microdata and CHIRPS climate data.

Table 31: Robustness: Scale and Propensity across Definitions of Low-Quality Housing

	Metal roofs deprived		*Metal roofs durable*		
	≥1 dim.	≥2 dim. (conservative)	≥1 dim. (main)	≥2 dim.	Nb. of deprived dimensions
	(1)	(2)	(3)	(4)	(5)
Panel A. Scale: definition-specific inflow per 100 urban residents					
Urban migration rate	0.94102***	0.30075***	0.28455***	0.13222***	-0.00382
	(0.04162)	(0.05247)	(0.06323)	(0.05065)	(0.00469)
First-stage F			29.65		
AR 95% CI	[0.86, 1.04]	[0.18, 0.40]	[0.13, 0.40]	[0.01, 0.23]	[-0.02, 0.00]
Effect of 1-SD drought (% mean)	+7.5%	+5.3%	+4.3%	+4.2%	−1.1%
Mean outcome	0.02275	0.01034	0.01196	0.00566	0.00065
Observations			241,540		
Panel B. Propensity: outcome measured among in-migrants					
Urban migration rate	0.41202	1.05043*	1.21591**	0.04763	0.94933
	(0.36910)	(0.56732)	(0.59790)	(0.40178)	(1.06501)
First-stage F			15.50		
AR 95% CI	[-0.32, 1.34]	[0.06, 2.69]	[0.17, 2.94]	[-0.87, 0.92]	[-1.27, 3.52]
Effect of 1-SD drought	+0.12 pp	+0.32 pp	+0.37 pp	+0.01 pp	+0.0029
Mean outcome	0.8508	0.3475	0.4064	0.1637	0.6620
Observations			72,723		
Origin-Destination FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Dist. to census control	Yes	Yes	Yes	Yes	Yes

Notes: Each column re-estimates both channels under one definition of low-quality housing (contemporaneous instrument, main specification; SEs clustered at `admin_origin`). Column (3) is the paper's main definition, the UN-Habitat operational criterion (≥1 deprivation, metal roofing durable). Panel A: inflow per 100 urban residents, unwinsorized. Panel B: share among in-migrants, unweighted. Column (5) uses the mean number of deprived dimensions instead of a share, so its coefficients and 1-SD effects are in dimension units. The first stage is common to all columns within a panel. Water is excluded from every definition: the harmonized IPUMS water-supply codes in these censuses cannot reliably separate deficient from adequate access. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 32: Crowding-Out: Does Drought-Induced Migration Degrade the Housing of Earlier Cohorts?

	Concurrent cohort		Earlier cohorts	
	Full sample	Restricted	Low-quality housing	Total inflow
	(1)	(2)	(3)	(4)
Urban migration rate	0.28455***	0.19956	-1.01066	0.24737
	(0.06323)	(0.12759)	(0.82838)	(1.38003)
First-stage F	29.65	10.80	10.80	10.80
AR 95% CI	[0.130, 0.398]	[-0.287, 0.396]	[-4.499, 0.138]	[-4.944, 2.403]
Mean outcome	0.01196	0.01252	0.04644	0.10137
Observations	241,540	219,589	219,589	219,589
OD pairs	16,846	16,846	16,846	16,846

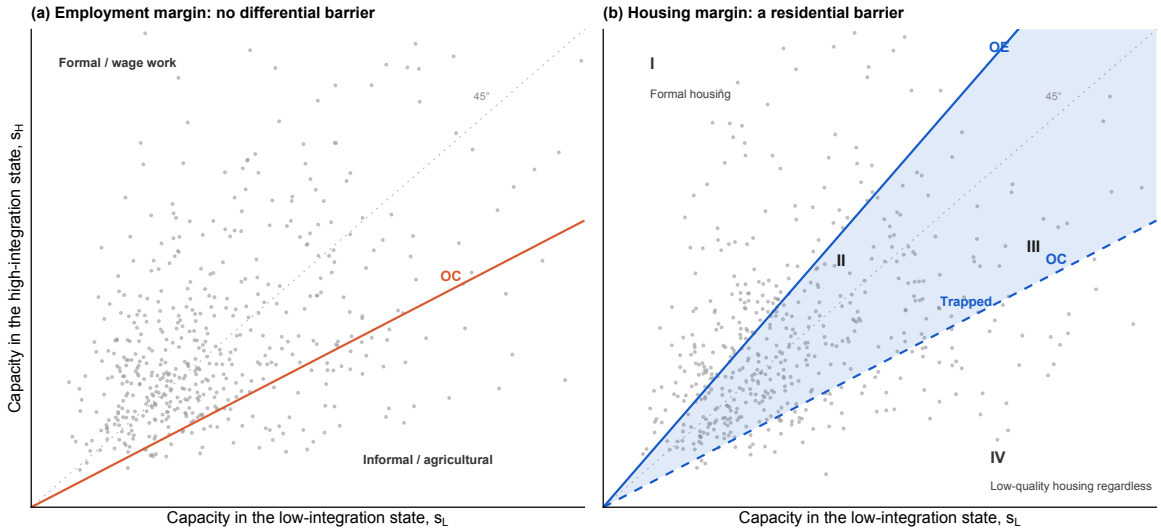
Notes: Test of whether drought-induced migration crowds previously settled migrants out of decent housing. All columns are the paper's main specification – the urban migration rate instrumented by drought intensity at origin, origin–destination pair and arrival-year fixed effects, distance-to-census control, standard errors clustered at the origin district. Column (1) is the headline scale estimate (Table 5). Column (2) repeats it on the corridors that have at least one earlier arrival cohort, the sample used in columns (3)–(4). Column (3) replaces the outcome with the low-quality-housing count of migrants who arrived on the same corridor in *earlier* years, per 100 urban residents; those migrants arrived before the shock and cannot be its compliers, so a positive coefficient would be crowding-out. Column (4) is a placebo on their total number, which no mechanism should move. The Anderson–Rubin interval in column (3) excludes any effect above 0.138 per marginal migrant. Note that this channel does not enter the headline outcome, which counts only the concurrent cohort: a null here shows that crowding-out does not spill onto settled migrants, not that it is absent within the arrival cohort (Table 33). $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 33: Crowding-Out within the Arrival Cohort: Evidence from Migrants Whose Inflow Does Not Respond to Drought

	Less than primary (1)	Primary (2)	Secondary or more (3)	All migrants (4)
Panel A: Does drought change how many of them arrive?				
Drought intensity	0.000142	0.000532***	0.000597***	0.003618***
	(0.000102)	(0.000107)	(0.000119)	(0.000664)
p-value	0.163	0.000	0.000	0.000
Panel B: Does their low-quality-housing propensity rise with drought-induced migration?				
Urban migration rate	1.25066	1.52444*	1.72953*	1.21591**
	(0.79559)	(0.82700)	(1.01358)	(0.59790)
First-stage F	13.78	11.57	6.15	15.50
AR 95% CI	[-0.210, 3.526]	[0.192, 4.448]	[0.324, 9.009]	[0.178, 2.942]
Baseline share	0.531	0.395	0.247	0.406
Observations	28,734	37,688	34,071	72,723

Notes: Test of crowding-out *within* the arrival cohort – the channel that would bias the headline estimate. Education is measured at census among adults aged 25 and over. Panel A regresses each group's own arrival rate per 100 urban residents on drought intensity at origin: the inflow of the least-educated group does not respond to drought (column (1), $p = 0.163$), whereas the other two do, so arrivals of that group in a drought year are almost entirely always-takers. Panel B regresses each group's share in low-quality housing on the *total* urban migration rate, instrumented by drought, under the main specification. If compliers crowded always-takers out of decent housing, column (1) would be positive. It is positive and of the same order as the aggregate propensity effect, but its Anderson–Rubin interval contains both zero and substantial crowding-out: the test is *inconclusive*, and we do not read it as a null. Column (4) reproduces the aggregate propensity estimate of Table 6 for reference. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Figure 21: The Two-Margin Integration Model: Employment versus Housing



Notes: **The figure is an illustration, not an estimate.** The scattered points are simulated draws and the slopes are chosen for legibility; no parameter is taken from the data, and the width of the shaded region carries no quantitative content. Each panel is a two-state Roy diagram for one integration margin, in the space of a migrant's capacities (s_L, s_H) in the low- and high-integration states. The panels show distinct, margin-specific draws, positively correlated across margins. Absent barriers, a migrant chooses the high state whenever $w_H s_H > w_L s_L$, that is, above the ray OC of slope w_L/w_H; a proportional barrier τ reduces the effective high-state return to $(1-\tau)w_H$ and rotates the cutoff to the steeper ray OE of slope $w_L/[(1-\tau)w_H]$. Panel (a), employment margin: the estimates find no differential barrier across destinations – the conversion of drought-induced migration into employment does not vary with city size – so the cutoff does not rotate and there is no trapped region. Panel (b), housing margin: a residential barrier opens the shaded fan between OC and OE, containing the migrants it misallocates. They clear the employment margin but remain in low-quality housing: the working poor of Table 11. Region I integrates despite the barrier, Region IV would take low-quality housing in any case, and the 45° line splits the fan into Region II, whose comparative advantage lies in the high state, and Region III, whose does not. Two features of the drawing are conventions rather than findings: the position of Region II above the 45° line presumes $(1-\tau)w_H < w_L < w_H$, and reading the shaded fan as the working poor presumes capacities are positively correlated across margins, which the cross-tabulation supports but the figure does not impose. Migrants below OC may also be employed and badly housed, but optimally so. In the paper's estimates the barrier is indexed by the destination city and by nothing else: neither climate status nor education indexes it, both having been tested and rejected in Section 8.

Source: Authors' illustration.

S1 Supplementary Appendix

S1.1 Exclusion Restriction: Further Tables

Table S1: Exclusion Restriction: Destination Shocks and the Propensity Margin

	Destination drought		Destination conflict		Destination crop prices	
	(1)	(2)	(3)	(4)	(5)	(6)
	No control	With control	No control	With control	No control	With control
Share of urban in-migrants in low-quality housing						
Urban migration rate	1.21591**	1.00212	1.83083**	1.82955**	1.13951*	1.16502*
	(0.59790)	(0.72609)	(0.90150)	(0.90228)	(0.59312)	(0.60277)
First-stage F	15.50	13.47	9.47	9.46	15.70	15.64
AR 95% CI	[0.17, 2.94]	[-0.44, 2.93]	[0.49, 5.69]	[0.49, 5.69]	[0.08, 2.81]	[0.11, 2.86]
Observations	72,723	72,723	53,168	53,168	72,109	72,109

Notes: 2SLS of the urban migration rate instrumented by contemporaneous drought intensity at origin, under the paper's main specification (origin–destination-pair and arrival-year fixed effects, distance-to-census control), with standard errors clustered at the origin district and cluster-robust Anderson–Rubin 95% intervals. **The instrument is origin drought in every column; every control is measured at the destination.** Each control forms a two-column group estimated on *its own* subsample, with the odd column reporting that subsample without the control: the comparison within a group is therefore a pure control effect, while differences across groups are differences of coverage. Destination drought is the instrument's construction applied to the destination unit (100% of the panel). Destination conflict is an indicator for any battle, act of violence against civilians or remote-violence event, plus $\log(1 + \text{fatalities})$, from ACLED matched to IPUMS units by point-in-polygon; ACLED begins in 1997 and reaches 70% of the panel. Crop prices are $\sum_c \text{share}_{cd} \log(P_{ct})$ and its first difference, with CROPGRIDS harvested-area shares for nine traded crops and World Bank Pink Sheet world prices (99.5% of the panel). The outcome is a share bounded in $[0, 1]$, so no ratio to the outcome mean is reported. **This is the paper's more fragile margin and the paired columns show why.** Within each group the control changes little – 1.216 to 1.002 under destination drought, 1.831 to 1.830 under conflict, 1.140 to 1.165 under crop prices – but the no-control columns differ sharply *across* groups, from 1.216 on the full panel to 1.831 on the ACLED subsample, so most of the apparent movement is coverage rather than confounding. The first-stage F on this sample is lower than on the scale sample in every column, between 9 and 16. Stars: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

S1.2 Multiple-Hypothesis Correction

S1.3 Robustness: Level of Clustering

S1.4 The Structure of Drought Shocks

Figure S1 documents the shock process behind Section 3.5 country by country. Panel (a) shows how many origin districts a drought reaches in a given year and how hard; panels (b) and (c) give the full distribution across destinations of the two synchrony measures summarised by their medians in Table 2.

Table S2: Exclusion Restriction: All Three Destination Controls Entered Jointly

	All three controls jointly	
	(1)	(2)
	No control	With controls
Panel A. Low-quality housing inflow		
Urban migration rate	0.24498***	0.17473**
	(0.05553)	(0.08422)
First-stage F	18.78	11.38
AR 95% CI	[0.11, 0.35]	[-0.11, 0.32]
Observations	168,820	168,820
Panel B. Employed inflow		
Urban migration rate	0.54363***	0.58392***
	(0.05424)	(0.07979)
First-stage F	18.78	11.38
AR 95% CI	[0.45, 0.69]	[0.46, 0.87]
Observations	168,820	168,820
Panel C. Share in low-quality housing		
Urban migration rate	1.75015**	3.05267*
	(0.88962)	(1.77002)
First-stage F	9.58	5.25
AR 95% CI	[0.40, 5.47]	[0.66, 20.72]
Observations	52,593	52,593

Notes: As Table 13, with destination drought, destination conflict and destination crop prices entered together, on the subsample where all three are observed (N = 168,820; the binding constraint is ACLED's 1997 start). **This is a stress test rather than an informative control.** The housing coefficient falls from 0.245 to 0.175 and stays significant at the 5% level, but the first-stage F drops from 13.88 to 11.38, and in the reduced form of Supplementary Appendix Table 14 the instrument itself loses significance on this specification ($p = 0.144$) while destination drought gains it ($p = 0.001$), having been indistinguishable from zero on its own ($p = 0.501$). The three destination shocks are correlated with origin drought – rainfall is spatially correlated and cropping shares are not independent of climate – so entered together they absorb a sizeable share of the identifying variation and trade significance among themselves. The employment result is unaffected (0.544 to 0.584). Stars: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table S3: Multiple-Hypothesis Correction on the Secondary Family of Destination Contrasts

	Contrast	Raw p	Romano–Wolf p	Bonferroni p
Panel A. First stage: volume of drought-induced migration ($\times 10^{-3}$)				
Past built-up expansion, 1975–1990	+3.05	0.0002	0.0050***	0.0038
Corridor distance	-4.83	0.0003	0.0075***	0.0054
Housing saturation at destination	-2.48	0.0083	0.0920*	0.1485
Baseline migration share at destination	+1.93	0.0155	0.1484	0.2784
Pre-period corridor network	+2.67	0.0383	0.2809	0.6896
Industrial employment share at destination	+1.89	0.0486	0.3098	0.8739
Panel B. 2SLS conversion into low-quality housing (per 100 marginal migrants)				
Industrial employment share at destination	+0.189	0.2350	0.8406	1.0000
Baseline migration share at destination	-0.052	0.5932	0.9890	1.0000
Corridor distance	+0.105	0.7255	0.9900	1.0000
Pre-period corridor network	-0.041	0.7658	0.9900	1.0000
Housing saturation at destination	+0.028	0.8527	0.9900	1.0000
Past built-up expansion, 1975–1990	-0.005	0.9615	0.9900	1.0000
Panel C. 2SLS conversion into employment (per 100 marginal migrants)				
Pre-period corridor network	+0.234	0.0256	0.1934	0.4600
Past built-up expansion, 1975–1990	-0.140	0.1232	0.5977	1.0000
Corridor distance	-0.194	0.2320	0.8406	1.0000
Baseline migration share at destination	+0.060	0.4231	0.9715	1.0000
Housing saturation at destination	-0.057	0.5483	0.9890	1.0000
Industrial employment share at destination	+0.054	0.5793	0.9890	1.0000

Notes: The family comprises 18 tests: six destination characteristics × three targets (the first stage and the two 2SLS conversions). Each contrast is the difference between the two bins of the saturated interacted IV of Section 6.1.3, split at the within-country median. **The city-size contrast is deliberately excluded**: it is the paper's pre-specified primary hypothesis (Section 6.1.2), tested on a single dimension chosen ex ante, and it does not belong to the exploratory family over which a stepdown correction is appropriate. Romano–Wolf p-values are computed by a stepdown procedure on 2,000 pairs-bootstrap draws resampling the 354 origin-district clusters, with the same cluster draw applied to every test within a replication so that the correlation across tests is preserved; this is what makes the correction markedly less punitive than Bonferroni, which is reported for comparison. Two contrasts survive at the 5% level, both on the volume margin: past built-up expansion and corridor distance. Housing saturation survives at 10%. None of the twelve conversion contrasts survives, which strengthens rather than weakens the paper's reading: the conversion of arrivals into low-quality housing and into employment does not vary with any destination characteristic in this family, and correcting for multiplicity can only reinforce a null. Stars refer to the corrected p-value: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table S4: Robustness of the Main Results to the Level of Clustering

	(1) admin_origin (main)	(2) Origin × year	(3) Two-way (origin + year)	(4) od_pair
Panel A. Scale: inflow per 100 urban residents at destination				
Low-quality housing inflow	0.28455***	0.28455***	0.28455***	0.28455***
	(0.06323)	(0.05909)	(0.08450)	(0.04932)
	[< 0.0001]	[< 0.0001]	[0.0018]	[< 0.0001]
Employed inflow	0.42760***	0.42760***	0.42760***	0.42760***
	(0.04330)	(0.04976)	(0.09043)	(0.03607)
	[< 0.0001]	[< 0.0001]	[< 0.0001]	[< 0.0001]
First-stage F	29.65	24.66	8.08	90.48
Observations		241,540		
Panel B. Propensity: share among urban in-migrants				
Share in low-quality housing	1.21591**	1.21591*	1.21591*	1.21591**
	(0.59790)	(0.64169)	(0.70369)	(0.55078)
	[0.0427]	[0.0582]	[0.0923]	[0.0273]
Share employed	1.39452**	1.39452**	1.39452*	1.39452**
	(0.61688)	(0.63927)	(0.71685)	(0.58343)
	[0.0244]	[0.0292]	[0.0594]	[0.0169]
First-stage F	15.50	12.44	7.53	32.49
Observations		72,723 / 71,503		
Number of clusters	354	5,410	354 / 38	16,846

Notes: Each cell reports the 2SLS coefficient on the urban migration rate instrumented by contemporaneous drought intensity at origin, under the paper's main specification (origin–destination-pair and arrival-year fixed effects, distance-to-census control), with standard errors in parentheses and p-values in brackets. The only thing that varies across columns is the level at which standard errors are clustered, so **the coefficients are invariant by construction**; column (1) is the specification used throughout the paper. The scale results are robust at $p < 0.0001$ under every clustering level. The housing propensity is the most fragile margin, moving from $p = 0.043$ in column (1) to $p = 0.092$ under two-way clustering, consistent with it being the paper's least robust result; the employment propensity holds at the 10% level throughout. Column (2) clusters at the origin-district × year cell: it is *finer* than column (1), permitting arbitrary correlation within an origin-year but imposing independence across years within an origin, which rainfall autocorrelation makes difficult to defend; we report it because it is a natural question, not as an alternative to column (1). Column (3) is the demanding check, since it additionally allows arbitrary correlation across all origins within a year – the spatial synchrony of shocks documented in Section 3.5, where the weighted pairwise correlation of drought across a destination's origins has a median of 0.67. All four results survive it. The first-stage F falls to 8.08 in column (3) not because the instrument weakens but because the year dimension supplies only 38 clusters, near the limit at which cluster-robust inference is reliable. Column (4) clusters at the corridor, which ignores correlation across corridors sharing an origin and is reported only as a lower bound on the standard errors. Stars: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

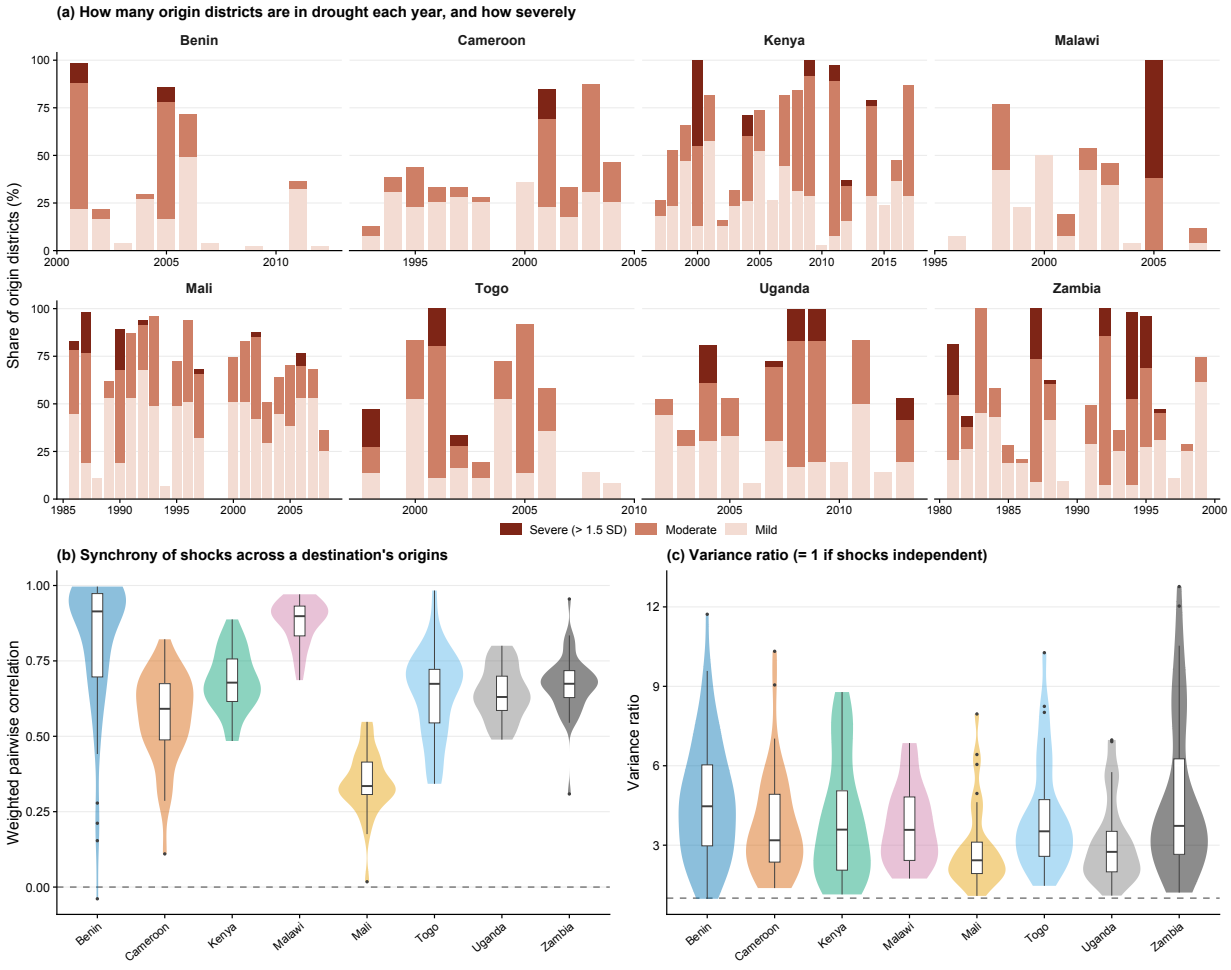


Figure S1: The Extent, Severity and Synchrony of Drought Shocks Across a Country's Origin Districts

Notes: Panel (a) reports, for each country and each year of its retrospective window, the share of origin districts recording a rainfall deficit, split by severity: *mild* is $0 < \text{drought}_{ot} \leq 0.643$, the median of the strictly positive values of the instrument; *moderate* is $0.643 < \text{drought}_{ot} \leq 1.5$; *severe* is $\text{drought}_{ot} > 1.5$, the threshold already used to define severe drought in the construction of the climate table. Bar height is therefore the extent of the shock and the stacking its intensity. The complement of each bar is the share of origins at or above normal rainfall, which is 50.1% of origin-years across the panel because the instrument is censored at zero; mild, moderate and severe account for 25%, 21% and 4%. Nine country-years reach 100% of origins in drought. Panels (b) and (c) report, by country, the distribution across destinations of the two synchrony measures of Table 2: the $w_{od}w_{o'd}$-weighted average pairwise correlation of drought across a destination's origins, and the variance ratio $\text{Var}_t(\sum_o w_{od}\,\text{drought}_{ot})/\sum_o w_{od}^2\,\text{Var}_t(\text{drought}_{ot})$, which equals one when origin shocks are independent; boxes are interquartile ranges with medians, violins are kernel densities of equal width, and the dashed lines mark the independence benchmarks of 0 and 1. Every country lies well above both benchmarks. Mali is the least synchronous on both measures, which is why its drought-attributable migration is the least temporally concentrated in Figure 1.

S1.5 Descriptive Statistics

S1.5.1 Migration Data: Fifteen-Country Sample

Figure S2 reports average in-migration rates over time for the full set of fifteen IPUMS countries, using five-year migration brackets for cross-country comparability; the yearly rates for the eight-country estimation sample are in Appendix Figure 6.

S1.5.2 Urbanization Data

Figure S3 reports the cumulative distributions of the census urban share and of Africapolis city size at destination, separately by country. The height of a country's curve at the 0.15 and 0.55 thresholds gives the share of its destinations in each urbanization bin; the curves confirm that the lower bins are populated by all countries rather than driven by any single one, with Malawi and Mali most concentrated at low urban shares.

S1.6 Heterogeneity by the Census Urban Share

This section repeats the destination heterogeneity of Section 6.1.2 on the census urban share instead of Africapolis agglomeration size, as a complementary check. Destinations are split into three bins by their urban population share at the most recent census: rural-like (< 0.15), urbanizing cities ($[0.15, 0.55)$) and main cities (≥ 0.55). The specification is identical to the one in the body.

Figure S4 and Table S5 report the estimates. The direction matches the size-based result: the coefficient declines as destinations urbanize, from 0.324 in rural-like destinations to 0.258 in urbanizing cities and 0.131 in main cities, the last significant only at the 10% level. The magnitudes are also consistent with the body, since the rural-like and urbanizing bins together cover much of what Africapolis classifies as small towns and secondary cities. What this classification does not deliver is inference: no pairwise contrast along the gradient is significant – urbanizing versus rural-like $p = 0.515$, urbanizing versus main cities $p = 0.128$, rural-like versus main cities $p = 0.119$ – and the urbanizing bin does not differ from the other two pooled ($p = 0.596$). For employment the ordering is different, with absorption lowest in

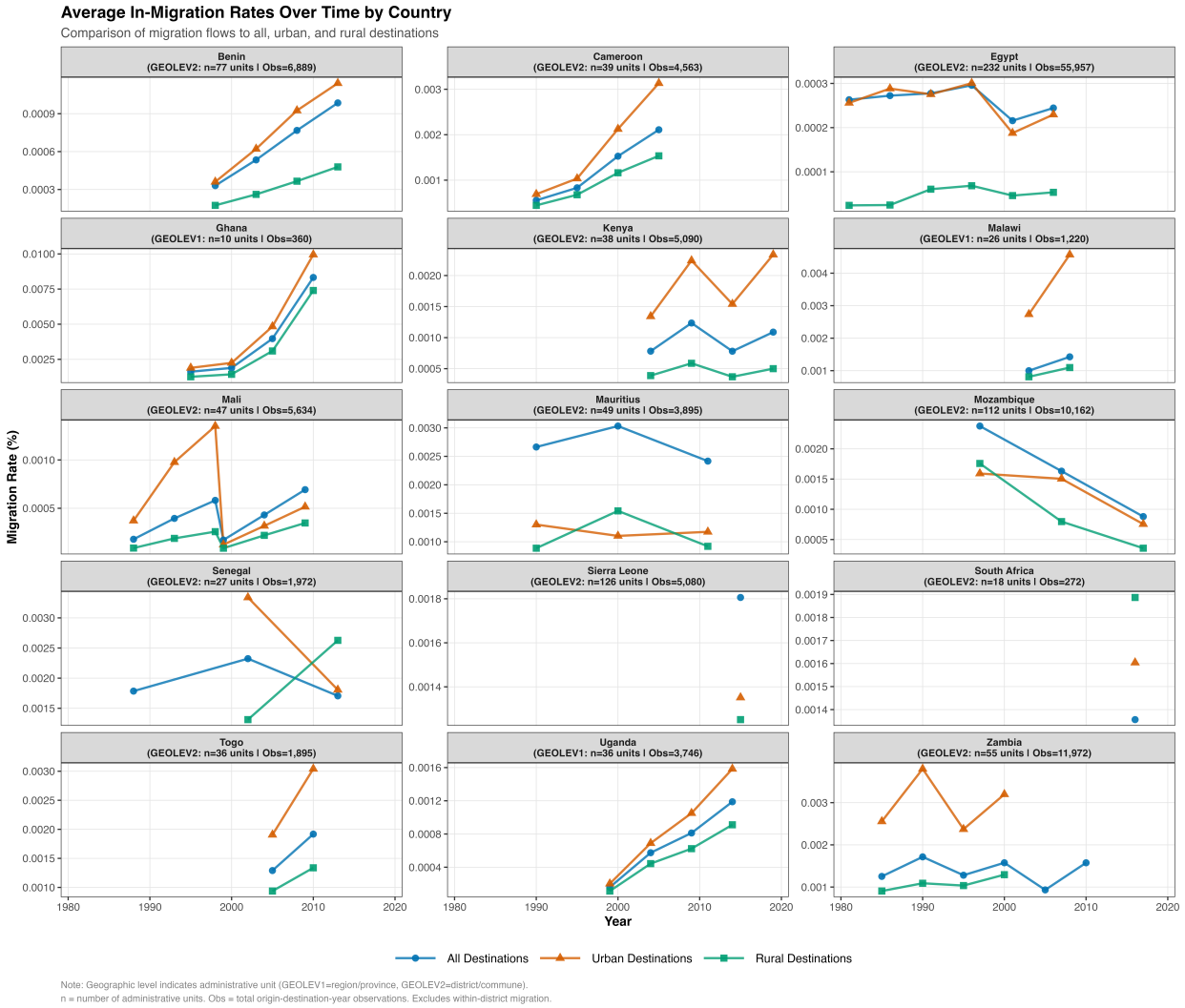
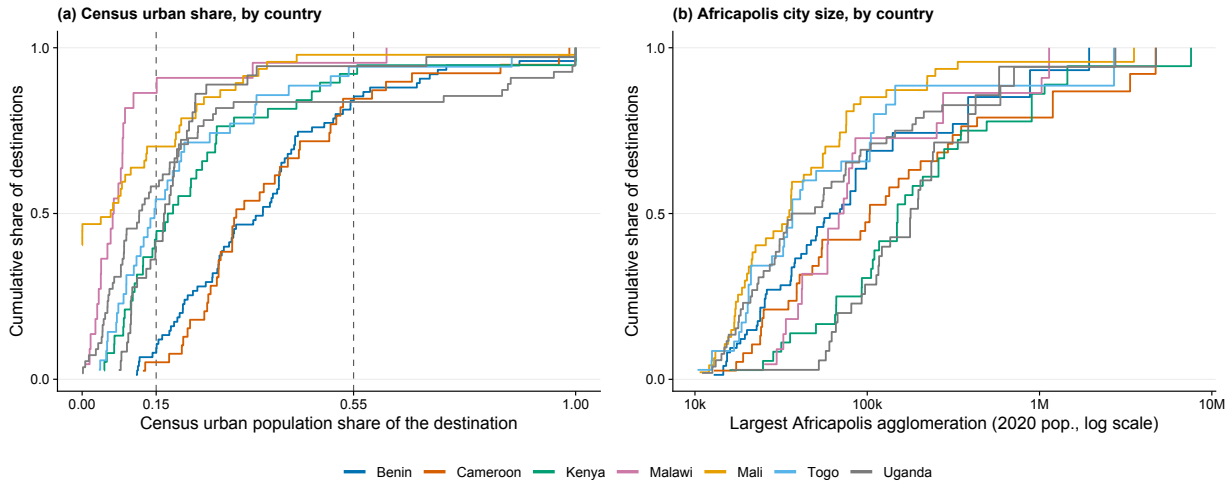


Figure S2: Average In-Migration Rates Over Time by Country.

Notes: This figure displays mean in-migration rates across administrative units for 15 African countries, disaggregated by destination type. Migration flows are calculated from IPUMS census microdata, measuring the number of in-migrants (weighted by `perwt`) divided by the destination population, aggregated into 5-year brackets. The panel titles show the geographic level (GEOLEV1 = region/province, GEOLEV2 = district/commune), the number of administrative units, and total observations. Note that Y-axis scales vary across countries to facilitate within-country comparisons. Countries with a single census year (Sierra Leone, South Africa) provide limited temporal variation. The figure reveals substantial heterogeneity in migration patterns, with urban destinations consistently attracting higher rates than rural areas, particularly in rapidly urbanizing countries like Ghana, Togo, and Uganda.

Source: Authors' elaboration based on IPUMS census data.

Figure S3: Distributions of Census Urban Share and Africapolis City Size, by Country



Notes: The figure reports empirical cumulative distribution functions across destinations, separately by country. Panel (a) shows the destination's urban population share in the most recent census. At each dashed threshold, 0.15 and 0.55, the height of a country's curve gives the proportion of its destinations below that threshold. Panel (b) shows the 2020 population of the largest agglomeration in the Africapolis 2024 database located within, or within 10 km of, each destination unit, on a logarithmic scale. Destinations without a matched Africapolis agglomeration of at least 10,000 inhabitants are excluded from panel (b). Four Africapolis agglomeration records reporting conurbation-cluster totals are corrected as described in the notes to Table 19.

Source: Authors' computations using IPUMS census data and the Africapolis 2024 database.

Table S5: Heterogeneity by Destination Urbanization: 3-Bin Interacted IV Estimates

	Low-quality housing inflow			*Employed inflow*		
	Rural-like†	Urbanizing cities	Main cities	Rural-like†	Urbanizing cities	Main cities
	(1)	(2)	(3)	(4)	(5)	(6)
Urban migration rate	0.3239***	0.2579***	0.1305*	0.4732***	0.3651***	0.4947***
	(0.0994)	(0.0629)	(0.0779)	(0.0714)	(0.0497)	(0.0602)
SW conditional F	14.01	24.17	11.62	14.01	24.17	11.62
AR 95% CI	[0.067, 0.524]	[0.102, 0.372]	[-0.123, 0.267]	[0.358, 0.703]	[0.263, 0.474]	[0.381, 0.674]
Mean outcome	0.0153	0.0096	0.0116	0.0122	0.0086	0.0139
Observations	91,019	122,821	27,700	91,019	122,821	27,700
Urbanizing − rural-like		$p = 0.515$			$p = 0.152$	
Urbanizing − main cities		$p = 0.128$			$p = 0.043$	
Urbanizing − rest		$p = 0.596$			$p = 0.092$	

Notes: 3-bin interacted IV: the urban migration rate interacted with each destination-urbanization bin (share urban at the most recent census; zones: rural-like < 0.15, urbanizing cities $[0.15, 0.55)$, main cities ≥ 0.55), instrumented by contemporaneous drought intensity interacted with the same indicators; origin–destination and arrival-year × bin fixed effects, bin-specific `dist_census` control, SEs clustered at `admin_origin`. Outcomes per 100 urban residents (no outcome is winsorized). SW F: Sanderson–Windmeijer conditional first-stage F; † weak first stage ($F < 10$) – the AR interval carries the inference there. The first stage is common to both outcomes (identical sample); per-bin coefficients 4.23 / 2.75 / 3.12 ($\times 10^{-3}$). Contrast p-values from the same system's clustered covariance; "rest" pools the rural-like and main-cities bins. Sample: corridors with non-missing destination urbanization ($N = 193{,}290$). $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

urbanizing cities (0.365 against 0.473 and 0.495), and the urbanizing–main-cities contrast is the one significant difference here ($p = 0.043$).

The reason we do not use this classification in the body is that its inference depends on where the rural/urban cutoff is placed, and the urban-share distribution offers no break to place it at. Across 351 destinations the share rises smoothly to a mode near 0.09–0.10 and declines slowly thereafter; the largest gap between consecutive destination shares below 0.35 is 0.014, against a median gap of 0.0008, so there is no separation anywhere in the range. Moving the lower cutoff therefore moves the contrasts without any data-driven reason to prefer one position:

Lower cutoff	0.05	0.10	0.15	0.20	0.25
Secondary bin (N)	186,843	151,097	122,821	93,884	71,652
Housing: secondary − main cities	+0.167	+0.183	+0.127	+0.146	+0.117
p-value	0.065	0.040	0.128	0.099	0.194
Employment: secondary − main cities	−0.068	−0.077	−0.130	−0.170	−0.139
p-value	0.325	0.220	0.043	0.006	0.026

The housing contrast crosses the 5% level at one cutoff only, and the employment contrast does so at a different and disjoint set of cutoffs. We therefore report the census-urban-share bins at the conventional 0.15 threshold, treat the declining gradient as directional evidence consistent with the body, and rest the inference on the size-based classification, which requires no such choice. The rural-like bin is also weakly identified here (Sanderson–Windmeijer conditional $F = 14.0$ against 24.2 in the urbanizing bin), so its Anderson–Rubin interval carries the inference.

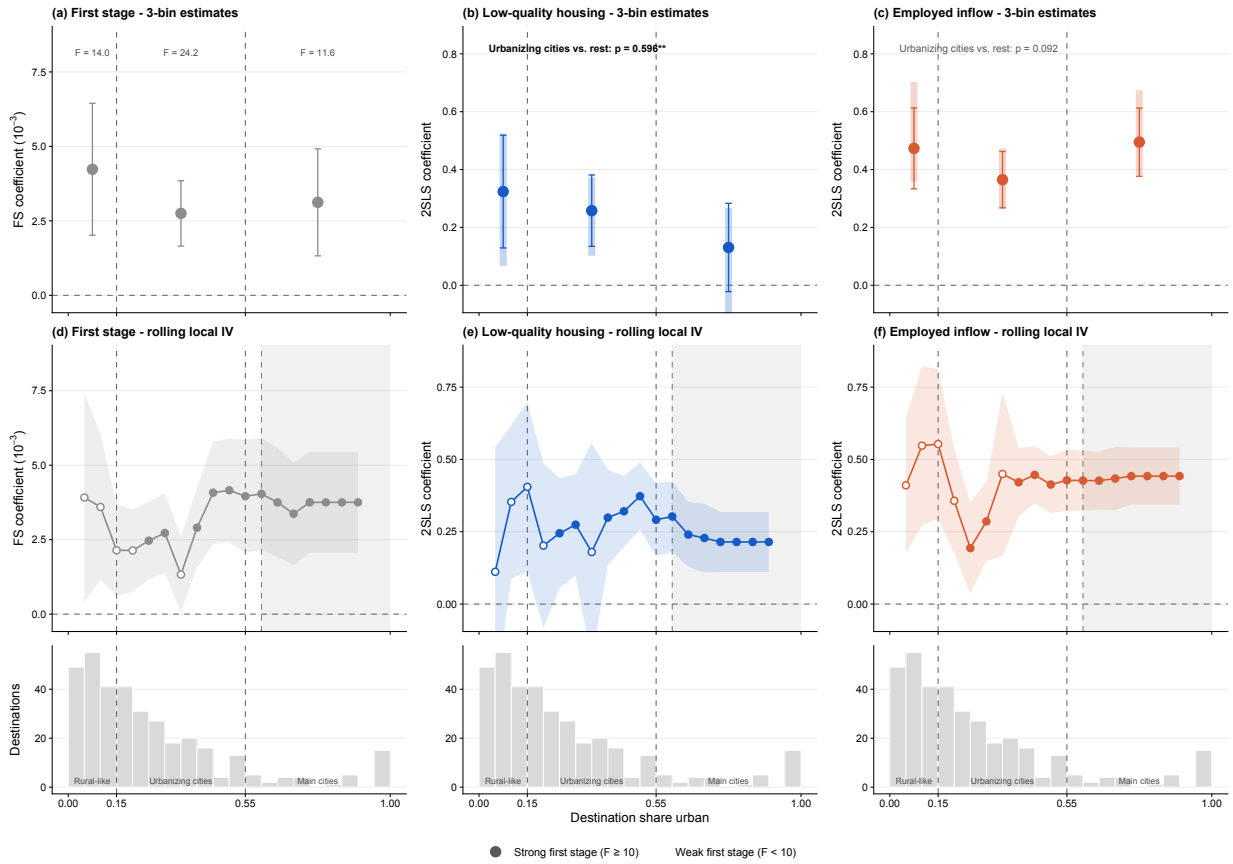
Figure S5 reports the urbanization heterogeneity for the propensity (sorting) margin.

Job Type and Sector of Drought-Induced Migrants - Figure S6 plots the composition gaps of Tables 9 and 10: for each class of worker and each sector, the difference in percentage points between the job-type composition of the marginal drought-induced migrants and that of the average urban in-migrant of the same sample. Two gaps are distinguishable from zero – less unpaid family work and more agriculture – and the wage/self-employed split is not.

Heterogeneity by Destination Characteristic: Full Estimates - Table S6 reports, for each of the six destination characteristics summarised in Figure 5, the first-stage and 2SLS coefficients in each bin and the contrast between them. Figures 2–S15 then give the full estimates characteristic by characteristic, each with its first stage, its two reduced forms, a rolling local-window counterpart and the distribution of destinations.

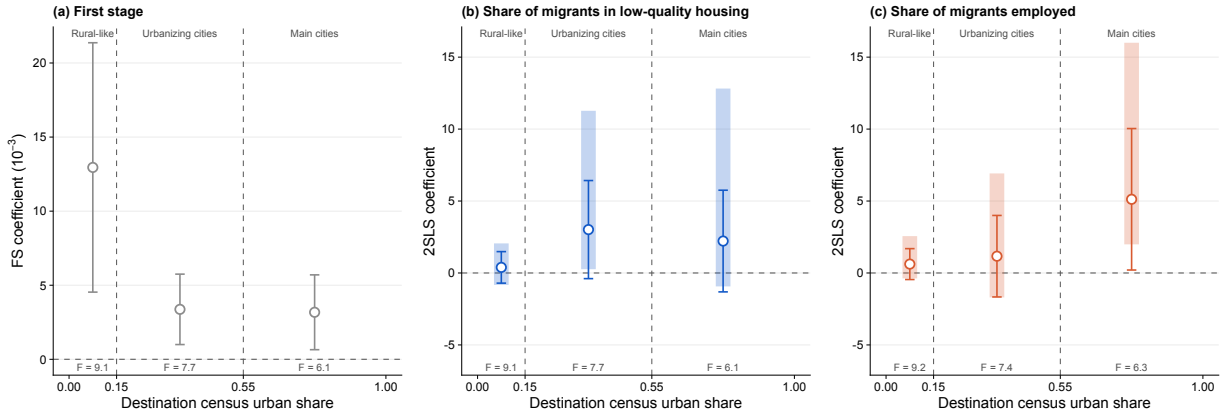
Conversion by Destination Characteristic - Figures S11–S13 report the 2SLS counterparts of the reduced forms of Section **??**: for each of the three destination characteristics, the number out of 100 marginal drought-induced migrants who enter low-quality housing and who are in employment, by half of the within-country distribution. No contrast is significant on either outcome. The industrial split is the informative null, both halves having a first stage above ten; the saturated half of the housing split and the long-distance half of the

Figure S4: Heterogeneity by Destination Urbanization: Low-Quality Housing and Employment



Notes: Heterogeneity by destination urbanization (urban population share at the most recent census; zones: rural-like < 0.15, urbanizing cities $[0.15, 0.55)$, main cities ≥ 0.55). Panels (a)-(c): first-stage and 2SLS estimates from a single specification saturated by urbanization bin -the migration rate interacted with each bin indicator, instrumented by drought intensity interacted with the same indicators, with origin–destination and arrival-year $\times$ bin fixed effects and bin-specific distance-to-census controls; standard errors clustered at the origin district. Estimates are plotted at bin midpoints; whiskers are 95% Wald intervals and shaded bars Anderson-Rubin 95% intervals; outcomes in panels (b)–(c) are the low-quality-housing and employed inflows per 100 urban residents. For housing, no pairwise contrast is significant: urbanizing $-$ rural-like $p = 0.515$, urbanizing $-$ main cities $p = 0.128$, rural-like $-$ main cities $p = 0.119$; the urbanizing bin does not differ from the other two pooled ($p = 0.596$). For employment, the urbanizing bin lies significantly below main cities ($p = 0.043$); the other contrasts are not significant. Panels (d)–(f): rolling-window local IV – the main specification re-estimated on corridors into the $\sim$60 destinations nearest each urban-share grid point; pointwise 95% Wald bands; indicative only, as windows overlap. The shaded area marks where the window no longer slides (too few destinations above 0.55); the binned estimates in panels (a)–(c) carry the inference there. Hollow points flag a weak first stage (Sanderson–Windmeijer conditional $F < 10$; bin values 14.0 / 24.2 / 11.6). Bottom row: distribution of destination urban shares with the three zones labeled. $N = 241{,}540$ corridor-years (91,019 / 122,821 / 27,700 across bins).

Figure S5: Heterogeneity by Destination Urbanization: Migrant Propensity



Notes: The figure reports estimates from a three-bin, fully interacted IV specification examining heterogeneity in migrant propensity by destination urbanization. Propensity outcomes are defined only for corridor-years with positive migrant flows; the samples contain $N = 55{,}698$ observations for housing and $N = 55{,}323$ for employment. Panel (a) reports the first-stage estimates and the corresponding Sanderson–Windmeijer conditional F-statistics. The conditional F-statistic is below 10 in each bin, as indicated by hollow markers; estimates in panels (b) and (c) should therefore be interpreted cautiously. Whiskers report conventional 95% Wald confidence intervals, while shaded bars report weak-instrument-robust 95% Anderson–Rubin confidence intervals. Intervals extending beyond the displayed axis range are truncated. The Anderson–Rubin interval excludes zero only for the share of migrants living in low-quality housing in urbanizing cities, with a confidence interval of $[0.81, 14.2]$. This result is consistent with the scale heterogeneity reported in Figure S4, although none of the pairwise contrasts is statistically significant. The specification otherwise matches that used in Figure S4.

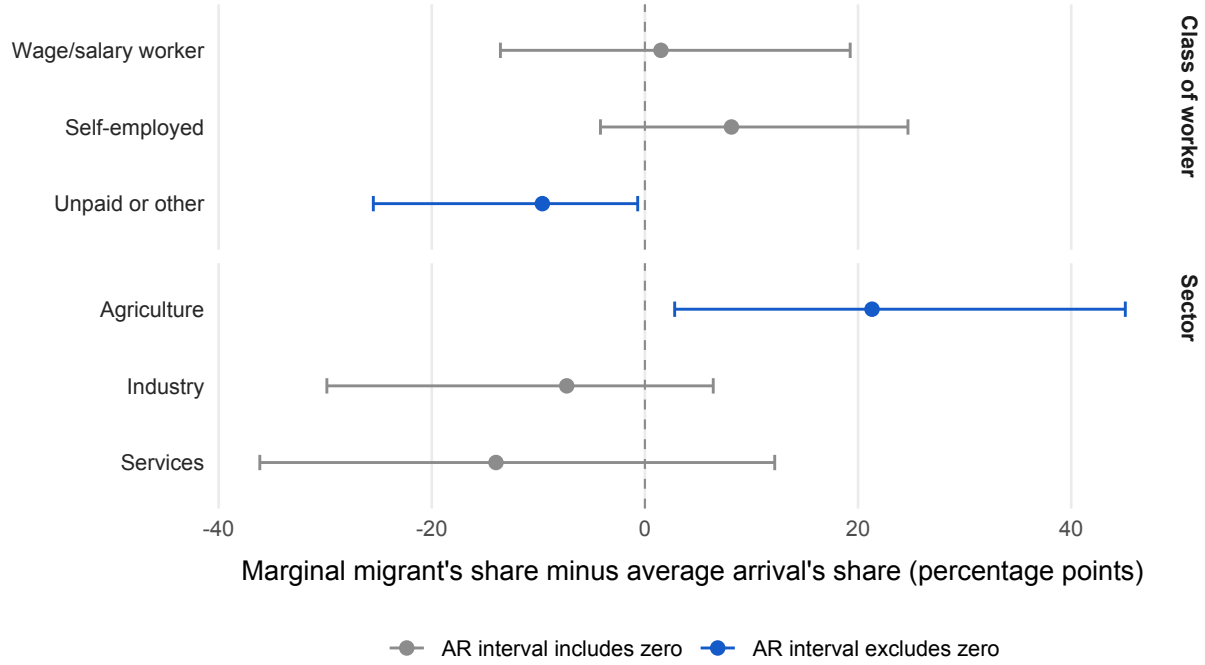
Source: Authors' computations using IPUMS census and CHIRPS climate data.

Table S6: Destination Characteristics: Volume Moves, Conversion Does Not

	First stage ($\times 10^{-3}$)				2SLS, low-quality housing (per 100)		
Destination characteristic	Low	High	Contrast	p	Low / High	Contrast	p
City size: metropolises vs. rest	3.35	5.21	+1.86*	(0.095)	30.6 / 10.0	-20.6**	(0.032)
Distance: far vs. near	5.79	0.95	-4.83***	(0.000)	18.6 / 29.1	+10.5	(0.726)
Housing saturation: high vs. low	4.62	2.14	-2.48***	(0.008)	24.2 / 27.0	+2.8	(0.853)
Industrial base: high vs. low	2.64	4.53	+1.89**	(0.049)	16.9 / 35.8	+18.9	(0.236)
Past built-up growth: fast vs. slow	2.11	5.16	+3.05***	(0.000)	28.8 / 28.3	-0.5	(0.962)
Baseline migration share: high vs. low	2.67	4.59	+1.93**	(0.016)	31.7 / 26.5	-5.2	(0.594)

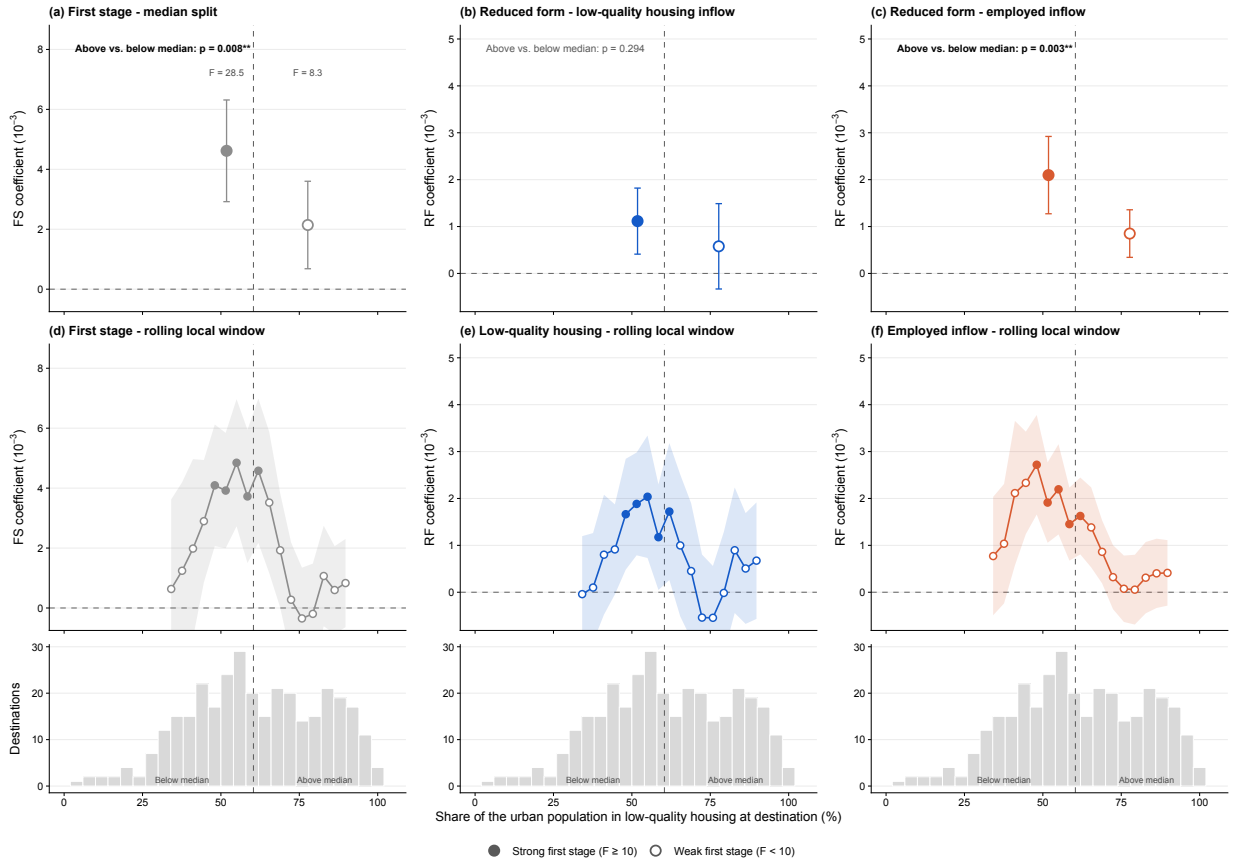
Notes: Each row is a two-bin saturated interacted IV on one destination characteristic (instrument × bin, origin–destination and arrival-year × bin fixed effects, bin-specific distance-to-census control, SEs clustered at the origin district). Splits are at the within-country median, except city size, where the pre-specified contrast is metropolises (≥ 1M, Africapolis) against the other two zones pooled. Columns 2–5 report the first-stage coefficient in each bin and their difference: how much drought-induced migration a destination of that type receives. Columns 6–8 report the 2SLS coefficient in each bin and their difference: how many of 100 marginal migrants enter low-quality housing there. The employment conversion, not shown, is insignificant in every row ($p = 0.85$, 0.23, 0.55, 0.58, 0.12 and 0.42 in the order of the table). Samples: N = 240,340; 157,556; 230,676; 215,556; 241,540; 241,540. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Figure S6: Do Drought-Induced Migrants Hold a Different Kind of Job?



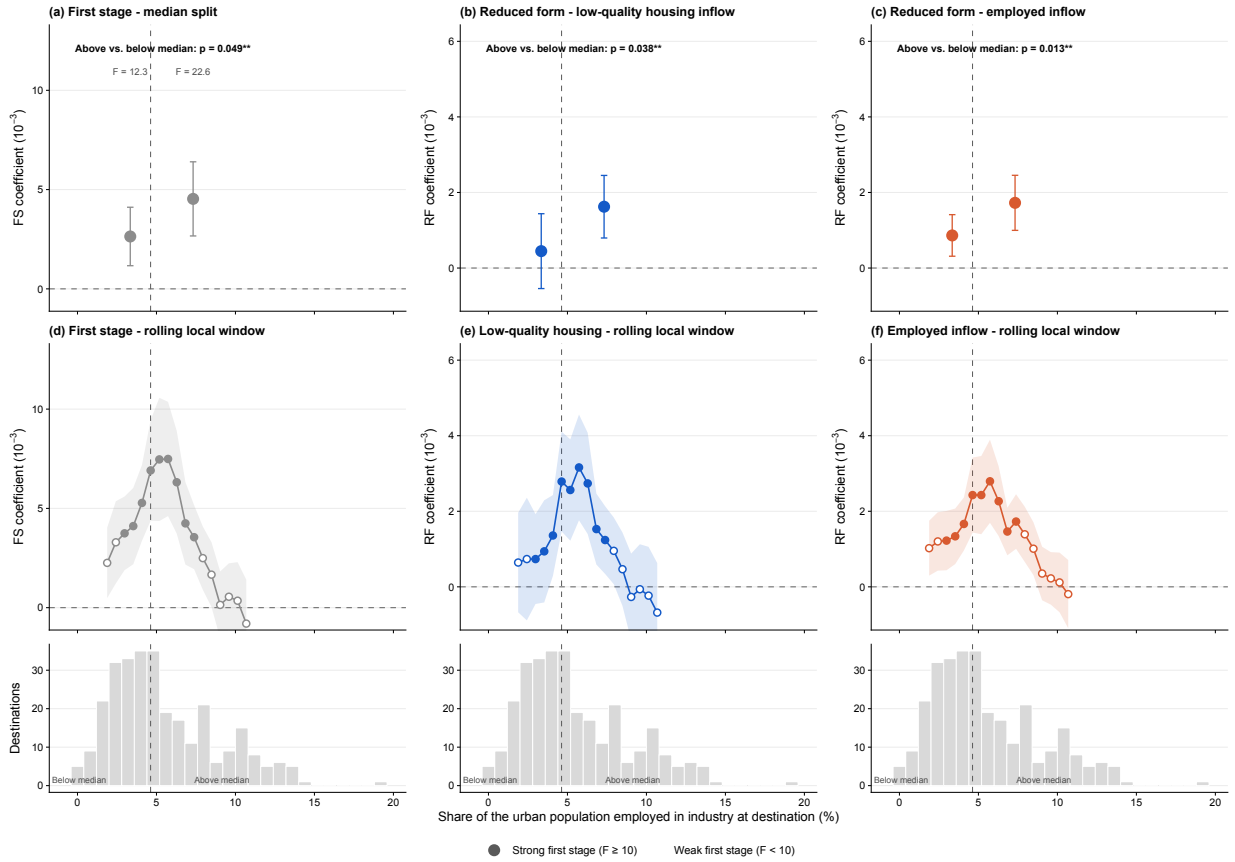
Notes: Difference, in percentage points, between the job-type composition of the marginal (drought-induced) migrants and that of the average urban in-migrant. The marginal share is the ratio of the category's 2SLS coefficient to the coefficient on the category-known total, both from the regressions of Tables 9 and 10; the average share is the migrant-weighted share among all urban in-migrants of the same sample. Instrument: contemporaneous drought intensity at origin; origin–destination-pair and arrival-year fixed effects, `dist_census` control, SEs clustered at `admin_origin`. Class of worker: 10 census-samples, $N = 224{,}668$, $F = 15.85$; sector: 8 census-samples, $N = 209{,}548$, $F = 10.97$. The sector panel is suggestive: dropping Zambia or Mali leaves the implied composition unchanged but reduces the first stage to $F = 2.66$ and 3.71.
Source: Authors' computations using IPUMS census and CHIRPS climate data.

Figure S7: Where Drought Sends Migrants: Destinations by Housing Saturation



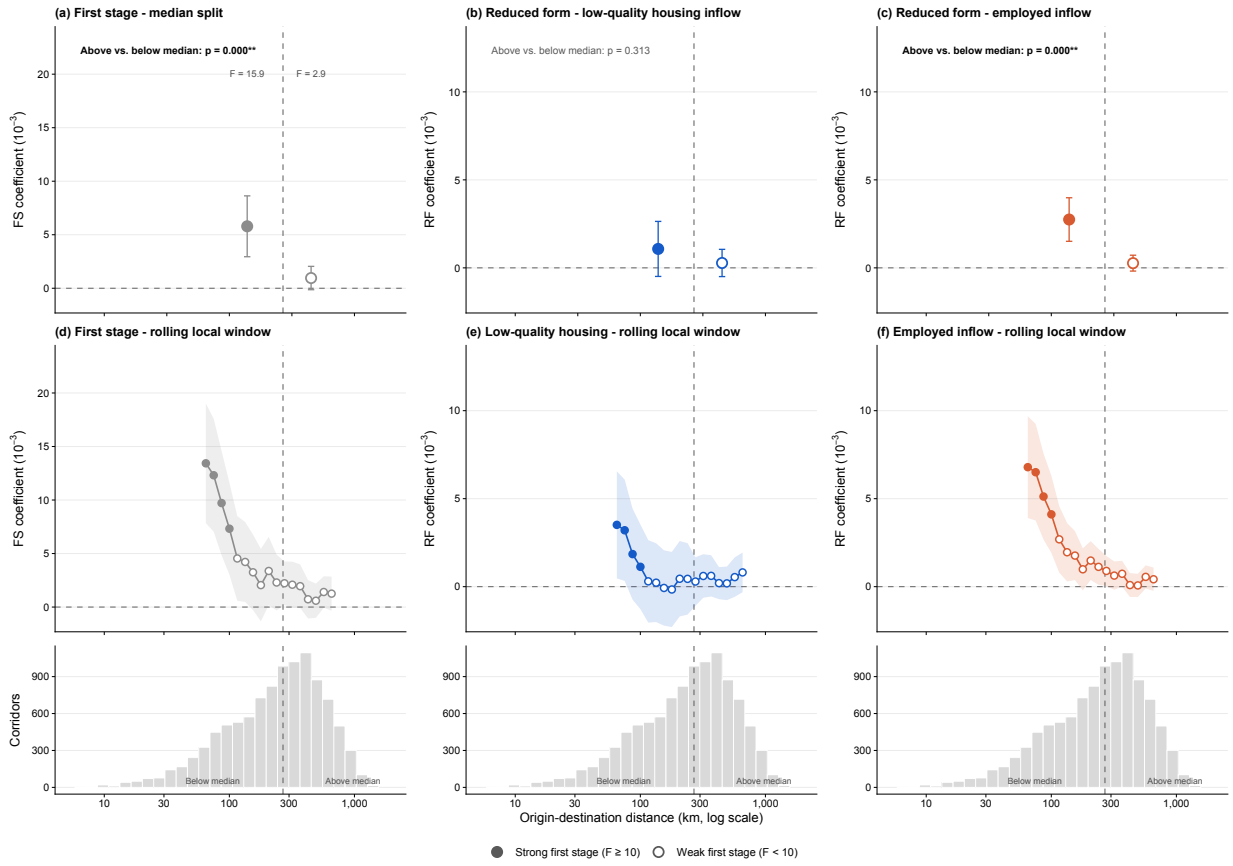
Notes: Destinations are split at the *within-country* median of the share of their urban population living in low-quality housing (UN-Habitat definition, at least one dimension deprived, metal roofs treated as durable), measured on all urban residents at the last usable census of each country (Zambia 2000 rather than 2010, whose urban variable is entirely missing). Panel (a): first stage, the effect of drought intensity at origin on the urban migration rate. Panels (b) and (c): reduced forms on the inflow of urban in-migrants in low-quality housing and on the inflow of employed urban in-migrants, both per 100 urban residents at destination. All three come from a two-bin saturated interacted model (instrument × bin, origin–destination and arrival-year × bin fixed effects, bin-specific distance-to-census control, SEs clustered at the origin district); coefficients are reported $\times 10^{-3}$ and whiskers are 95% Wald intervals. Panels (d)–(f) re-estimate the same three quantities on a rolling local window – at each grid point, the 60 destinations whose saturation rate is closest to it – with pointwise 95% Wald bands; the window slides over the whole grid, and these estimates are indicative only, the inference being carried by the binned contrasts. Hollow points: weak first stage ($F < 10$), reported once per half in panel (a); with disjoint bin indicators the Sanderson–Windmeijer conditional F has $\mathrm{df}_1 = L - K + 1 = 1$ and coincides with it. The dashed vertical line marks the pooled median and is indicative only, the split being within country. $N = 230{,}676$ corridor-years, 331 destinations, 8 countries. Bottom row: distribution of the classifier across destinations.
Source: Authors' computations using IPUMS census and CHIRPS climate data.

Figure S8: Where Drought Sends Migrants: Destinations by Industrial Employment Share



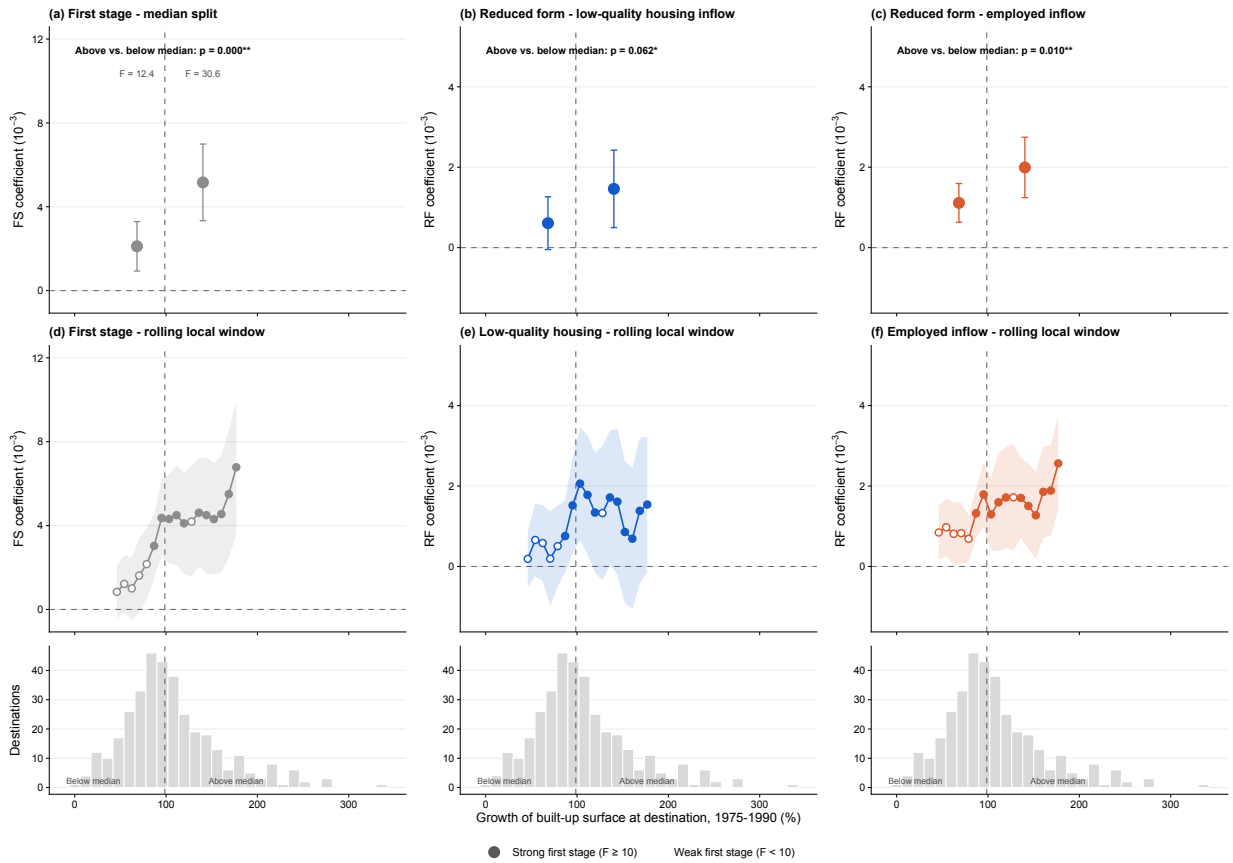
Notes: Destinations are split at the *within-country* median of the share of their urban population employed in industry (IPUMS `INDGEN` codes 20/30/40/50) among those with known employment status (`EMPSTAT` 1–3), at the last usable census of each country. Uganda is absent: its 2014 census carries a class-of-worker variable but no industry code. Zambia 2000 has industry coverage for 81% of employed urban residents, against 97–101% elsewhere. Panels, specification, rolling window and conventions as in Figure S7. Neither half is weakly identified here. $N = 215{,}556$ corridor-years, 295 destinations, 7 countries. *Source:* Authors' computations using IPUMS census and CHIRPS climate data.

Figure S9: Where Drought Sends Migrants: Corridors by Origin–Destination Distance



Notes: Corridors are split at the *within-country* median of the great-circle distance between the centroids of the origin and destination administrative units. This classifier is at the corridor rather than the destination level, so the bottom row counts corridors. Distance is available for 10,375 of 16,846 corridors ($N = 157{,}556$ of 241,540 corridor-years); on this sub-sample the pooled low-quality-housing reduced form is already insignificant (0.75×10^{-3}, $p = 0.115$, against 1.03×10^{-3}, $p = 0.004$ on the full panel) while the employment margin is unchanged, so panel (b) carries little information. The rolling window retains, at each grid point, the 2,075 corridors nearest to it – the same 20% of units as the 60-destination window of Figure S7. Horizontal axis on a logarithm scale. Other panels, specification and conventions as in Figure S7.
Source: Authors' computations using IPUMS census and CHIRPS climate data.

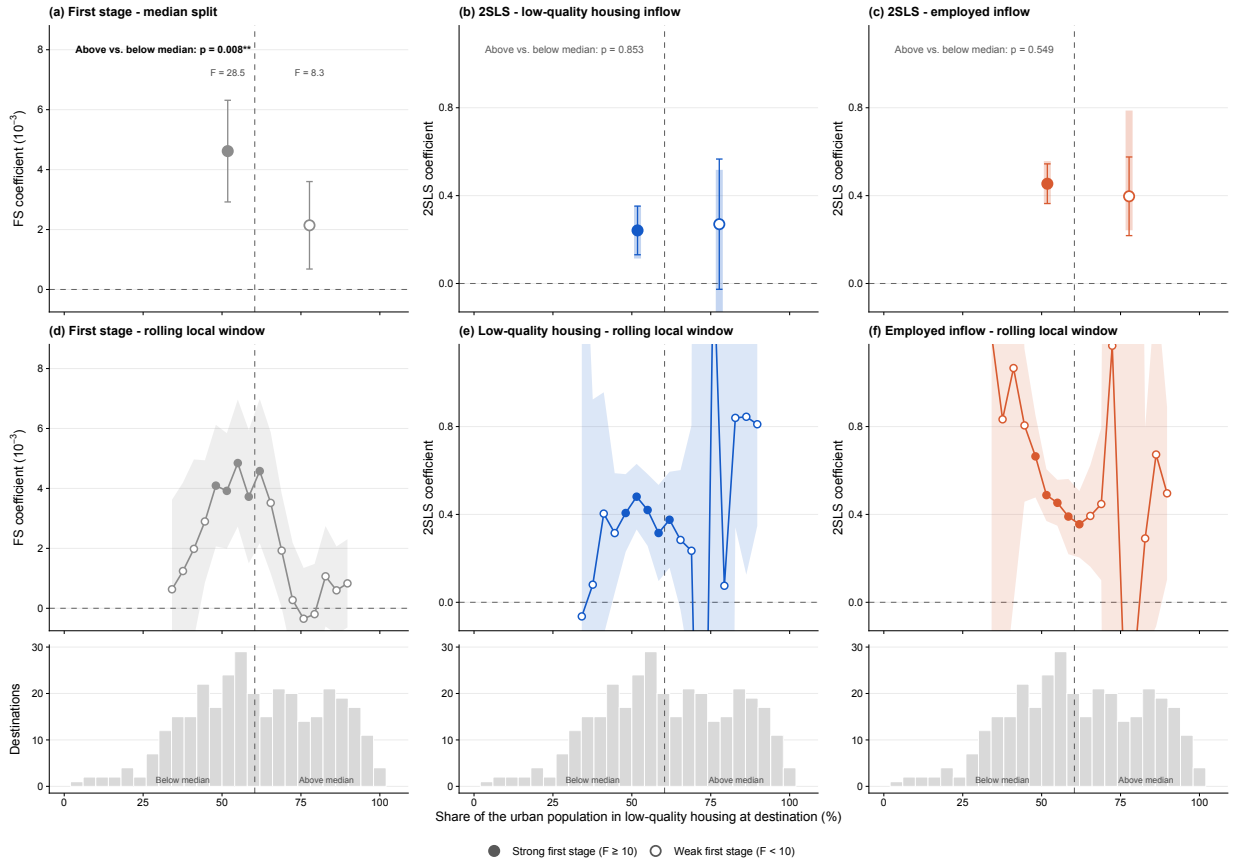
Figure S10: Where Drought Sends Migrants: Destinations by Past Built-Up Expansion



Notes: Destinations are split at the *within-country* median of the growth of their built-up surface between 1975 and 1990, from the Global Human Settlement Layer (GHS-BUILT-S, release R2023A; Pesaresi and Politis, 2023), supplied already aggregated to IPUMS administrative units; for Malawi and Uganda, which enter the panel at the first administrative level, the nested second-level units are summed, an aggregation that reproduces the published first-level totals to within 0.001%. All 351 destinations and all $N = 241{,}540$ corridor-years are matched, so this is the only one of the four splits estimated on the full canonical panel. Panels, specification, rolling window and conventions as in Figure S7. Neither half is weakly identified. The variable is a proxy for the destination's past capacity to build, measured before the migration flows and before the drought shocks that instrument them; it is not a supply elasticity in the sense of Saiz (2010), and it mixes construction capacity with the demand that was already being accommodated. The effect runs opposite to a supply-constraint prediction: both the volume and the housing inflow are larger where past expansion was *faster*.
Source: Authors' computations using IPUMS census, CHIRPS climate and GHSL built-up data.

distance split are weakly identified and their Anderson–Rubin intervals are correspondingly wide, so those two are an absence of evidence rather than evidence of absence. Specification, rolling window and conventions are those of Figures S7–S9; the vertical range of each panel is set from the Wald intervals and from the rolling windows whose first stage exceeds ten, so bands from weakly identified windows extend beyond the plotting range and are truncated.

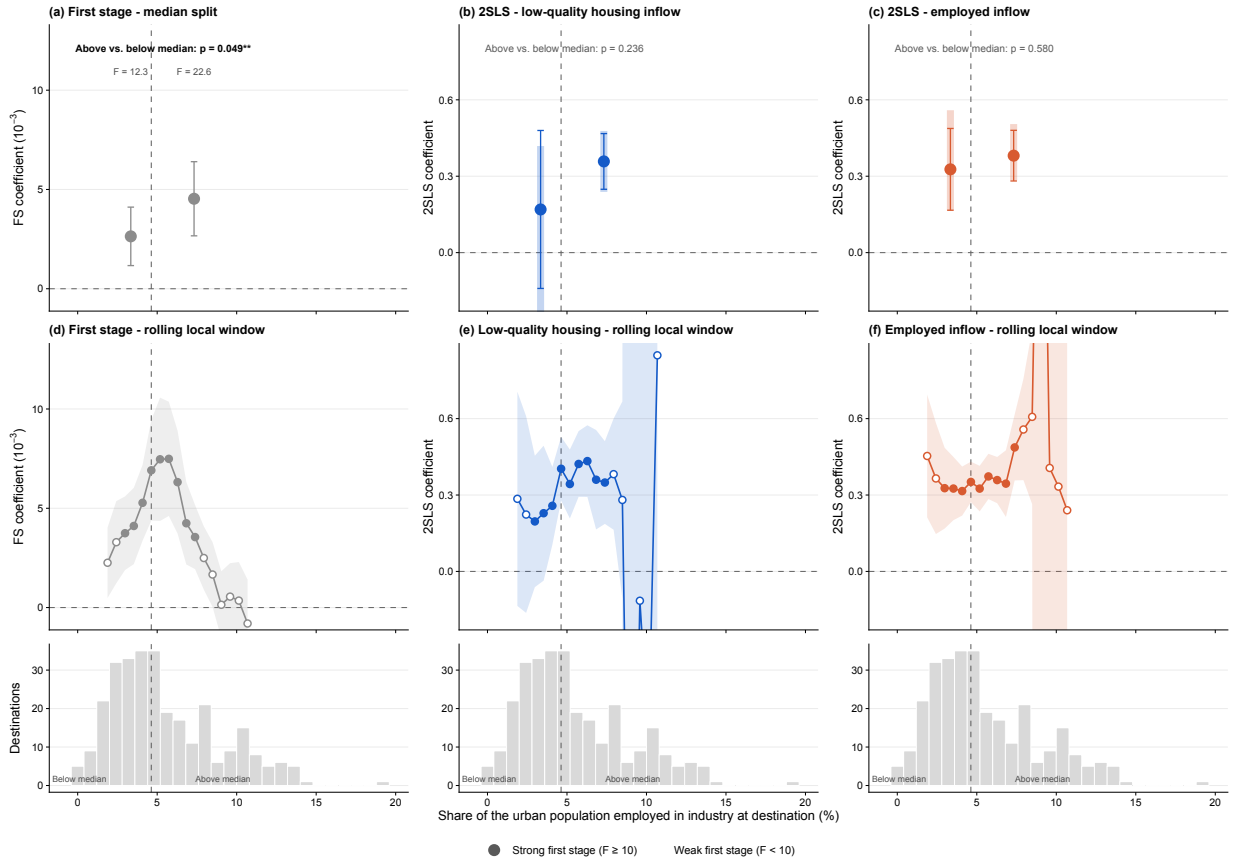
Figure S11: Conversion into Low-Quality Housing and Employment, by Destination Housing Saturation



Notes: Panel (a) reproduces the first stage of Figure S7. Panels (b) and (c) report 2SLS coefficients from the two-bin saturated interacted model, reading as the number, out of 100 marginal drought-induced migrants, in low-quality housing and in employment respectively. Whiskers: 95% Wald intervals; shaded bars: cluster-robust Anderson–Rubin 95% intervals computed on the just-identified subsample. Panels (d)–(f) are the rolling local-window counterparts. Hollow points mark a weak first stage ($F < 10$), reported once per half in panel (a). Neither contrast is significant ($p = 0.853$ for housing, $p = 0.549$ for employment), but the above-median half is weakly identified ($F = 8.3$) and its housing Anderson–Rubin interval, $[-0.39, +0.52]$, contains both zero and twice the below-median estimate.
Source: Authors' computations using IPUMS census and CHIRPS climate data.

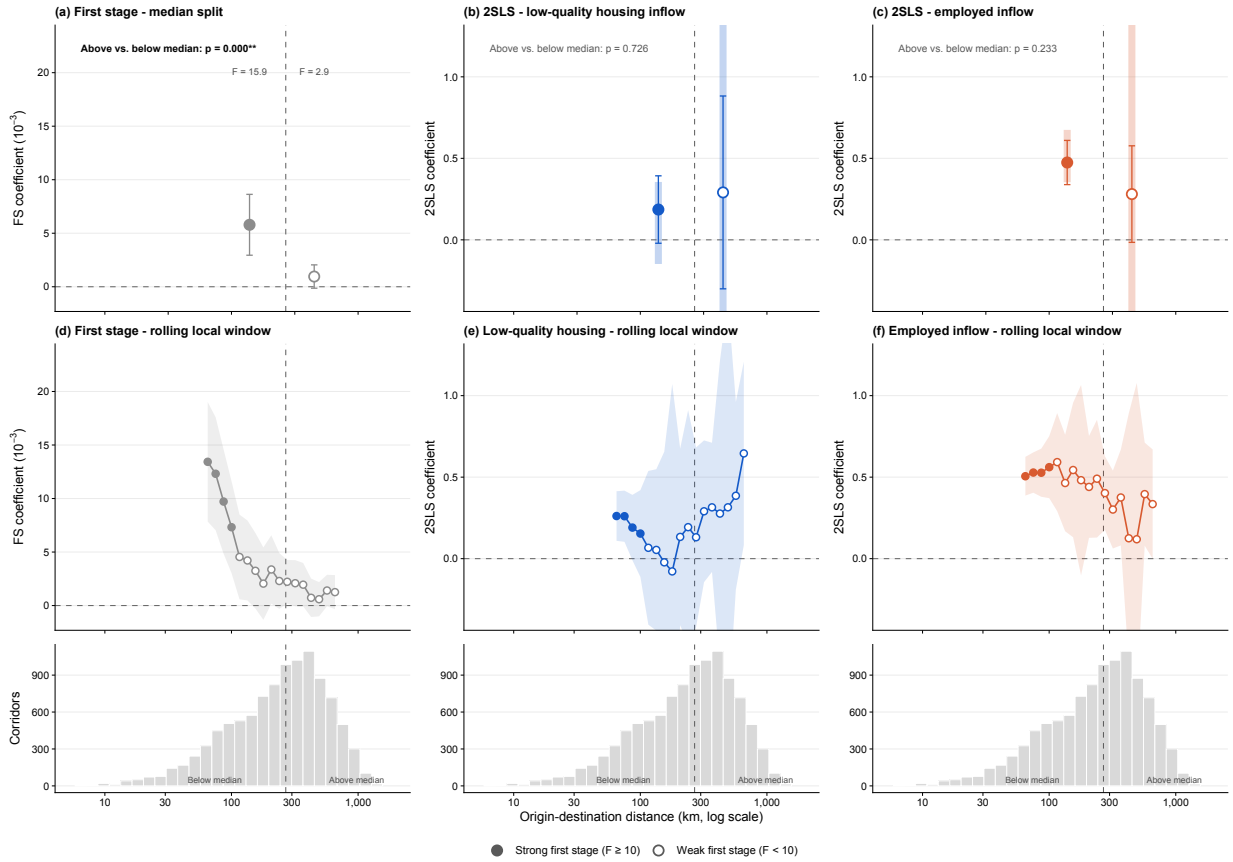
Heterogeneous Effects by Migrant Profile - Tables S7–S10 report the full estimates behind Appendix Figures 17 and 18: the main effect separately by migrant subgroup, on the scale margin (the subgroup's low-quality-housing inflow) and the propensity margin (the share of the subgroup's migrants in low-quality housing), each with subgroup-specific migration rates instrumented by drought under the main specification; the reference column reproduces the

Figure S12: Conversion into Low-Quality Housing and Employment, by Destination Industrial Employment Share



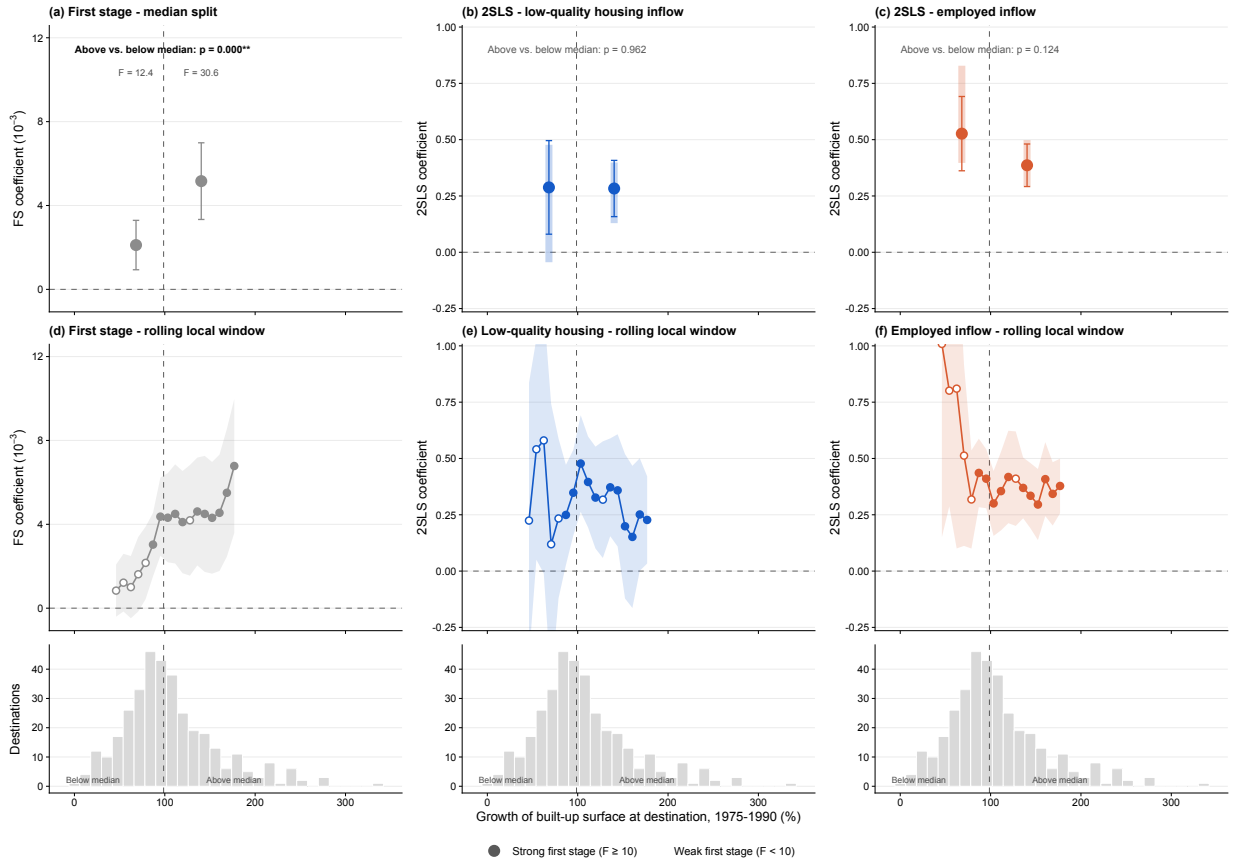
Notes: Panels, specification and conventions as in Figure S11; the classifier is that of Figure S8. Both halves are strongly identified ($F = 12.3$ and 22.6) and neither contrast is significant ($p = 0.236$ for housing, $p = 0.580$ for employment), which makes this the most informative of the three conversion nulls. The below-median Anderson–Rubin interval on housing, $[-0.40, +0.42]$, nonetheless remains compatible with a difference of the plotted size.
Source: Authors' computations using IPUMS census and CHIRPS climate data.

Figure S13: Conversion into Low-Quality Housing and Employment, by Origin–Destination Distance



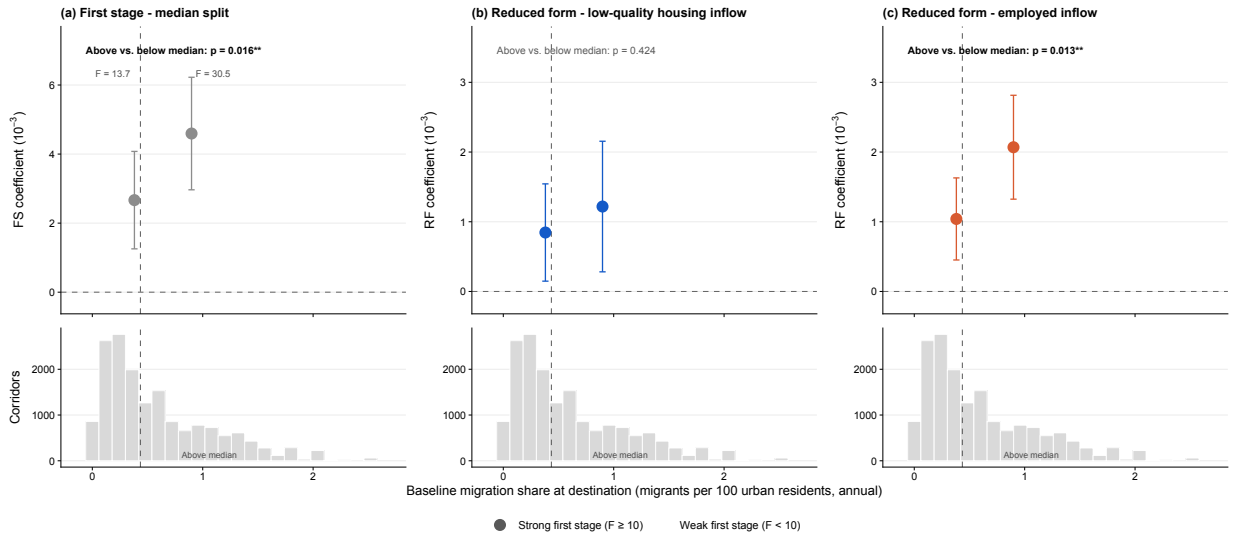
Notes: Panels, specification and conventions as in Figure S11; the classifier is that of Figure S9. The above-median half has a first stage of 2.9 and Anderson–Rubin intervals of $[-3.32, +3.91]$ on housing and $[-1.53, +2.09]$ on employment; the contrasts of $p = 0.726$ and $p = 0.233$ are therefore mechanically non-rejectable and carry no information. The below-median half is well identified ($F = 15.9$), and its housing estimate of 0.186 has an Anderson–Rubin interval including zero, consistent with the pooled housing result already being unidentified on the 65% of corridor-years for which distance is available.
Source: Authors' computations using IPUMS census and CHIRPS climate data.

Figure S14: Conversion into Low-Quality Housing and Employment, by Past Built-Up Expansion



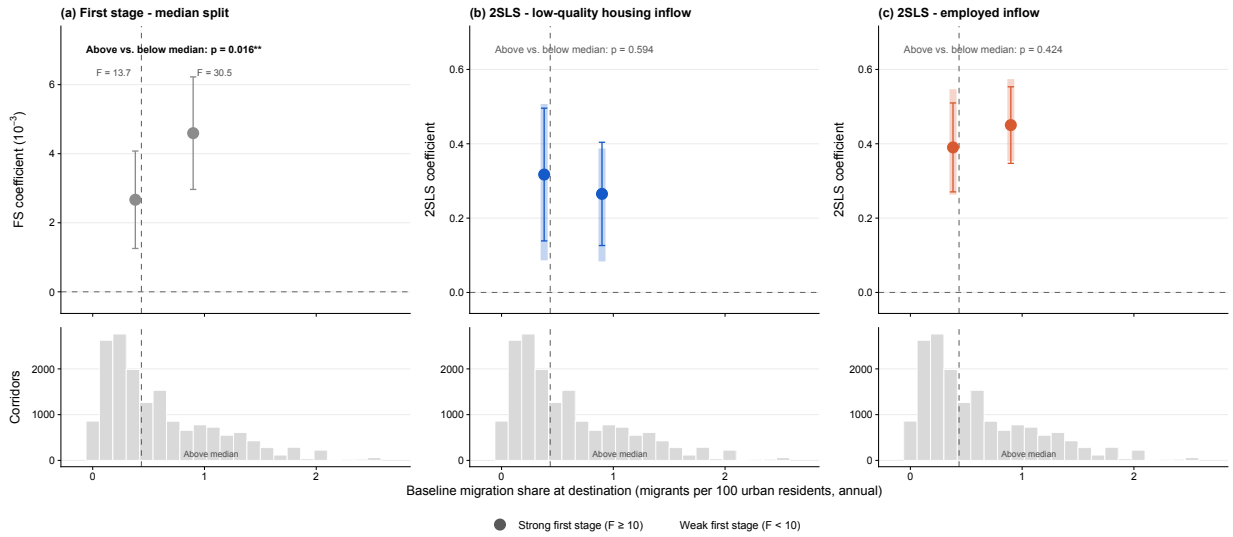
Notes: Panels, specification and conventions as in Figure S11; the classifier is that of Figure S10. Both halves are strongly identified ($F = 12.4$ and 30.6) and the housing conversion is all but identical across them – 0.288 against 0.283 per 100 marginal migrants, $p = 0.962$, with Anderson–Rubin intervals of $[-0.04, +0.48]$ and $[+0.13, +0.40]$. The employment conversion is lower where past expansion was faster (0.527 against 0.386) but the contrast is not significant ($p = 0.124$). Together with the first stage of Figure S10, which is two and a half times larger in the fast-expanding half, this is the clearest case of the pattern described in Section **??**: the destination moves the number of arrivals, not what becomes of them.
Source: Authors' computations using IPUMS census, CHIRPS climate and GHSL built-up data.

Figure S15: Where Drought Sends Migrants: Corridors by the Destination's Baseline Migration Share



Notes: Corridors are split at the *within-country* median of their destination's baseline migration share: the number of urban in-migrants the destination received from *other* origins over the three earliest years of its recall window, per year and per 100 of its urban residents. The leave-own-origin-out construction means a corridor never enters its own classifier, so those three years remain in the estimation sample; a single value per corridor, taken from its earliest census cycle, is applied throughout, which keeps the origin–destination fixed effects nested within bins. This is a descriptive rather than a predetermined characteristic: a destination's baseline migration share is itself the product of everything that makes it attractive. $N = 241{,}540$ corridor-years, 16,846 corridors. Panels and specification as in Figure S7, without the rolling local-window row. Both halves are strongly identified ($F = 13.7$ and 30.5). Drought sends 1.7 times more migrants to destinations that were already migrant-intensive ($p = 0.016$), and the employed-inflow reduced form follows ($p = 0.013$), while the low-quality-housing reduced form does not separate ($p = 0.310$).
Source: Authors' computations using IPUMS census and CHIRPS climate data.

Figure S16: Conversion into Low-Quality Housing and Employment, by the Destination's Baseline Migration Share



Notes: Corridors are split at the *within-country* median of their destination's baseline migration share: the number of urban in-migrants the destination received from *other* origins over the three earliest years of its recall window, per year and per 100 of its urban residents. The leave-own-origin-out construction means a corridor never enters its own classifier, so those three years remain in the estimation sample; a single value per corridor, taken from its earliest census cycle, is applied throughout, which keeps the origin–destination fixed effects nested within bins. This is a descriptive rather than a predetermined characteristic: a destination's baseline migration share is itself the product of everything that makes it attractive.
$N = 241{,}540$ corridor-years, 16,846 corridors. Panels, specification and conventions as in Figure S11. Both halves are strongly identified, so these are informative nulls: the housing conversion is 0.317 against 0.265 per 100 marginal migrants ($p = 0.594$, Anderson–Rubin intervals $[+0.09, +0.51]$ and $[+0.08, +0.39]$) and the employment conversion 0.390 against 0.450 ($p = 0.424$).
Source: Authors' computations using IPUMS census and CHIRPS climate data.

aggregate anchors. Both margins are broadly uniform across gender and age - no within-dimension contrast is significant. The only gradient is educational: on the scale margin, the effect is twice as strong for primary-educated adults as for those with secondary education or above ($p = 0.036$), suggestive given that attainment is measured at census. The oldest and least-educated adults are AR-only on both margins - drought barely moves their migration, itself informative - and the propensity first stages are weak throughout (F between 8 and 13), so propensity inference rests on the Anderson–Rubin intervals: the sorting shift is detected among men and adults, and imprecise for women and children.

Table S7: Heterogeneous Effects by Migrant Profile: Low-Quality Housing Inflows (Scale)

		Gender		*Age at arrival*			*Education, 25+*		
	All	Women	Men	<15	15+	55+	<Prim.	Primary	Sec.+
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Subgroup urban migration rate	0.3106***	0.2799***	0.2862***	0.3242***	0.2535***	AR-only	AR-only	0.3928***	0.2158***
	(0.0858)	(0.0645)	(0.0757)	(0.0926)	(0.0574)			(0.0898)	(0.0606)
First-stage F (SW cond.)	23.86	29.70	25.88	19.96	28.66	0.10	1.95	24.71	25.00
AR 95% CI	[0.08, 0.46]	[0.12, 0.40]	[0.10, 0.42]	[0.08, 0.49]	[0.11, 0.36]	[-16.9, 17.6]	[-1.80, 3.37]	[0.19, 0.57]	[0.09, 0.35]
Contrast p (within dimension)		0.916			0.378			0.092	
Share of migrants (%)	100.0	50.4	49.6	28.5	71.5	1.4	7.8	20.5	22.8
Dist. control	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
OD-pair FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean outcome	0.0123	0.0061	0.0059	0.0050	0.0070	0.0003	0.0019	0.0021	0.0010
Observations	245,292	241,540	241,540	241,540	241,540	241,540	241,540	241,540	241,540
OD pairs	16,888	16,846	16,846	16,846	16,846	16,846	16,846	16,846	16,846

Notes: Heterogeneous effect of drought-induced migration on low-quality housing inflows (main definition), scale margin, by migrant profile. Each column is a separate 2SLS: the outcome is the subgroup's low-quality-housing inflow per 100 urban residents (unwinsorized; country), the endogenous variable its migration rate, instrumented by drought at origin – canonical specification (od-pair and year FE, *dist_census* control, SEs clustered at `admin_origin`). Coefficients read as the number per 100 additional drought-induced migrants of the subgroup settling in low-quality housing; column (1) is the aggregate reference and reproduces the published anchor (0.228). Contrast p-values from the fully saturated stacked interaction. Effects are broadly uniform across gender and age – no significant contrast; the only gradient is educational (primary > secondary+, $p = 0.036$), suggestive since education is measured at census and may post-date arrival. Columns with conditional first-stage F below 8 (55+, less than primary) are AR-only – drought barely moves their migration, itself informative. Propensity-margin counterpart: Table S8. Age at arrival = census age minus years since migration. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Saturated First Stage by Profile - Table S11 reports the fully saturated first stage across the twelve gender × age-at-arrival × education cells - the additivity check behind Section 6.2: selection into migration runs along the educational gradient alone, with all less-than-primary cells near zero across gender and age.

Does Drought Select on Destination Type? - Table S12 reports the linear projection of a causal forest fitted on the first stage, a data-driven check on whether drought directs migrants toward a particular kind of destination. Because the treatment is rainfall at origin, unconfoundedness holds and the forest recovers how the migration response varies with

Table S8: Heterogeneous Effects by Migrant Profile: Share in Low-Quality Housing (Propensity)

		Gender		*Age at arrival*			*Education, 25+*		
	All	Women	Men	<15	15+	55+	<Prim.	Primary	Sec.+
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Subgroup urban migration rate	1.2159**	0.9462	3.1375**	0.0117	3.9751**	AR-only	AR-only	AR-only	AR-only
	(0.5979)	(1.0923)	(1.5943)	(1.2202)	(1.6423)				
First-stage F (SW cond.)	15.50	19.48	10.11	12.53	11.27	2.33	0.70	5.16	7.27
AR 95% CI	[0.17, 2.94]	[-1.19, 3.59]	[0.61, 9.29]	[-3.06, 2.64]	[1.6, 10.6]	[-450.4, 568.2]	[-367.9, 430.1]	[1.6, 91.4]	[1.7, 38.9]
Contrast p (within dimension)		0.177			0.039			–	
Share of migrants (%)	100.0	50.4	49.6	28.5	71.5	1.4	7.8	20.5	22.8
Dist. control	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
OD-pair FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean outcome	0.406	0.413	0.394	0.424	0.395	0.438	0.531	0.395	0.247
Observations	72,723	57,284	58,921	46,618	64,621	8,305	28,734	37,688	34,071
OD pairs	10,537	9,337	9,573	8,478	9,981	2,948	6,657	7,272	6,712

Notes: Heterogeneous effect of drought-induced migration on the propensity margin, by migrant profile. Each column is a separate 2SLS: the outcome is the share of the subgroup's migrants in low-quality housing (unweighted, not winsorized), on corridor-years with a positive subgroup flow (per-column N); the endogenous variable is the subgroup's migration rate, instrumented by drought at origin – canonical specification (od-pair and year FE, *dist_census* control, SEs clustered at `admin_origin`). Column (1) is the aggregate reference and reproduces the published anchor (1.081). Contrast p-values from the fully saturated stacked interaction; the education contrast is not estimable (both columns below the F gate). Propensity shifts are broadly uniform across gender and age – no significant contrast; the shift is detected among men and adults (AR intervals excluding zero) and imprecise for women and children. First stages are weak throughout (F 8–13): inference rests on the cluster-robust Anderson–Rubin intervals. Columns with conditional F below 8 are AR-only – 55+ and less than primary are not identified on this margin either (drought barely moves their migration, itself informative), and all education columns are AR-only here. Education is measured at census and may post-date arrival. Scale-margin counterpart: Table S7. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table S9: Heterogeneous Effects by Migrant Profile: Employed Inflows (Scale)

		Gender		*Age at arrival*			*Education, 25+*		
	All	Women	Men	<15	15+	55+	<Prim.	Primary	Sec.+
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Subgroup urban migration rate	0.4276***	0.3455***	0.5155***	0.0768***	0.6775***	AR-only	AR-only	0.7932***	0.9139***
	(0.0433)	(0.0515)	(0.0524)	(0.0269)	(0.0564)			(0.0805)	(0.0403)
First-stage F (SW cond.)	29.65	29.70	25.88	19.96	28.66	0.10	1.95	24.71	25.00
AR 95% CI	[0.35, 0.53]	[0.25, 0.47]	[0.42, 0.64]	[0.02, 0.14]	[0.57, 0.81]	[-54.6, 52.1]	[-1.84, 3.31]	[0.63, 0.97]	[0.83, 1.00]
Contrast p (within dimension)		0.003			0.000			0.175	
Share of migrants (%)	100.0	50.4	49.6	28.5	71.5	1.4	7.8	20.5	22.8
Dist. control	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
OD-pair FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean outcome	0.0106	0.0042	0.0064	0.0009	0.0097	0.0002	0.0017	0.0032	0.0031
Observations	241,540	241,540	241,540	241,540	241,540	241,540	241,540	241,540	241,540
OD pairs	16,846	16,846	16,846	16,846	16,846	16,846	16,846	16,846	16,846

Notes: Heterogeneous effect of drought-induced migration on employed inflows, scale margin, by migrant profile. Each column is a separate 2SLS: the outcome is the subgroup's employed inflow per 100 urban residents (unwinsorized), the endogenous variable its migration rate, instrumented by drought at origin – canonical specification (od-pair and year FE, *dist_census* control, SEs clustered at `admin_origin`; N = 245,292 per row). Column (1) reproduces the published anchor (0.339). Contrast p-values from the fully saturated stacked interaction. The age contrast is MECHANICAL – children are rarely employed, so their employed inflow is near zero by construction – and should not be read as economic heterogeneity. Employment absorption is otherwise uniform: no significant gender contrast, and no educational gradient ($p = 0.787$), in contrast with the housing margin (Table S7). Columns with conditional first-stage F below 8 (55+, less than primary) are AR-only – drought barely moves their migration, itself informative. Education is measured at census and may post-date arrival. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table S10: Heterogeneous Effects by Migrant Profile: Employed Share (Propensity)

		Gender		*Age at arrival*			*Education, 25+*		
	All	Women	Men	<15	15+	55+	<Prim.	Primary	Sec.+
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Subgroup urban migration rate	1.3945**	2.4854**	1.2001	2.4288**	1.0167	AR-only	AR-only	AR-only	AR-only
	(0.6169)	(1.0848)	(1.3858)	(0.9511)	(1.0656)				
First-stage F (SW cond.)	15.51	19.29	10.17	16.33	11.17	2.29	0.67	4.98	7.26
AR 95% CI	[0.40, 3.32]	[0.69, 5.62]	[-1.78, 5.11]	[0.86, 5.27]	[-0.96, 4.31]	[-665.5, 526.8]	[-284.0, 249.7]	[-3.0, 66.6]	[-4.4, 17.4]
Contrast p (within dimension)		0.434			0.314			–	
Share of migrants (%)	100.0	50.4	49.6	28.5	71.5	1.4	7.8	20.5	22.8
Dist. control	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
OD-pair FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean outcome	0.505	0.385	0.614	0.143	0.638	0.461	0.592	0.702	0.796
Observations	71,503	56,113	57,537	42,415	64,593	8,300	28,717	37,671	34,057
OD pairs	10,452	9,248	9,459	8,047	9,979	2,945	6,652	7,271	6,710

Notes: Heterogeneous effect of drought-induced migration on the employment propensity margin, by migrant profile. Each column is a separate 2SLS: the outcome is the employed share among the subgroup's migrants with known labor-force status (unweighted, not winsorized), on corridor-years with a positive status-known subgroup flow (per-column N); the endogenous variable is the subgroup's migration rate, instrumented by drought at origin – canonical specification. Column (1) reproduces the published anchor (1.319). Contrast p-values from the fully saturated stacked interaction; the education contrast is not estimable (both columns below the F gate). No contrast is significant. The under-15 column is child employment, conceptually thin. First stages are weak throughout (F 9–13): inference rests on the cluster-robust Anderson–Rubin intervals. Columns with conditional F below 8 are AR-only (55+, less than primary, and all education columns on this margin). Education is measured at census and may post-date arrival. Scale counterpart: Table S9. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table S11: Saturated First Stage by Migrant Profile: Who Does Drought Displace?

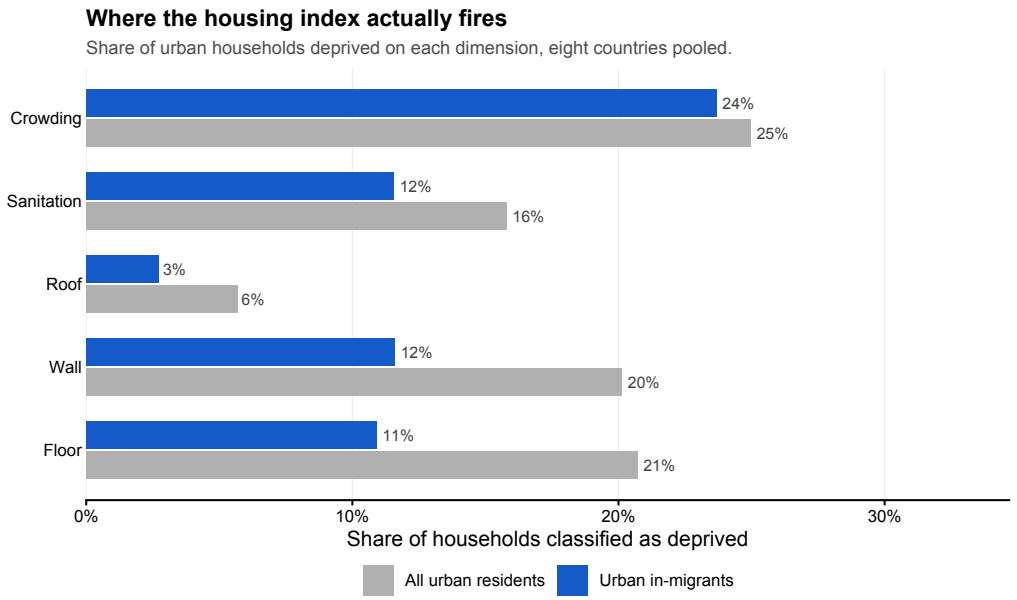
	Share of migrants (%)	Relative first stage (1-SD drought, % of cell mean)	(SE)
Panel A: Secondary education or above			
Women, arrived <30	13.7	8.38***	(1.99)
Women, arrived 30+	4.6	13.56***	(3.59)
Men, arrived <30	16.9	5.92***	(2.01)
Men, arrived 30+	9.4	8.16***	(2.25)
Panel B: Primary completed			
Women, arrived <30	13.7	5.62***	(1.71)
Women, arrived 30+	6.0	7.30***	(2.69)
Men, arrived <30	12.5	5.06***	(1.70)
Men, arrived 30+	8.0	5.14**	(2.03)
Panel C: Less than primary (the trapped margin)			
Women, arrived <30	4.6	2.77	(2.03)
Women, arrived 30+	4.2	3.82	(2.68)
Men, arrived <30	3.2	1.14	(2.93)
Men, arrived 30+	3.2	0.86	(2.66)
Equality of the 12 responses (Wald)		$p = 0.289$	
Interactions beyond additive margins (Wald, 7 terms)		$p = 0.955$	

Notes: Fully saturated first stage by migrant profile: the corridor-year migration rate of each of the twelve gender × age-at-arrival × education cells (urban migrants aged 25+ at census with known attainment), regressed on drought intensity at origin under the canonical specification (od-pair and year FE, *dist_census* control, SEs clustered at `admin_origin`; $N = 245{,}292$ corridor-years per cell). Because rainfall is exogenous, each cell's OLS first stage is valid regardless of instrument strength. The reported effect is the RELATIVE first stage - a 1-SD drought as a % of the cell's own mean migration rate - the scale comparable across cells (raw coefficients mostly reflect cell size); the aggregate value is +6.9%. Joint tests from the stacked model on the standardized outcome, clusters shared. All four less-than-primary cells are near zero (first-stage $F \leq 0.75$) - the trapped-population result, robust across gender and age; its formal test is the pooled contrast ($p = 0.001$) reported in the main text. Education is measured at census and may post-date arrival; the top cell (women arrived 30+, secondary+) is only 4.6% of migrants. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

corridor and destination characteristics; the same exercise would be invalid on the second stage, where the migration rate is endogenous. The response does vary across corridors – the calibration test rejects homogeneity – but only corridor distance and origin urban share have projection coefficients distinguishable from zero. No destination characteristic does, including the urban share that defines the bins of Figure S4. The forest also recovers, without being told, the distance decay imposed by hand elsewhere in the paper: the predicted response falls from 0.0061 in the nearest within-country distance tercile to 0.0032 and 0.0019.

Composition of the Housing Index - Figures S17 to S20 open up the index dimension by dimension. Figure S17 pools the eight countries and shows which dimensions actually drive the measure; Figures S18 and S19 report, country by country, which material and facility categories fall on the deprived side of each rule; Figure S20 shows the distribution of persons per habitable room against the three-person threshold.

Figure S17: Which Dimensions Drive the Housing Index



Notes: Urban households, person-weighted. An in-migrant is a person whose unit of origin differs from the unit of enumeration and whose arrival year falls in that census's retrospective window. Water is omitted throughout: deprivation is nil in every sample, because these censuses distinguish only types of piped water while UN-Habitat counts boreholes and protected dug wells as improved sources. Share of urban households deprived on each dimension, eight countries pooled. Crowding is the only dimension on which in-migrants and residents coincide; on every material dimension in-migrants are about half as often deprived.

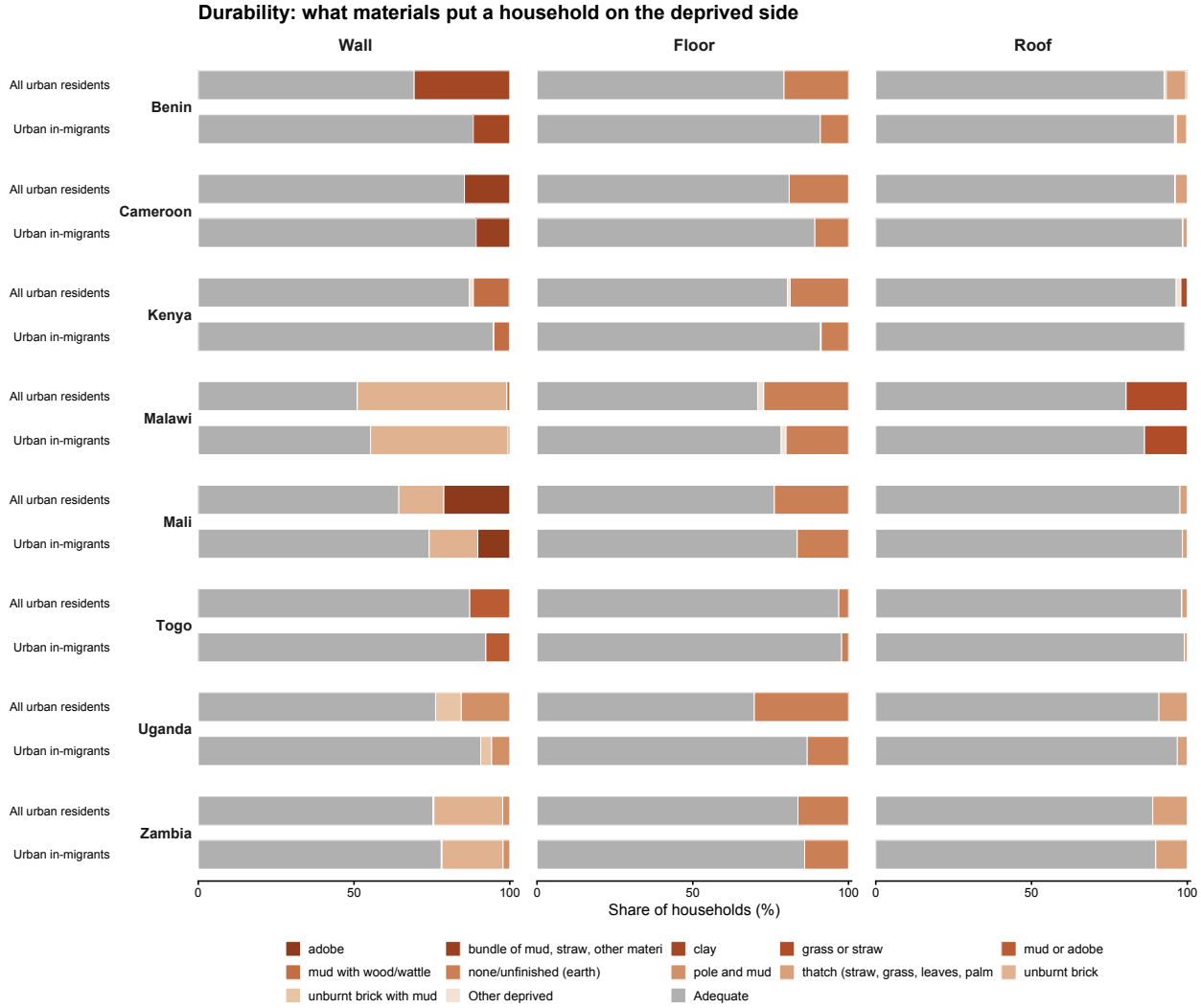
Source: Authors' computations from IPUMS census microdata.

Table S12: Does Drought Select on Destination Type? Linear Projection of the First-Stage Causal Forest

	Projection coefficient (per 1 SD, $\times 10^{-3}$)	(SE)	p-value
Corridor			
Distance, within-country percentile	-2.424**	(1.119)	0.030
Distance (km)	0.017	(0.900)	0.985
Origin			
Urban share	-1.691**	(0.738)	0.022
Growing-season rainfall (mm)	-1.268	(0.968)	0.190
Destination			
Urban share	-0.847	(0.549)	0.123
Baseline low-quality housing share	-1.081*	(0.561)	0.054
Baseline employed share	0.186	(0.582)	0.749
Growing-season rainfall (mm)	-0.061	(0.805)	0.940
Population density (per km^2)	0.055	(0.247)	0.823
Mean $\hat{\tau}(x)$	0.00369 (pooled first stage 0.00362)		
Calibration test, differential forest prediction	1.105 ($p = 1.4 \times 10^{-22}$)		
Corridor-years / corridors / origin clusters	241,540 / 16,846 / 354		
Trees	1000		

Notes: Best linear projection of the conditional average partial effect $\hat{\tau}(x)$ estimated by a causal forest (`grf`, 1000 trees) of drought intensity at origin on the corridor's urban migration rate, with standard errors clustered at `admin_origin`. The projection, not the forest's variable-importance ranking, is the object to interpret: importance splits credit arbitrarily between correlated covariates - the two distance measures and the two rainfall measures here - and ranked destination urban share second although its projection coefficient is indistinguishable from zero. The forest has no fixed effects, so the migration rate and drought intensity are first residualized on origin–destination pair and arrival-year fixed effects with the *dist_census* control and the forest is fitted on the residuals ($\hat{Y} = \hat{W} = 0$): the identifying variation stays within corridor, while the covariates, being corridor- and destination-level and time-invariant, carry the heterogeneity across corridors. As an anchor, the mean of $\hat{\tau}(x)$ reproduces the pooled first stage (0.00369 against 0.00362). Coefficients are reported per one standard deviation of the covariate and in units of 10^{-3}, because the raw units are not comparable (kilometres, millimetres, persons per km^2, and 0–1 shares); this rescaling is monotone and leaves p-values unchanged. The calibration test rejects the absence of heterogeneity ($p = 1.4 \times 10^{-22}$), so the response does vary across corridors - but only corridor distance and origin urban share have projection coefficients distinguishable from zero, and no destination characteristic does. The two destination composition covariates are measured among migrants, not residents, since the estimation panel contains no resident population: they are the low-quality-housing and employed shares of urban in-migrants arriving in the destination's earliest census window, computed leave-one-origin-out with a 30-observation floor, so their nulls are partly a measurement limitation rather than established absences of selection. This is a first-stage validation exercise - drought is exogenous, so unconfoundedness holds - and says which destinations drought directs migrants toward; it is not an estimate of the effect of migration on housing. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

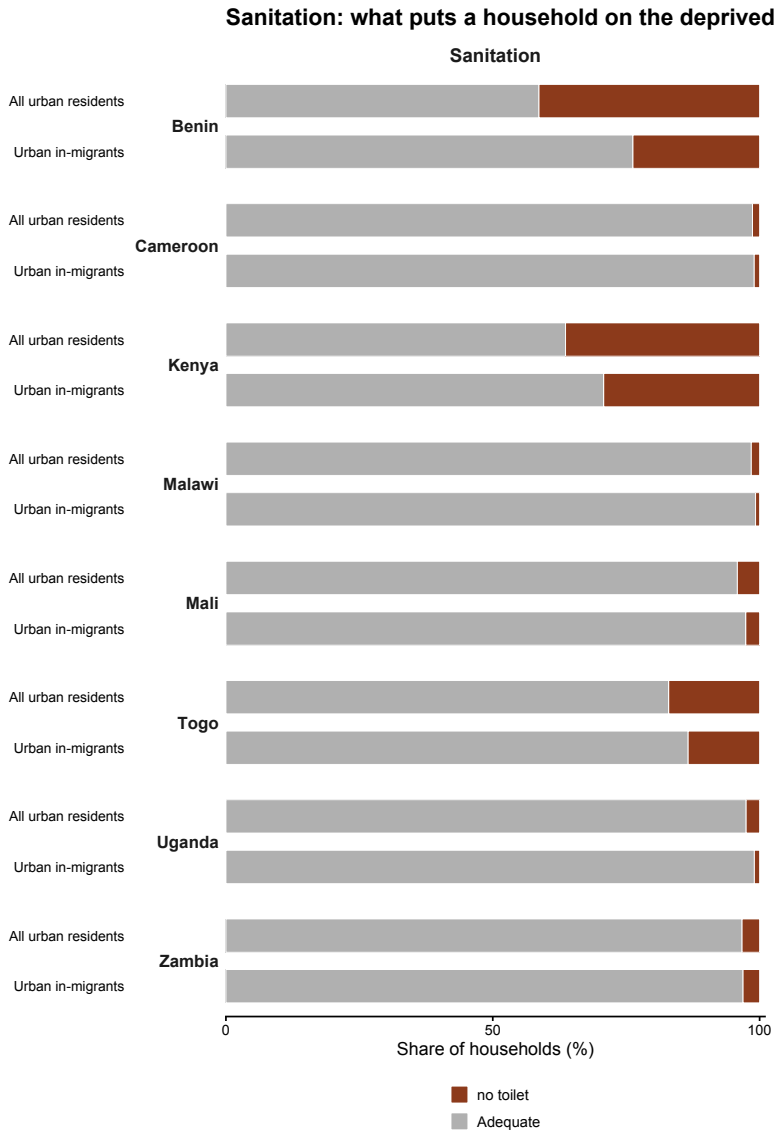
Figure S18: Durability: Materials on the Deprived Side of the Rule



Notes: Urban households, person-weighted. An in-migrant is a person whose unit of origin differs from the unit of enumeration and whose arrival year falls in that census's retrospective window. Water is omitted throughout: deprivation is nil in every sample, because these censuses distinguish only types of piped water while UN-Habitat counts boreholes and protected dug wells as improved sources. Each bar is the category composition of one dimension for one country and population. Deprived categories are named individually and shown in warm shades when they exceed 5 per cent of a bar; all adequate categories are pooled into a single grey, since the question here is where the deprivation cut falls. Non-permanent wall materials are the unburnt-brick, adobe, mud and plant families; burnt or stabilised brick, cement-stabilised earth, stone, concrete, metal and wood count as permanent. Sheet-metal roofs count as adequate.

Source: Authors' computations from IPUMS census microdata.

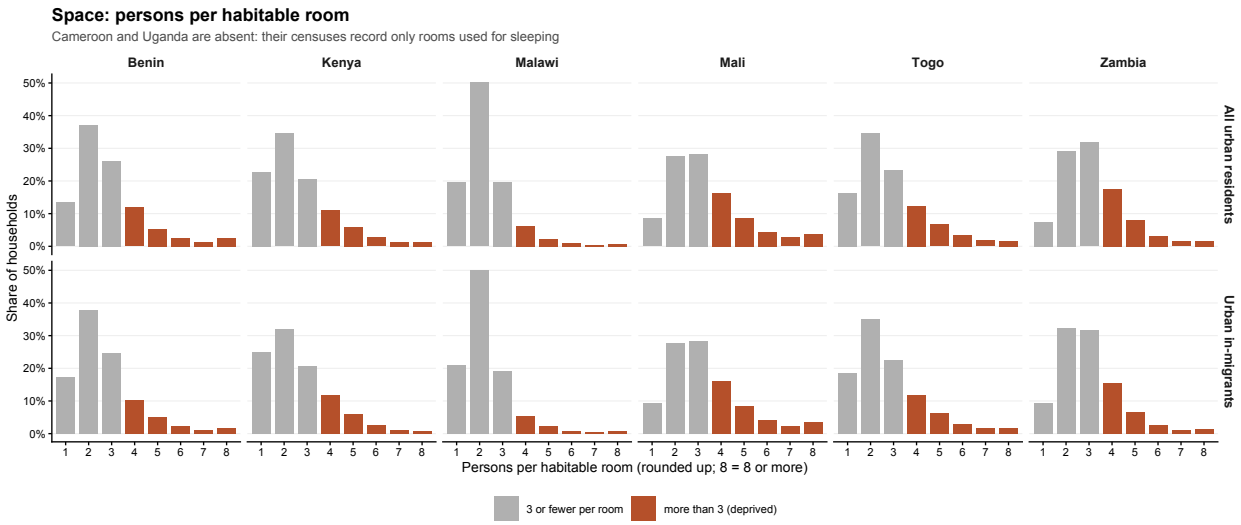
Figure S19: Sanitation: Facilities on the Deprived Side of the Rule



Notes: Urban households, person-weighted. An in-migrant is a person whose unit of origin differs from the unit of enumeration and whose arrival year falls in that census's retrospective window. Water is omitted throughout: deprivation is nil in every sample, because these censuses distinguish only types of piped water while UN-Habitat counts boreholes and protected dug wells as improved sources. Only the complete absence of a toilet counts as deprivation. Pit latrines, which account for roughly half of the pooled sample, are not counted: IPUMS records them in a single undifferentiated category, and UN-Habitat treats an improved pit latrine with a slab as adequate sanitation, so the conservative reading is adopted.

Source: Authors' computations from IPUMS census microdata.

Figure S20: Living Space: Persons per Habitable Room



Notes: Urban households, person-weighted. An in-migrant is a person whose unit of origin differs from the unit of enumeration and whose arrival year falls in that census's retrospective window. Water is omitted throughout: deprivation is nil in every sample, because these censuses distinguish only types of piped water while UN-Habitat counts boreholes and protected dug wells as improved sources. A household is deprived when more than three people share a habitable room, the UN-Habitat threshold. Habitable rooms exclude kitchens, corridors, verandas and bathrooms. Cameroon 2005 and Uganda 2014 are absent: their censuses record only the number of rooms used for sleeping, which is not comparable to a room count, so crowding is left unmeasured there rather than approximated.

Source: Authors' computations from IPUMS census microdata.